Building Cloud Infrastructure

Use **VCF Operations** and the **vSphere Client** as a VI administrator to add and configure infrastructure for consumer and management workloads in the VCF private cloud after you deploy or converge to a VCF fleet and a VCF Instance with its management domain.

- You can start adding workload domains to a VCF Instance and individual clusters and hosts to these workload domains.
  VCF Operations deploys the vCenter and NSX Manager instances for the added workload domains. You can also enable vSphere Supervisor on the default cluster of the workload domain.
- You can import vCenter instances with their managed ESX hosts as workload domains so that they are included in identity and access management, certificate management, and lifecycle management operations performed centrally in VCF Operations .
- You can also add NSX network services to the management domain and the workload domains of a VCF Instance by deploying NSX Edge clusters, transit gateways, and virtual private clouds (VPCs). Then you can configure workloads to use these network services.
  You can place VCF Operations and VCF Automation on NSX overlay or VLAN segments so that they can use the NSX load balancer, or to support planned migration and disaster recovery, or IP mobility.
  You can configure consumer workloads to use the network services either in VCF Automation or directly in the vSphere Client.
- If you want to implement vSAN stretched clusters in a workload domain, you can use the SDDC Manager API for automated stretched cluster configuration.

In This Chapter

- Managing ESX Hosts in VMware Cloud Foundation
- Managing VCF Domains in VMware Cloud Foundation
- Stretching vSAN Clusters in VMware Cloud Foundation
- Managing Network Connectivity in vCenter
- Managing Virtual Private Clouds in vCenter

Getting to Know VCF

To apply this documentation, you must be acquainted with the VCF Product Overview, VCF Design, and VCF Release Notes.

Deploying or Converging to VCF

For information about deploying or converging to a VCF fleet or VCF Instance, see Deployment, Convergence, and Upgrade.

SDKs, APIs, and CLI for VCF Administration

You can build, operate, and manage your VCF private cloud by using the VCF SDK and APIs, and VCF PowerCLI. See Administration SDKs, APIs, and CLI.

Managing ESX Hosts in VMware Cloud Foundation

Managing ESX hosts in VMware Cloud Foundation involves commissioning hosts into the global inventory, assigning them to workload domains, and decommissioning them when they are no longer needed.

To add ESX hosts to the global inventory, you must first create a network pool or expand the default network pool created during deployment. After you have a suitable network pool, you can commission ESX hosts for use in a VCF domain.

The ESX host lifecycle in VMware Cloud Foundation consists of the following stages:

1. Create or expand a network pool. A network pool provides the IP addressing and Layer 2 network specifications for VMkernel adapters on ESX hosts. See Network Pool Management.
2. Commission ESX hosts. Commissioning associates ESX hosts with a network pool and adds them to the global inventory. Newly commissioned ESX hosts appear in the global inventory with a host state of *UNASSIGNED*. The global inventory is the central repository of all ESX hosts that are available to VMware Cloud Foundation, regardless of whether they are assigned to a VCF domain. See Commission ESX Hosts.
3. Assign hosts to a VCF domain. When an ESX host is added to a workload domain, an IP address from the network pool IP range is assigned to it. See View ESX Host Inventory.
4. Decommission ESX hosts. Decommissioning removes ESX hosts from the global inventory. If you want to reuse a host in a different VCF domain, you must decommission, re-image, and commission the host before adding it to the VCF domain. See Decommission ESX Hosts.

See VMware Configuration Maximums for information about ESX host maximums.

Note:

VCF does not support stateless ESX hosts. All ESX hosts must have a local installation of the ESX operating system that persists across reboots.

Network Pool Management

Network pools in VCF manage the IP addressing and Layer 2 network specifications for VMkernel adapters on ESX hosts. Use the following guidance to create, size, and manage network pools for vSAN, Network File System (NFS), Internet Small Computer System Interface (iSCSI), and vMotion traffic.

When you create a workload domain or add an ESX host or vSphere cluster to an existing VCF domain, you only need to enter an IP address manually for the vmk0 management interface. Network pools automatically assign static IP addresses to vSAN, NFS, iSCSI, and vMotion VMkernel ports.

Overview

A network pool in VCF is a logical container that manages the IP addressing and Layer 2 network specifications, including virtual local area network (VLAN) and maximum transmission unit (MTU), for the VMkernel adapters on ESX hosts.

Depending on the storage option, a network pool includes information about subnets reserved for the vMotion and vSAN, vSAN Storage Client, NFS, or iSCSI networks that are required for adding a host to the global inventory.

Table 983: Information Required for a Network Pool

Storage Being Used	Required Networks in Network Pool
vSAN	vMotion and vSAN
vSAN compute	vMotion and vSAN
vSAN storage (formerly known as vSAN Max)	vMotion, vSAN, vSAN Storage Client
NFS	vMotion and NFS
vSAN and NFS	vMotion, vSAN, and NFS
VMFS on FC	vMotion only or vMotion and NFS
vVols on FC*	vMotion only or vMotion and NFS
vVols on iSCSI*	vMotion and iSCSI
vVols on NFS*	vMotion and NFS

* Starting with VMware Cloud Foundation 9.0 and vSphere Foundation 9.0, the vSphere Virtual Volumes capability, also known as vVols, is deprecated and will be removed in a future release of VMware Cloud Foundation and vSphere Foundation. Support for vSphere Virtual Volumes will continue for critical bug fixes only for versions of vSphere 8.x, VMware Cloud Foundation and vSphere Foundation 5.x, and other supported versions until end-of-support for the respective release.

The network pool contains a range of IP addresses for each network. IP addresses from the ranges are assigned to the VMkernel ports on the ESX host. The use of IP ranges allows you to limit the IP addresses that are consumed from a given subnet. You can add more IP ranges to expand the use of the provided subnet.

A default network pool is created when you deploy VMware Cloud Foundation. This network pool is automatically associated with the management domain.

To create a workload domain with hosts in a different layer-2 network than the management domain, you must create a new network pool. Also, if you want to use external NFS or VMFS on FC storage, you must create a new network pool. A network pool can contain both vSAN and NFS networks.

When you create a workload domain with multiple vSphere clusters, those vSphere clusters can share a network pool or each cluster can have its own network pool.

Dual-Stack Configuration

In VCF 9.1, network pools are expanded to support IPv6-only configuration for vmkernel adapters for storage and vMotion traffic, allowing for greater scalability and flexibility in how storage and vMotion traffic is routed.

Traffic Type	Supported IP Version	IP Address Assignment Mode	Description
Storage	IPv4 or IPv6	IPv6 - Static, DHCP, SLAAC IPv4 - Static	Must be dedicated to a single protocol family per cluster (all hosts use IPv6 for vSAN).
vMotion	IPv4 or IPv6	IPv6 - Static, DHCP, SLAAC IPv4 - Static	Allows offloading bulk mobility traffic to IPv6 networks.
NSX Overlay (TEP)	IPv4 Only	Static, DHCP	VCF 9.1 does not support IPv6 for Tunnel Endpoints (TEPs).
Management	IPv4 or Dual-Stack	Static	Management interfaces support static IP assignment.

You cannot mix IP versions for the same traffic type within a cluster. For example, if you assign an IPv6 vSAN network pool to a cluster, all hosts in that cluster must use IPv6 for vSAN.

For Dual-Stack management pools, you must ensure that your DNS servers have both Forward (A/AAAA) and Reverse (PTR) records configured for the components before deployment.

While the Management network can be Dual-Stack (carrying both IPv4 and IPv6 traffic), vSAN and vMotion networks are typically configured as IPv6-Only or IPv4-Only to optimize performance and simplify routing.

Size a Network Pool

Properly sizing a network pool is critical to prevent future issues in the environment due to insufficient IP addresses. Care must be taken when defining the subnets for a network pool as the subnet cannot be changed after it is deployed. The scope of IP addresses used from the defined subnet can be limited by the definition of one or more inclusion ranges. Thus, it is recommended that you begin with defining a larger subnet than what is initially required and utilize the inclusion ranges to limit use. This will provide you the capability to grow with demand as needed.

You begin sizing a network pool by determining the number of ESX hosts that you will have in each vSphere cluster. The minimum number of ESX hosts varies based on the storage type and deployment model. A vSphere cluster can be expanded to the supported maximum number of ESX hosts.

Allocate a minimum of one IP address per ESX host plus enough additional IP addresses to account for growth and expansion of the environment. Ensure that the subnet defined provides enough unused IP addresses and that appropriate inclusion ranges are defined. Note that some of the IP addresses within the subnet will be used for other purposes, such as defining the gateway address, firewalls, or other entities. Use care not to conflict with these addresses.

Here are some important considerations for determining the size of your network pool:

- Type of network architecture.
- Number of hosts that can be placed in each rack or layer-2 network domain.
- Number of hosts required in a vSphere cluster.

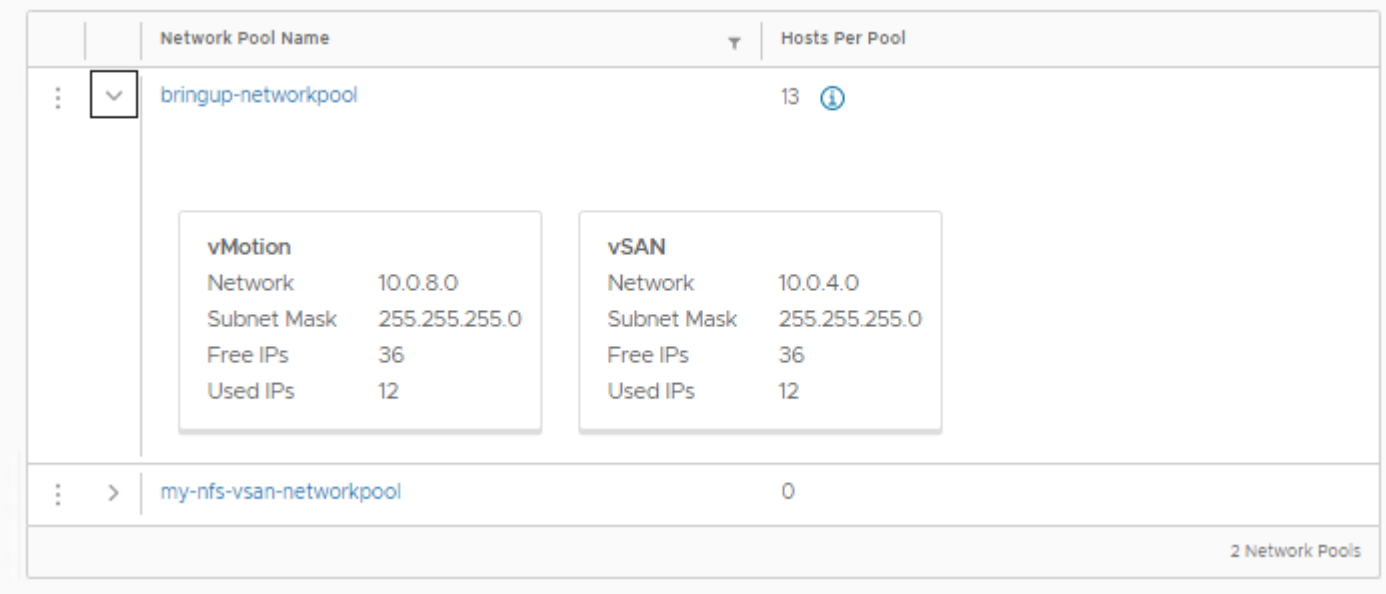
- Number of workload domains and cluster you plan on creating.

View Network Pool Details

You can view the network details for a network pool as well as the total number of used and available (if IPv4) IP addresses.

To perform this task in the vSphere Client, you must have access to the management domain vCenter, or your VCF Instance must be configured with VCF SSO and vCenter linking. See Configuring VCF Single Sign-On and Linking vCenter instances in VCF Operations. As an alternative, you can perform this task using the SDDC Manager UI (https://*<sddc-manager-fqdn>*).

1. In the vSphere Client, click **Global Inventory Lists** > **Hosts** > **Network Pools**.
2. Click the arrow to the left of the pool name.



   A summary view of the network pools details is displayed.

3. Click the name of a network pool.
   A detailed view of the network pool details is displayed, including network types and included IP ranges.

Create a Network Pool

A network pool must include the vMotion network information. Depending on the type of storage you are using, you must provide network information for vSAN, NFS, and iSCSI.

Within a VCF instance, a given subnet must be defined in only one network pool per network type. Defining the same subnet in two network pools is not supported, even if the included IP ranges do not overlap.

To perform this task in the vSphere Client, you must have access to the management domain vCenter, or your VCF Instance must be configured with VCF SSO and vCenter linking. See Configuring VCF Single Sign-On and Linking vCenter instances in VCF Operations. As an alternative, you can perform this task using the SDDC Manager UI (https://*<sddc-manager-fqdn>*).

> **Important:**
>
> Starting with VMware Cloud Foundation 9.0 and vSphere Foundation 9.0, the vSphere Virtual Volumes capability, also known as vVols, is deprecated and will be removed in a future release of VMware Cloud Foundation and vSphere Foundation. Support for vSphere Virtual Volumes will continue for critical bug fixes only for versions of vSphere 8.x, VMware Cloud Foundation and vSphere Foundation 5.x, and other supported versions until end-of-support for the respective release.

1. In the vSphere Client, click **Global Inventory Lists** > **Hosts** > **Network Pools**.
2. Click **Create Network Pool**.
3. Enter a name for the network pool.
4. Select the network type(s).
   - For vSAN ESA/OSA and NFS, you can include both vSAN and NFS network information in the same network pool or create separate network pools for vSAN and NFS.
   - For vSAN storage, select vSAN, vSAN Storage Client, and vMotion.
   - For vSAN compute, select vSAN and vMotion.
   - For VMFS on FC, select vMotion only.
   - For vVols on FC, select vMotion only or vMotion and NFS.
   - For vVols on iSCSI, select iSCSI.
   - For vVols on NFS, select vMotion and NFS.
5. Provide the following network information for the selected network type(s).

Option	Description
IP Scheme	IPv4 or IPv6
VLAN ID	Enter a VLAN ID between 1 and 4094.
MTU	Enter an MTU between 1500 and 9000.
(IPv6 Only) IP Assignment Method	Select between static, DHCP, or SLAAC.
Default Gateway	For IPv6 networking, this options is only available if you select static address assignment. Enter the default gateway (CIDR notation).
Included IP Ranges	For IPv6 networking, this options is only available if you select static address assignment. Enter an IP address range from which an IP address can be assigned to hosts that are associated with this network pool. The IP address range must be from within the specified subnet. You cannot include the IP address of the default gateway in the IP address range. You can enter multiple IP address ranges. **Note:** Ensure that you have entered the correct IP address range. IP ranges cannot be edited after the network pool is created.

6. Click **Save**.

Add a Network Pool IP Address Range

You can edit a network pool to add a new IP address range.

To perform this task in the vSphere Client, you must have access to the management domain vCenter, or your VCF Instance must be configured with VCF SSO and vCenter linking. See Configuring VCF Single Sign-On and Linking vCenter instances in VCF Operations. As an alternative, you can perform this task using the SDDC Manager UI (https://*<sddc-manager-fqdn>*).

1. In the vSphere Client, click **Global Inventory Lists** > **Hosts** > **Network Pools**.
2. Click the vertical ellipsis (three dots) in the network pool row you want to edit and then click **Add additional Network Pool IP Range**.
3. Enter the start and end IP addresses and click **Add**.

   For IPv6, this option is available only when address assignment is set to static.

Rename a Network Pool

You can rename an existing network pool.

To perform this task in the vSphere Client, you must have access to the management domain vCenter, or your VCF Instance must be configured with VCF SSO and vCenter linking. See Configuring VCF Single Sign-On and Linking vCenter instances in VCF Operations. As an alternative, you can perform this task using the SDDC Manager UI (https://*<sddc-manager-fqdn>*).

1. In the vSphere Client, click **Global Inventory Lists** > **Hosts** > **Network Pools**.
2. Click the vertical ellipsis (three dots) in the network pool row you want to edit and then click **Rename Network Pool**.
3. Enter a new name for the network pool.
4. Click **Rename**.

Delete a Network Pool

You can delete a network pool if none of the ESX hosts with an IP address from the pool belong to a workload domain. The default pool created during deployment cannot be deleted.

Ensure that the ESX hosts in the network pool are not assigned to a workload domain. To verify this, view the network pool details and confirm that the **Used IPs** for the network pool is 0.

To perform this task in the vSphere Client, you must have access to the management domain vCenter, or your VCF Instance must be configured with VCF SSO and vCenter linking. See Configuring VCF Single Sign-On and Linking vCenter instances in VCF Operations. As an alternative, you can perform this task using the SDDC Manager UI (https://*<sddc-manager-fqdn>*).

1. In the vSphere Client, click **Global Inventory Lists** > **Hosts** > **Network Pools**.
2. Click the vertical ellipsis (three dots) in the network pool row you want to edit and then click **Delete**.
3. Click **Delete**.

View ESX Host Inventory

The Hosts page in the vSphere Client displays details about all the ESX hosts in the global inventory, including CPU utilization and memory usage across all hosts, as well as the total number of hosts used and unallocated.

For each ESX host, the Hosts page displays details, including name, status, vSphere cluster, and CPU and memory usage.

The Hosts page also provides controls for commissioning hosts.

1. In the vSphere Client, click **Global Inventory Lists** > **Hosts**.
2. Click **Assigned Hosts** to see a list of ESX hosts that are assigned to a workload domain.

   You can view details about each ESX host, including name, vSphere cluster, and CPU and memory usage. Click the host name to view more details about the host in the vSphere inventory. Click the vSphere cluster name to view more details about the cluster in the vSphere inventory.
3. Click **Unassigned Hosts** to see a list of commissioned ESX hosts that are available to create a new workload domain or add to an existing workload domain.

   You can view details about commissioned ESX hosts and also commission and decommission ESX hosts from the **Unassigned Hosts** tab.

Commission ESX Hosts

Adding ESX hosts to the global inventory is called commissioning. You must commission ESX hosts before you can use them to create a new workload domain or add them to an existing VCF domain.

Prepare your hosts. See Preparing ESX Hosts for VCF or vSphere Foundation.

Ensure that each ESX host you commission meets the following criteria:

- Hosts for vSAN-based workload domains are vSAN-compliant and certified on the VMware Hardware Compatibility Guide.

- Hosts for NFS-based workload domains are certified on the VMware Hardware Compatibility Guide.
- Hosts for VMFS on FC-based workload domains are certified on the VMware Compatibility Guide. In addition, the hosts must have supported FC cards (Host Bus Adapters) and drivers installed and configured. For compatible FC cards, see the VMware Compatibility Guide.

  Important:
  The VMFS datastore must be mounted on all the hosts before commissioning.
- Hosts for vVols-based workload domains are certified on the VMware Hardware Compatibility Guide.
  - For vVols on FC-based workload domains, ensure that all ESX hosts have access to the FC array before launching the workflow.
  - For vVols on NFS-based workload domains, ensure that all ESX hosts must be able to reach the NFS server from the NFS network assigned in the IP pool.
  - For vVols on iSCSI-based workload domains, ensure that the iSCSI software initiator must be enabled on each ESXi host and the VASA provider URL must be listed as the dynamic target.

  Important:

  Starting with VMware Cloud Foundation 9.0 and vSphere Foundation 9.0, the vSphere Virtual Volumes capability, also known as vVols, is deprecated and will be removed in a future release of VMware Cloud Foundation and vSphere Foundation. Support for vSphere Virtual Volumes will continue for critical bug fixes only for versions of vSphere 8.x, VMware Cloud Foundation and vSphere Foundation 5.x, and other supported versions until end-of-support for the respective release.
- Each host must have the correct number of physical NICs for the desired network configuration for the host. Hosts can have a single physical NIC or multiple physical NICs. For hosts with multiple physical NICs, only one can be connected to a standard switch, and the NIC must support all management traffic.
- Management IP address is configured on a VMkernel interface connected to the Management Network on the standard switch.
- Host has the drivers and firmware versions specified in the VMware Hardware Compatibility Guide.
- A supported version of ESX is installed on the host. See the *VMware Cloud Foundation Release Notes* for information about supported versions.
- Has both forward and reverse lookup entries (FQDN) configured on the DNS server.
- Host name must be same as the FQDN.
- Self-signed certificate regenerated based on FQDN of host. See Regenerate the Self-Signed Certificate on the ESX Hosts.
- Hardware health status is healthy without any errors.
- All disk partitions on HDD and SSD are deleted.
- A network pool must be created and available before host commissioning.
- The network pool must support the storage type you select for a host (vSAN, vSAN ESA, vSAN Storage Client, NFS, VMFS on FC, vVols). See Network Pool Management.
- Host is configured with an appropriate gateway. The gateway must be part of the management subnet.

To perform this task in the vSphere Client, you must have access to the management domain vCenter, or your VCF Instance must be configured with VCF SSO and vCenter linking. See Configuring VCF Single Sign-On and Linking vCenter instances in VCF Operations. As an alternative, you can perform this task using the SDDC Manager UI (https://`<sddc-manager-fqdn>`).

You can add ESX hosts individually or use a JSON template to commission multiple hosts at once. You can also run multiple commission hosts tasks at the same time.

After you specify host details and select the network pool to associate the hosts with, the platform validates and commissions each host. Each host is added to the free pool and is available for VCF domains and vSphere clusters.

- Hosts that use vSAN storage can only be used with vSAN-based clusters.
  - Hosts that are commissioned for vSAN HCI with ESA enabled can only be used for vSAN ESA clusters.
  - Hosts that are commissioned for vSAN HCI without ESA enabled can only be used for vSAN OSA clusters.
  - Hosts that are commissioned for vSAN Storage (formerly vSAN Max) can only be used for vSAN storage clusters (a subset of vSAN ESA).
  - Hosts that are commissioned for vSAN Compute Cluster can only be used for vSAN compute clusters.

- Hosts that use NFS storage can only be used with NFS-based clusters.
- Hosts that use VMFS on FC storage can only be used with VMFS on FC-based clusters.
- Hosts that use vVols storage can only be used with vVols-based clusters.

See VMware Configuration Maximums for information about the maximum number of hosts you can commission at one time and the maximum number of commission hosts tasks that you can run in parallel.

If you used ESX hosts with certificates generated by an external CA for deployment, hosts used for expanding VCF domains or for creating workload domains must also use certificates generated by an external CA. See Configure ESX Hosts with Signed Certificates.

Note:

When you add a host during host commissioning, the platform temporarily activates SSH on the host in order to retrieve the host key. After it retrieves the host key, the platform deactivates SSH on the host.

With VMware Cloud Foundation 9.0.1 and later, you can use non-certified ESA disks during vSAN ESA cluster configuration for proof-of-concept purposes only. For more information, see kB 408300.

1. In the vSphere Client, click **Global Inventory Lists** > **Hosts** > **Unassigned Hosts**.
2. Click **Commission Hosts**.
3. On the **Checklist** page, review the prerequisites and confirm by clicking **Select All**.
4. Click **Proceed**.
5. Select whether you want to add hosts one at a time or import multiple hosts at once from a JSON file.

Option	Description
Add new	Enter the following information for the host you want to add: • Host FQDN The host FQDN must match the host name (including capitalization) provided during DNS configuration. • Storage Type (vSAN, NFS, VMFS on FC, or vVol) for principal storage – If you select vSAN as the storage type: • Select the vSAN Type (vSAN HCI, vSAN Storage, or vSAN Compute Cluster). • For vSAN HCI choose whether or not to use the vSAN Express Storage Architecture (ESA). – If you select vVols as the storage type, specify the vVols storage protocol type (FC, iSCSI, or NFS). • Network Pool Name (select a network pool from the drop-down menu) • User Name and Password (root credentials for the ESX host) Click **Add**. You can now add more ESX hosts or proceed to the next step.
Import	1. To download the JSON template, click the link. 2. Open the JSON template file and enter information about the hosts to add. – Host FQDN – Username and password (ESX root credentials) – Storage type (VSAN, VSAN_REMOTE, VSAN_ESA, VSAN_MAX, NFS, VMFS_FC, VVOL) for principal storage **Note:** Use VSAN_REMOTE for a vSAN compute cluster. Use VSAN_MAX for a vSAN storage cluster. – Network pool name **Note:** You only need to provide the vVols storage protocol type (FC, ISCSI, or NFS) when you are using VVOLS as the storage type 3. To locate and select the JSON file containing host information, click **Browse**. 4. Click **Upload**.

The ESX host or hosts appear in the **Hosts Added** section.

6. Verify that the server fingerprint is correct for each host and then activate the **Confirm All Finger Prints** toggle.
7. Click **Validate All**.

   The platform validates the host information you provided. Each host is marked as **Valid** or **Invalid**.

   For invalid hosts, you can correct the problem and validate again, or select the host and click **Remove** to proceed with commissioning the valid hosts.
8. Click **Next**.
9. Activate or deactivate **Skip failed hosts during commissioning**.

   This setting is enabled by default. When enabled, hosts that fail are skipped and commissioning continues on the remaining hosts.
10. Click **Commission**.

    The Hosts page appears, and the status of the commission task is displayed in the Tasks window.

    Note:
    Multiple VMkernels are created to test the vMotion network. If MAC address filtering is enabled on your physical infrastructure, this may cause issues such as vMotion network connectivity validation failure.

The commissioned hosts are added to the Unassigned Hosts table.

Decommission ESX Hosts

Removing ESX hosts from the global inventory is called decommissioning. If you want to re-use a host in a different VCF domain, you must decommission, re-image, and commission the host before adding it to the VCF domain.

The hosts that you want to decommission must not be assigned to a VCF domain. If a host is assigned to a VCF domain, you must remove it from the VCF domain before you can decommission it.

To perform this task in the vSphere Client, you must have access to the management domain vCenter, or your VCF Instance must be configured with VCF SSO and vCenter linking. See Configuring VCF Single Sign-On and Linking vCenter instances in VCF Operations. As an alternative, you can perform this task using the SDDC Manager UI (https://*<sddc-manager-fqdn>*).

See VMware Configuration Maximums for information about the maximum number of hosts you can decommission at one time and the maximum number of decommission hosts tasks that you can run in parallel.

1. In the vSphere Client, click **Global Inventory Lists** > **Hosts** > **Unassigned Hosts**.
2. Select one or more hosts and click **Decommission Selected Hosts**.
3. Activate or deactivate **Skip failed hosts during decommissioning**.

   This setting is enabled by default. When enabled, hosts that fail are skipped and decommissioning continues on the remaining hosts.
4. Click **Confirm**.

If the decommissioned host was previously part of a VCF domain, you must re-image it before commissioning it again and adding it to a VCF domain. If the decommissioned host was not part of a VCF domain, you do not have to re-image it before commissioning it again.

Managing VCF Domains in VMware Cloud Foundation

A VCF domain represents a logical unit of application-ready infrastructure that groups ESX hosts managed by a vCenter instance with specific characteristics according to VMware recommended practices. VCF supports two types of VCF domains; the management domain and workload domains.

The management domain is created during the deployment or convergence process by VCF Installer and contains the management components for a VCF instance, including SDDC Manager, vCenter, NSX Manager, and ESX hosts. After the management domain is created, you can deploy or import workload domains to run customer workloads. Workload

domains include a vCenter and NSX Manager (which can be shared with other workload domains), as well as one or more vSphere clusters with ESX hosts.

For more information about VCF domains, see the VMware Cloud Foundation Overview and Design Guide.

After creating or importing a workload domain in VCF Operations, you can add or remove vSphere clusters and ESX hosts using the vSphere Client to support the resource and availability needs of your organization.

Certain operations related to VCF domains, such as expanding or reducing workload domains, stretching or unstretching a cluster, changing the primary storage, creating or deleting datastores, can be performed outside of the VCF Operations UI, which can create a configuration drift in VCF Operations and SDDC Manager. VCF 9.1 introduces specific guardrails which allow making such out-of-band changes without affecting the configuration state of the environment in VCF Operations.

Sharing a vSphere Distributed Switch Across Clusters in a VCF Domain

With VCF 9.1.1 and later, multiple vSphere clusters within a VCF domain can share a single vSphere Distributed Switch (VDS) instead of maintaining a dedicated VDS for each cluster. This capability centralizes network management, optimizes network deployment, and ensures uniform policy application across clusters.

Supported configurations for sharing a VDS between clusters:

- Layer 2 (L2) clusters
- Clusters in the management domain or workload domains.
- Clusters in the same datacenter and the same network pool.
- For full VDS reuse, clusters in the same network pool. For policy reuse and hybrid, clusters can be in different network pools.

Unsupported configurations for sharing a VDS between clusters:

- Layer 3 (L3) clusters
- vSAN stretched clusters, including both the new cluster and the source cluster.
- Partial reuse of port groups. All port groups within a VDS must either be shared or distinct.

Create a New Workload Domain

Creating a new workload domain adds a logical pool of compute, network, and storage infrastructure to a VCF instance.

Requirements for a workload domain with a vSphere cluster (full deployment with cluster):

- Hosts must be commissioned with the target principal storage type. See Managing ESX Hosts in VMware Cloud Foundation.
- If the hosts in the management domain have an express patch applied, the hosts for the new workload domain must be imaged with the same express patch.
- If you are using vVols principal storage type, add a VASA provider to the inventory. See Add a VASA Provider for vVols Storage in Workload Domains.

  Important:

  Starting with VMware Cloud Foundation 9.0 and vSphere Foundation 9.0, the vSphere Virtual Volumes capability, also known as vVols, is deprecated and will be removed in a future release of VMware Cloud Foundation and vSphere Foundation. Support for vSphere Virtual Volumes will continue for critical bug fixes only for versions of vSphere 8.x, VMware Cloud Foundation and vSphere Foundation 5.x, and other supported versions until end-of-support for the respective release.
- If you are using vSAN principal storage type, SSD disks or NVMe disks without pre-existing disk partitions must be available with each host.
- If you are using vSAN as the principal storage type, a vSphere Lifecycle Manager image must exist in the VCF image library before you create the workload domain. See Managing vSphere Lifecycle Manager Images for VMware Cloud Foundation.
- A static IP pool or a DHCP server configured and advertising IP addresses on the workload domain's NSX host overlay (HOST TEP) VLAN.

Note:

With VCF 9.1.1 and later, this is not required for VLAN backed VPC workload domains.

- Each host must have at least one physical NIC, connected to a standard switch.
- A network pool with free IP addresses must be available.
- The IP addresses and Fully Qualified Domain Names (FQDNs) for the vCenter and NSX Manager instances must be resolvable over DNS.
- At least one IP address and a Fully Qualified Domain Names for the Virtual Network Appliance. If you are deploying a cluster of Virtual Network Appliances, you need two IP addresses and FQDNs. Required only when you select **Full Stack VPC** with **Distributed Connectivity** on the Networking page. See Create a Virtual Networking Appliance Cluster.
- Starting with VCF 9.1.1 and later, the first vSphere cluster in a workload domain cannot share a vSphere Distributed Switch (VDS) because no existing VDS is available to reuse. Subsequent clusters that you add to the workload domain can share a VDS with the first cluster.

Requirements for a workload domain without a vSphere cluster (domain infrastructure only):

- A static IP pool or a DHCP server configured and advertising IP addresses on the workload domain's NSX host overlay (Host TEP) VLAN.

  Note:

  Starting with VCF 9.1.1, this requirement does not apply when you deploy a TEP-less workload domain. In TEP-less mode, you realize subnets with layer 2 support using a VLAN-backed VPC, and the Host TEP infrastructure is not required.
- The IP addresses and Fully Qualified Domain Names (FQDNs) for the vCenter and NSX Manager instances must be resolvable over DNS.

Requirements for a dual-stack workload domain:

- To deploy a workload domain with dual-stack (IPv4 and IPv6) networking, SDDC Manager must be configured in dual-stack mode. If SDDC Manager is IPv4-only, the option to create a dual-stack workload domain is not available.
- All ESX hosts for the new workload domain cluster must use the same IP version configuration.
- You can configure vSAN and vMotion to run strictly on IPv6.
- The NSX Overlay network does not support IPv6 or dual-stack networking. It requires a dedicated IPv4 range in your network pool.

For newly-deployed workload domains, NSX Manager and vCenter are deployed on the management domain primary cluster.

1. Log in to VCF Operations.
2. On the VCF Operations top navigation bar, click **Operate** and in the left navigation pane, select **Overview** > **Inventory**.
3. Expand **VCF Instances** and select the VCF instance where you are creating a workload domain.
4. From the **Add workload domain** drop-down menu, select **Create new**.
5. Review the prerequisites, click **Select All**, and click **Proceed.**

   The **Workload Domain** wizard opens.
6. On the **General information** page, configure the details and click **Next**.

Option	Description
Workload Domain Name	Type a name for the workload domain, such as `sfo01`. The name must contain between 3 and 20 characters. It is good practice to include location information in the name since resource object names (such as vSphere Distributed Switch names) are generated based on the workload domain name.
Full deployment with cluster	Deploys a fully provisioned workload domain with an initial vSphere cluster.

Option	Description
Domain infrastructure only	Create a workload domain without a vSphere cluster. The workflow deploys and configures a vCenter instance and deploys and configures a new or joins an existing NSX Manager instance. The new vCenter and NSX Manager instances are deployed in the management domain. No unassigned ESX hosts are required for creating this workload domain. See Create a New Workload Domain without a vSphere Cluster.
Enable vSphere Supervisor	This setting is activated by default, but can be deactivated. When activated, the workload domain is enabled for vSphere Supervisor with NSX VPC Networking, creating a Supervisor, a single Supervisor control plane VM, and a single vSphere zone. For VCF 9.0.x, after the workload domain deployment is complete, you must deploy an NSX Edge cluster with an tier-0 gateway to complete Supervisor activation, unless you are using an existing NSX Manager instance with a compatible NSX Edge cluster. See Configure Centralized Network Connectivity with Edge Clusters for information about deploying an NSX Edge cluster. With VCF 9.1 and later, you can deploy a VNA cluster as part of the workload domain deployment to provide NAT and load-balancing capabilities to vSphere Supervisor. Use the vSphere Client to scale out, upgrade, monitor, and configure vSphere Supervisor after deployment. Click **Learn More** and review the vSphere Supervisor documentation for more information.
Dual-Stack (IPv4 and IPv6) Networking	Enables IPv6 for management, vSAN, and vMotion traffic. Note that NSX Overlay (TEP) traffic remains IPv4-only.
SSO Domain Name	Enter the vCenter Single Sign-On (SSO) domain name, for example, `example.domain.local`. This creates a workload domain that is isolated from the other workload domains in your VCF instance.
Password Creation	VCF automatically generates random complex passwords for any appliances that get deployed during workload domain creation. You can retrieve auto-generated passwords using the SDDC Manager API.

7. On the **vCenter** page, configure the vCenter details and click **Next**.

Option	Description
vCenter FQDN	Enter the FQDN for the workload domain's vCenter. **Note:** The vCenter for a workload domain deployed in the management domain.
vCenter IP Address	Enter the IPv4 and / or IPv6 address of the vCenter.
vCenter Gateway	Enter the IPv4 and / or IPv6 address of the vCenter gateway.

8. On the **Cluster** page, configure the cluster details and click **Next**.

Option	Description
Cluster name	Enter a name for the vSphere cluster.
Manage this cluster by using vSphere Lifecycle Manager images	If you transitioned from vSphere Lifecycle Manager baselines to vSphere Lifecycle Manager images, select this check box.

Option	Description
vSphere Zone Name	If you enabled vSphere Supervisor, enter a vSphere Zone name. The vSphere Zone is created and the cluster is added to it. **Note:** This workflow creates a single vSphere Zone. You can use the vSphere Client to create additional zones.

9. On the **Image** page, configure the settings and click **Next**.

Option	Description
Select an image from the catalog	Select an existing vSphere Lifecycle Manager image from the VCF instance catalog. **Note:** If the cluster image contains a different version of a vendor add-on or component than what is installed on the ESX hosts you add to the cluster, the hosts will be remediated to use the cluster image during cluster creation.
Extract an image from a host in this VCF instance	You can use an existing (assigned or unassigned) ESX host as a reference host to create a new vSphere Lifecycle Manager image (if the corresponding image does not exist or reuse if the image already exists in the VCF image library).

10. On the **Networking** page, configure the NSX Manager details and click **Next**.

    Choose whether to create a new NSX Manager instance or join an existing NSX Manager instance. If you already have an NSX Manager instance for a different workload domain, you can reuse that NSX Manager instance or create a new one.

    If the shared NSX instance is deployed in an IPv4-only VCF domain, the new workload domain must use IPv4 for communication between the components, even if the workload domain itself is configured for dual-stack networking.

    If you are using an existing NSX Manager instance, only the following configurations are supported. In case host transport nodes are not on version 9.1, new NSX 9.1 features remain inactive until all ESXi hosts in all attached workload domains are upgraded to the matching version.

Shared NSX Version	Workload Domain Version	State
NSX 9.1	vCenter 9.1	Full features
	vCenter 9.0.x	Pinned state (features limited to vCenter 9.0)
	vCenter 8.0.x	Pinned state (features limited to vCenter 8.0)
NSX 9.0.x	vCenter 9.0.x	Full features

Shared NSX Version	Workload Domain Version	State
NSX 4.2.x	vCenter 8.0.x	Legacy mode

Option	Description
NSX Manager Instance Options	**Join Existing NSX Manager Instance**: Connects this workload domain to an existing NSX Manager cluster. Choose a compatible NSX Manager from the table. Only NSX Managers meeting the version compatibility requirements for the workload domain are displayed. The workload domain vCenter is automatically configured as a compute manager in the selected NSX instance. In VCF 9.1, you can select the management domain NSX Manager as a shared NSX manager for this workload domain. In VCF 9.1.1 and later, when you join an existing NSX Manager instance, the full stack VPC is shared across all full stack VPC or NSX TEP configured workload domains associated with that NSX instance.
	Create New NSX Manager Instance: Deploys a new, dedicated NSX Manager cluster for this workload domain. You must provide the FQDNs for the new appliances.
Deployment Size	• **Simple**: Deploys single-node NSX Manager requiring fewer resources. • **High-Availability** (recommended): Deploys a three-node NSX Manager cluster to enhance the availability of management and control plane functions, as well as minimize downtime during upgrades.
Appliance Size	• Medium • Large • Extra Large
Appliance 1 FQDN	Enter the FQDN for the first NSX Manager node.
Appliance 2 FQDN	Enter the FQDN for the second NSX Manager node. This is not applicable if you chose **Standard** deployment size.
Appliance 3 FQDN	Enter the FQDN for the third NSX Manager node. This is not applicable if you chose **Standard** deployment size.
Appliance Cluster FQDN	Enter the NSX Manager cluster FQDN.
VPC Network Configuration (This option is only available for VCF 9.1.1 and later)	**Full Stack VPC**: Provides the complete VPC feature set, including all routing, switching, and gateways, to manage the virtual network environment. This option uses TEP-based overlay networking and supports both Distributed Connectivity and Centralized Connectivity models. Select this option to use the standard NSX VPC networking model.
	VLAN-Backed VPC: Provides essential VPC services without TEPs, prioritizing the existing VLAN infrastructure. This option enables TEP-less VPC networking. If you select this option, the **VPC Gateway Connectivity** and **VNA Cluster Details** sections are not displayed unless you select Supervisor configuration. Private CIDRs are not required for the VLAN-backed VPC domain.
VPC Gateway Connectivity (This option is only available if you selected **Full Stack VPC** for your VPC network configuration.)	**Centralized Connectivity**: Routes all north-south traffic through edge node clusters for connectivity to external networks. If you select **Centralized Connectivity**, configure it after the workload domain is created to make it VPC-ready. See Configure Centralized Network Connectivity with Edge Clusters.

Option	Description
	Distributed Connectivity: Routes traffic among the hypervisor hosts using the Virtual Network Appliance. The Tier-0 Gateway and the distributed edge services are not used, because they provide NSX overlay-specific functionality. If you select **Distributed Connectivity**, configure the following: • **VLAN ID**: A dedicated VLAN network is required for the distributed gateway's external connectivity. • **Gateway CIDR IPv4 Address**: The gateway CIDR IP address should be the same as the VLAN default gateway configured in the external router (TOR). • **External IP Block**: Advertised CIDRs that allow outside connectivity to VPC workloads either on public subnets or through external IPs. The external IP block must be the same as, or a sub-block of, the configured Gateway CIDRs. • **Private - Transit Gateway IP Block**: Private CIDRs that are available for inter-VPC (Virtual Private Cloud) communication and are not advertised by the Transit Gateway to the outside datacenter. For Distributed Transit Gateways to provide stateful NAT and load-balancing services, you must later deploy a Virtual Network Appliance cluster. If vSphere Supervisor is turned on, you must configure the Virtual Network Appliance (VNA) Cluster: • Enter the FQDN of the first Virtual Network Appliance. • Optionally enter the FQDNs for the second Virtual Network Appliance. After the workload domain with distributed connectivity is created, it is VPC-ready. For more information about VPCs, see Managing Virtual Private Clouds in vCenter. **Note:** VPC does not currently support IPv6.

11. On the **Storage** page, configure the storage details and click **Next**.

Option	Description
vSAN	Select to configure a vSAN-based workload domain. Select **Enable vSAN ESA** to configure a vSAN ESA workload domain. If you do not select this option, the workload domain is configured with vSAN OSA. **Note:** vSAN storage clusters require vSAN ESA
NFS	Select to configure an NFS-based workload domain.
VMFS on FC	Select to configure a Fiber Channel-based workload domain
vVol	Select to configure a vVol-based workload domain.

a) On the **vSAN** page, configure the vSAN details and click **Next**.

This screen only displays if you selected vSAN storage. The vSAN storage options are different for vSAN OSA and vSAN ESA.

Table 984: vSAN OSA Options

Option	Description
vSAN Cluster Type	• vSAN HCI: Provides storage and compute resources. • vSAN Compute Cluster: Provides compute resources only. Supports a two-node configurations without a witness host.
Data-in-transit Encryption	• vSAN HCI: data-in-transit encryption is a cluster-wide setting. All data and metadata between hosts in the cluster is encrypted. • vSAN Compute Cluster: data-in-transit encryption encrypts all data and metadata between server and client clusters.
Rekey Interval	The rekey interval sets the time duration before vSAN automatically regenerates the host-to-host communication encryption keys. vSAN does not re-encrypt data at rest or reformat disk after the interval expires. The rekey interval is defined in minutes, with the default being 1440 minutes (24 hours).
Failures to Tolerate	Specify the level of availability you want configured for the cluster. Your selection determines the minimum number of required ESX hosts.
Space efficiency	Select the check box to enable vSAN deduplication and compression

Table 985: vSAN ESA Options

Option	Description
vSAN Cluster Type	• vSAN HCI: Provides storage and compute resources. • vSAN Storage**:** Provides storage resources only. You can mount a vSAN Storage datastore on other vSAN ESA or vSAN compute clusters
Network Option (vSAN Storage only)	Select the **vSAN Storage Client Network** checkbox to isolate client or application network traffic from vSAN Storage internal traffic by utilizing dedicated VMkernel ports backed by physical network ports.
Storage Type	Local vSAN datastore. This setting cannot be modified.
Storage Policy	Based on the type of cluster and number of hosts, vSAN creates and assigns a default datastore policy for best capacity utilization after the cluster configuration is completed. Policy details can be viewed in the vSphere Client (**Policies and Profiles** > **VM Storage Policies**).
Data-in-transit Encryption	• vSAN HCI: data-in-transit encryption is a cluster-wide setting. All data and metadata between hosts in the cluster is encrypted. • vSAN Compute Cluster: data-in-transit encryption encrypts all data and metadata between server and client clusters.
Rekey Interval	The rekey interval sets the time duration before vSAN automatically regenerates the host-to-host communication encryption keys. vSAN does not re-encrypt data at rest or reformat disk after the interval expires. The rekey interval is defined in minutes, with the default being 1440 minutes (24 hours).

b) On the **NFS Storage** page, configure the NFS storage details and click **Next**.

This screen only displays if you selected NFS storage.

Option	Description
Datastore Name	Enter a name for the NFS datastore. The datastore name must contain 3-80 characters and cannot contain any leading or trailing whitespace. If you using a previously mounted NFS share, you must use the exact original datastore name from the previous deployment.
Folder	Enter the path to the NFS share. The folder location must contain 3-255 characters and the path cannot contain any whitespace.
NFS Server IP Address	Enter the IPv4 and / or IPv6 addresses of the NFS server.

c) On the **FC Storage** page, configure the FC storage details and click **Next**.

This screen only displays if you selected FC storage.

Option	Description
Datastore Name	Enter the name of the VMFS for the FC datastore. The datastore name must contain from 3 through 80 characters and cannot contain any leading or trailing whitespace.

d) On the **FC Storage** page, configure the vVol storage details and click **Next**.

This screen only displays if you selected vVols storage.

Option	Description
VASA Protocol Type	Select a VASA protocol type. vVols supports FC, NFS, and iSCSI storage protocol types.
VASA Provider	Select a VASA provider name. The VASA protocol type determines the VASA Providers list.
Storage Container	Select a storage container. The selected VASA Provider determines the Storage Containers list. The list does not display storage containers that are already in use by other clusters.
VASA User	Select a VASA user. The selected VASA Provider determines the VASA User list.
Datastore Name	Enter a datastore name. The datastore name must contain 3-80 characters and cannot contain any leading or trailing whitespace.

12. On the **Hosts** page, select the ESX hosts to use for creating the workload domain and click **Next**.

The ESX hosts must be commissioned with the same storage type to be used as the primary cluster of the workload domain. For example, select hosts that are commissioned for vSAN ESA storage for a vSAN ESA workload domain.

13. On the **vSphere Distributed Switch** page, configure the vSphere Distributed Switch details, configure the network traffic types, and click **Create Distributed Switch**.
    a) To use a preconfigured profile, select one of the profiles and click **Next**.

Option	Description
Default	This is the recommended profile. The profile provides a unified fabric for all traffic types using a single vSphere Distributed Switch.
Storage Traffic Separation	This profile creates two vSphere Distributed Switches with separate physical NICs. One switch is dedicated for storage traffic, while the other is used for all other traffic.
NSX Traffic Separation	This profile creates two vSphere Distributed Switches with separate physical NICs. One switch is dedicated for NSX (Virtual Networking) traffic, while the other is used for all other traffic.
Storage Traffic and NSX Traffic Separation	This profile creates three vSphere Distributed Switches with separate physical NICs. One switch is dedicated for NSX (Virtual Networking) traffic, the second switch is dedicated for storage traffic, while the third is used for all other traffic.
Custom Switch Configuration	Copy from a preconfigured profile or create a new distributed switch configuration. Multiple vSphere Distributed Switches can be configured. Each vSphere Distributed Switch can hold one or more network traffic configurations. Some network traffic types are mandatory. Switch configuration is incomplete until these mandatory traffic types are configured. Management, vMotion, vSAN, and NSX network types can be configured only once for a cluster. NSX and Public network traffic types can be configured in multiple vSphere Distributed Switches.

    b) To create a custom configuration for the distributed switch, click **Create custom configuration**.

Option	Description
Distributed Switch name	Enter a name for the distributed switch.
MTU	Determines the largest size of a data packet that can pass through the distributed switch.
Uplinks / LAG configuration	
VDS Uplinks	VDS Uplinks represent the individual physical network adapter (VNICs) on an ESXi host connecting to the physical switch. In this mode, the VDS manages load balancing and failover using software-based algorithms like Route Based on Physical NIC Load (LBT) or Route Based on Originating Virtual Port ID. The physical switch ports do not need to be configured in a bond or channel; they operate as independent links.
Number of uplinks	The number of uplinks dedicated to this distributed switch.
Map uplinks to physical NICs on host	Connects your logical virtual network to your physical network hardware. This mapping pairs a logical uplink (a software object on the distributed switch), with a physical adapter (hardware on the ESXi host).
VDS Lag	VDS LAG bundles multiple physical uplinks into a single logical channel using the Link Aggregation Control Protocol (LACP). This requires a corresponding LACP configuration on the physical ToR (Top-of-Rack) switches. The physical switch and VDS negotiate traffic hashing.
LAG Name	The logical name that is assigned to this group of ports.

Option	Description
LACP Mode	• Active: The VDS actively sends LACP packets to negotiate the link with the physical switch. This is the recommended setting. • Passive: The VDS waits for the physical switch to initiate the LACP negotiation.
LAG Load Balancing	Defines the hashing algorithm used to distribute traffic across the physical links.
LACP Time Out	Determines how often LACP heartbeats (PDUs) are sent: • Slow: 30 seconds • Fast: 1 second
Number of ports	Defines how many physical network adapters (VMNICs) will be bundled into this single logical LAG per host.
Map ports to physical network adaptors on host	Maps the logical ports of the LAG to the actual physical hardware on the ESX hosts.
Traffic types for the distributed switch	• ESX Management • VM Management • vMotion • vSAN • NSX

c) To configure the traffic types for the distributed switch, turn on the toggle for each traffic type and configure the settings.

Option	Description
Distributed portgroup name	Name of the distributed portgroup for this traffic type.
MTU	Determines the largest size of a data packet that can pass through the distributed switch for this traffic type.
Load balancing	*Route Based on Originating Virtual Port (Default)*: the distributed switch maps port IDs to specific uplinks based on the load balancing source of the NSX uplink profile.
	Route Based on Physical NIC Load (LBT): the distributed switch monitors the saturation of the physical uplinks. If the primary uplinks exceeds a set amout of utilization (75% default) for 30 seconds, the distributed switch moves some VM traffic to another active uplink.
	Route Based on Source MAC Hash: the distributed switch uses the VM's MAC address and hashes it to an uplink, which creates even distribution for VMs that might change port IDs but keep MAC addresses.
	Route Based on IP Hash: the distributed switch uses the source and destination IP address of every packet to decide which uplink to use. This allows a single VM to use multiple uplinks simultaneously. **Note:** You must configure LACP (Link Aggregation Control Protocol) or static EtherChannel on the physical Top-of-Rack switches.
	Use Explicit Failover Order: you define the order in which uplinks are used in case of failure.

Option	Description
Uplink	*Active*: This traffic type uses the uplink immediately during normal operation. In case of multiple active uplinks, traffic of this type is distributed across all active uplinks simultaneously.
	Standby: A standby uplink is connected, but no traffic is sent through it as long as at least one "Active" uplink is healthy. If all active uplinks fail, the distributed switch moves the standby uplink to active status.
	Unused: Traffic of this type never uses this uplink, regardless of the status of the remaining uplinks.

14. On the **vSphere Supervisor** page, configure the vSphere Supervisor details and click **Next**.

    Note:
    This screen is not displayed if you deactivated the **Enable vSphere Supervisor** toggle.

    For more information about vSphere Supervisor, click the **Learn More** link.

Option	Description
Supervisor Name	Enter a Supervisor name.
Service CIDR	Enter an IP range for Kubernetes cluster services. The range must not overlap with any Supervisor-routable networks.
Use ESXi Management VMK settings	Select the check box to use the ESX Management VMK VLAN ID, subnet mask, gateway, and vSphere Distributed Switch for the management network.
VLAN ID	Enter a VLAN ID for the management network. This option does not appear if you selected **Use ESXi Management VMK settings**.
Control Plane IP Range	Enter an IP range with minimum 5 addresses for control plane VMs and management traffic.
Subnet Mask	Enter a network mask to define the IP range size. This option does not appear if you selected **Use ESXi Management VMK settings**. For example: `255.255.255.0`
Gateway	Enter a gateway IP for external communication. This option does not appear if you selected **Use ESXi Management VMK settings**
VDS	Select a vSphere Distributed Switch for the management network. This option does not appear if you selected **Use ESXi Management VMK settings**.
NSX Project	Select an NSX project for system and namespace VPC deployment. **Note:** This option is not configurable if you are deploying a new NSX Manager instance.
VPC Connectivity Profile	Select a profile for configuring VPC transit gateway connections. **Note:** This option is not configurable if you are deploying a new NSX Manager instance.

Option	Description
Private (Transit Gateway) CIDR	Enter an IP range for the private transit gateway. **Note:** This option is not available if you selected VLAN-backed VPC for your VPC network configuration.
Private CIDR	Enter a non-overlapping IP range for private subnet allocation. **Note:** This option is not available if you selected VLAN-backed VPC for your VPC network configuration.
Workload DNS	Enter DNS server addresses for all workload VMs.
Workload NTP	Enter routable NTP server addresses for workload time synchronization.

15. Review the information about the workload domain and click **Finish**.

    After successful validation, workload domain creation starts.

16. Click **Fleet Management** > **Tasks** to monitor progress.

If you enabled vSphere Supervisor with Centralized Transit Gateway option selected, you must deploy an NSX Edge cluster with a tier-0 gateway to complete Supervisor activation, unless you are using an existing NSX Manager instance with a compatible NSX Edge cluster. See Configure Centralized Network Connectivity with Edge Clusters for information about deploying an NSX Edge cluster.

You do not need to deploy an Edge cluster if you are using VNAs.

Create a New Workload Domain without a vSphere Cluster

Create a workload domain without a vSphere cluster. The workflow deploys and configures a vCenter instance and deploys and configures a new or joins an existing NSX Manager instance. The new vCenter and NSX Manager instances are deployed in the management domain. No unassigned ESX hosts are required for creating this workload domain.

The IP addresses and Fully Qualified Domain Names (FQDNs) for the vCenter and NSX Manager instances must be resolvable over DNS.

A workload domain without a vSphere cluster has the following limitations:

- A workload domain without a vSphere cluster cannot be a management domain.
- You cannot enable vSphere Supervisor for a workload domain without a vSphere cluster.
- You cannot patch or upgrade the components in the workload domain before adding a vSphere cluster. All workload domains that share an NSX Manager with a workload domain without a vSphere cluster cannot be patched or upgraded.

Requirements for a dual-stack workload domain:

- To deploy a workload domain with dual-stack (IPv4 and IPv6) networking, SDDC Manager must be configured in dual-stack mode. If SDDC Manager is IPv4-only, the option to create a dual-stack workload domain is not available.
- The NSX Overlay network does not support IPv6 or dual-stack networking. It requires a dedicated IPv4 range in your network pool.

1. Log in to VCF Operations.
2. On the VCF Operations top navigation bar, click **Operate** and in the left navigation pane, select **Overview** > **Inventory**.
3. Expand **VCF Instances** and select the VCF instance where you are creating a new workload domain.
4. From the **Add workload domain** drop-down menu, select **Create new**.
5. Review the prerequisites, click **Select All**, and click **Proceed.**

   The **Workload Domain** wizard opens.

6. On the **General information** page, configure the settings and click **Next**.

Option	Description
Workload Domain Name	Type a name for the workload domain, such as `sfo01`. The name must contain between 3 and 20 characters. It is good practice to include location information in the name since resource object names (such as vSphere Distributed Switch names) are generated based on the workload domain name.
Deploy infrastructure only	This option creates a workload domain without an ESX cluster. The domain is deployed with vCenter and NSX Manager instances only.
Dual-Stack (IPv4 and IPv6) Networking	This option is only available if SDDC Manager is deployed with dual-stack networking.

7. On the **vCenter** page, configure the vCenter details and click **Next**.

Option	Description
vCenter FQDN	Enter the FQDN for the workload domain's vCenter. **Note:** The vCenter for a workload domain is deployed in the management domain.
SSO Domain Name	Enter the vCenter Single Sign-On (SSO) domain name, for example, `example.domain.local`. This creates a workload domain that is isolated from the other workload domains in your VCF instance.

8. On the **Networking** page, configure the NSX Manager details and click **Next**.

   Choose whether to create a new NSX Manager instance or join an existing NSX Manager instance. If you already have an NSX Manager instance for a different workload domain, you can reuse that NSX Manager instance or create a new one.

   If the shared NSX instance is deployed in an IPv4-only VCF domain, the new workload domain must use IPv4 for communication between the components, even if the workload domain itself is configured for dual-stack networking.

   If you are using an existing NSX Manager instance, only the following configurations are supported. In case host transport nodes are not on version 9.1, new NSX 9.1 features remain inactive until all ESXi hosts in all attached workload domains are upgraded to the matching version.

Shared NSX Version	Workload Domain Version	State
NSX 9.1	vCenter 9.1	Full features
	vCenter 9.0.x	Pinned state (features limited to vCenter 9.0)
	vCenter 8.0.x	Pinned state (features limited to vCenter 8.0)
NSX 9.0.x	vCenter 9.0.x	Full features

NSX 4.2.x	vCenter 8.0.x	Legacy mode

Option	Description
NSX Manager Instance Options	• Join Existing NSX Manager Instance: Connects this workload domain to an existing NSX Manager cluster. Choose a compatible NSX Manager from the table. Only NSX Managers meeting the version compatibility requirements for the workload domain are displayed. The workload domain vCenter is automatically configured as a compute manager in the selected NSX instance. In VCF 9.1, you can select the management domain NSX Manager as a shared NSX manager for this workload domain. • Create New NSX Manager Instance: Deploys a new, dedicated NSX Manager cluster for this workload domain. You must provide the FQDNs, IP addresses, and administrative passwords for the new appliances.
Deployment Size	• **Simple**: Deploys single-node NSX Manager requiring fewer resources. • **High-Availability** (recommended): Deploys a three-node NSX Manager cluster to enhance the availability of management and control plane functions, as well as minimize downtime during upgrades.
Appliance Size	• Medium • Large • Extra Large
Appliance 1 FQDN	Enter the FQDN for the first NSX Manager node.
Appliance 2 FQDN	Enter the FQDN for the second NSX Manager node. This is not applicable if you chose **Standard** deployment size.
Appliance 3 FQDN	Enter the FQDN for the third NSX Manager node. This is not applicable if you chose **Standard** deployment size.
Appliance Cluster FQDN	Enter the NSX Manager cluster FQDN.
VPC Network Configuration (This option is only available for VCF 9.1.1 and later)	**Full Stack VPC**: Provides the complete VPC feature set, including all routing, switching, and gateways, to manage the virtual network environment. This option uses TEP-based overlay networking and supports both Distributed Connectivity and Centralized Connectivity models. Select this option to use the standard NSX VPC networking model.
	VLAN-Backed VPC: Provides essential VPC services without TEPs, prioritizing the existing VLAN infrastructure. This option enables TEP-less VPC networking. If you select this option, the **VPC Gateway Connectivity** and **VNA Cluster Details** sections are not displayed unless you select Supervisor configuration. Private CIDRs are not required for the VLAN-backed VPC domain.
VPC Gateway Connectivity (This option is only available if you selected **Full Stack VPC** for your VPC network configuration.)	**Centralized Connectivity**: Routes all north-south traffic through edge node clusters for connectivity to external networks. If you select **Centralized Connectivity**, configure it after the workload domain is created to make it VPC-ready. See Configure Centralized Network Connectivity with Edge Clusters.

Option	Description
	Distributed Connectivity: Routes traffic among the hypervisor hosts using the Virtual Network Appliance (TEP-less mode). The Tier-0 Gateway and the distributed edge services are not used, because they provide NSX overlay-specific functionality. If you select **Distributed Connectivity**, configure the following: • **VLAN ID**: A dedicated VLAN network is required for the distributed gateway's external connectivity. • **Gateway CIDR IPv4 Address**: The gateway CIDR IP address should be the same as the VLAN default gateway configured in the external router (TOR).

9. On the **Review** page, review the information about the workload domain and click **Finish**.

   After successful validation, workload domain creation starts.

10. Click **Fleet Management** > **Tasks** to monitor progress.

To add a vSphere cluster to the workload domain, see Add an SDDC Cluster to a VCF Domain. After you add a cluster to the domain, all lifecycle operations are available.

Import an Existing vCenter to Create a Workload Domain

If you have an existing vCenter instance, with or without NSX Manager, you can use VCF Operations to import it to a VCF Instance as a workload domain.

- Verify that your existing vSphere infrastructure satisfies the requirements and supported configurations that are listed below.
- Verify that SSH is enabled on the existing vCenter instance.
- Verify that all ESX hosts use FQDNs instead of short names in your vSphere inventory. See Broadcom knowledge base article 425102.
- Verify that SDDC Manager has access to all NSX Manager nodes on port 443.
- Verify that your NSX certificate contains the FQDNs of all NSX nodes and the NSX cluster VIP in the SAN field.
- If you are importing a vCenter without NSX, verify that you have available the NSX install bundle that corresponds to the version of the vCenter.
- If you plan to use an existing shared NSX Manager instance, verify that the version meets one of the following conditions.
  - If no workload domains currently share the instance, the version must be the same or later than the version of the management domain.
  - If one or more workload domains share the instance, the version must be the same or later than the version of all domains that share the instance.

Version Requirements

When you import existing virtual infrastructure, that infrastructure becomes a new workload domain in an existing VCF Instance. All vSphere clusters in the existing vCenter are imported as part of this process. You cannot select a subset of the clusters.

The versions of the imported VCF components determine which VCF features are available for that workload domain. The component with the earliest version defines the effective VCF version for the workload domain. VCF features introduced in later versions are available only after all components in the workload domain reach the target version. To access the full feature set of VCF 9.1, upgrade all components to version 9.1.

The versions of the components you import must be compatible with the version of the VCF instance where you are importing. Verify that all components meet the minimum version requirements:

Table 986: Minimum Component Version Requirements

Component	Version
VCF (instance in which you are importing)	9.0
VMware vCenter	8.0 Update 3a
VMware ESX	8.0 Update 3
(Optional) NSX Manager	4.2.1

Forward-Only Upgrade Requirement for Imported Components

VCF enforces strict upgrade path validation. You can only upgrade your VCF components to a target version that is released after the release date of your current version. The target version can be a major, minor, or a maintenance release. In general, you can upgrade directly to a patch release only if you can also upgrade to its parent maintenance release.

After the import, you can only upgrade your workload domain to a version that is released after the release date of the earliest version of the imported VCF components. If the imported NSX version is released after a specific VCF minor or maintenance release, this workload domain cannot be upgraded to that specific VCF minor or maintenance version.

Before you import, verify your target VCF upgrade paths in the VMware Interoperability Matrix to avoid deploying an NSX version newer than your target VCF release. VCF components might have further limitations, see Broadcom Interoperability Matrix.

NSX Considerations

- If your vCenter instance is earlier than version 9.0 and has no existing NSX registration, during the import procedure, you deploy a new three-node NSX Manager cluster. If your vCenter instance is version 9.0 or later and has no existing NSX registration, during the import procedure, you can choose to deploy a new single-node or three-node NSX Manager cluster. The vCenter instance is registered as Compute Manager in NSX and all eligible vSphere clusters in the domain are prepared for NSX. The version of the deployed NSX is the latest NSX version compatible with the vCenter version, see Broadcom Interoperability Matrix. To change the NSX version, see Broadcom knowledge base article 429205.
  - If the imported vCenter is hosted on the management domain vCenter, the NSX Manager appliances are deployed on the management vCenter.
  - If the imported vCenter is co-located on its own cluster, the NSX Manager appliances are deployed on the vCenter being imported.
- NSX 9.1 does not support vCenter 8.0 Update 3a or later. If your vCenter instance is without an existing NSX registration and you plan to use a shared NSX 9.1 instance or deploy a 9.1 NSX instance during the import procedure, first upgrade your vCenter instance to version 9.1.
- If your vCenter instance has an existing NSX registration, vSphere clusters that are not prepared for NSX are not reconfigured.
- NSX on Distributed Virtual Port Groups (DVPGs) during import:
  - With VCF 9.1.1 and later, the import worklflow no longer activates NSX on Distributed Virtual Port Groups (DVPGs).
  - For VCF 9.1.0.x, the import workflow automatically activates NSX on Distributed Virtual Port Groups (DVPGs). This action automatically activates the Distributed Firewall (DFW) on all DVPGs in the imported cluster, applying firewall rules to existing workloads. The default DFW rules allow all Layer 2 and Layer 3 traffic. After the import completes, you can deactivate NSX on DVPGs by using the NSX UI or the Transport Node Collection API.
- If you are importing an NSX Federation configuration, only the Local Manager is imported. To upgrade such an environment, see Upgrading NSX Global Manager Nodes in a Federated Environment.
- For VCF 9.1 and earlier, if you are importing a vCenter without NSX and do not have TEP traffic configured on your ESX hosts, you do not need to change the default MTU size of 1500.

Supported Import Configurations

The following table lists supported and not supported configurations for importing your existing infrastructure to a workload domain.

Table 987: Supported Configurations

Category	Supported	Not Supported
Storage	• Enough free space for the full deployment of VMware Cloud Foundation. See VMware Configuration Maximums. • Datastore shared across all hosts in the cluster, accessible from all hosts in the cluster, and writable by all hosts in the cluster. • Any supported vSphere storage type. If the default cluster includes multiple datastore types, VCF determines the primary datastore using the following priority: 1. vSAN 2. NFS v3 3. VMFS 4. NFS 4.1 5. iSCSI 6. vVols* **Note:** If you are using a `JSON` specification file to converge, you can use the `existingDatastoreName` property to override the priority order and use a specific datastore. • In VCF 9.1, vSAN Stretched Clusters (ESA, OSA, or ESA storage clusters) with at least 3 ESX hosts in each availability zone and a witness host deployed in a vSAN witness zone that is different from the availability zones • In VCF 9.1.1 and later, vSAN Stretched Clusters (ESA, OSA, or ESA storage clusters) with at least 2 ESX hosts in each availability zone and a witness host deployed in a vSAN witness zone that is different from the availability zones • Two-node vSAN clusters for remote offices/branch offices (ROBO) with two ESX hosts and one witness host for VCF simple deployments. **Note:** For two-node vSAN clusters, the admission control policy should be configured to >= 50% for both memory and CPU. • vSAN OSA clusters, where deduplication and compression are either both activated or both deactivated. • vSAN OSA clusters with activated compression only.	
Network	• vSphere Distributed Switch (VDS) 8.0 or later **Note:** You can move to a vSphere Distributed Switch. See Migrating from Standard to Distributed vSwitch. • Ports aligned with VMware Ports and Protocols • vCenter instance with a temporary IP address • Statically assigned VMkernel IP addresses, except NSX Host TEPs which can include DHCP assigned addresses • VMkernel adapters for different network traffic types (such as management and vMotion) sharing the same port group • Statically and dynamically allocated NSX Host TEP interfaces	• Cisco virtual switches • vCenter instances without a vSphere Distributed Switch • vCenter instance connected to a custom distributed port group • Custom vCenter ports for client connections • Environments using ports not aligned with VMware Ports and Protocols • Dynamically allocated VMkernel IP addresses **Note:** You can move to statically assigned IP addresses. See Edit a VMkernel Adapter Configuration. • Multiple NSX Manager instances managing a single vCenter instance

Category	Supported	Not Supported
Compute	• Virtual machine of vCenter instance hosted on one of the clusters that is managed by the same instance **Note:** You can move the virtual machine to a supported location by using vSphere vMotion. See Migrating Virtual Machines Between vCenter Systems. • Virtual machine of vCenter instance hosted on the default cluster in the management domain • Clusters using vSphere Configuration Profiles • Clusters using vSphere Lifecycle Manager images **Note:** You can convert clusters that use baselines to use images before upgrading to ESX 9.0. See Convert a Cluster or a Host That Uses Baselines Into a Cluster or a Host That Uses vSphere Lifecycle Manager Images. • Clusters using fully automated vSphere DRS • Standalone ESX hosts and single ESX host clusters, if your vCenter instance has an additional cluster that meets the cluster requirements **Note:** You can manually perform the lifecycle management of the standalone ESX hosts and single ESX host clusters by using the vSphere Client. • Clusters with vSphere Supervisor enabled • Clusters with vSphere Supervisor enabled and Avi Load Balancer	• vCenter instances with ELM **Note:** If your vCenter instance does not have existing NSX registrations, you can manually upgrade your vCenter instances to version 9.0 or later and then break ELM for all vCenter instances. See Deactivate Enhanced Link Mode for Standalone vCenter Instances Using the cmsso-util break-elm Utililty. • vCenter instances with a shared NSX Manager in different VCF Instances. • Clusters using baselines for lifecycle management • Clusters using manual or partially automated vSphere DRS **Note:** You can set the automation level to fully automated. See Edit Cluster Settings. • Partial import of clusters – All clusters managed by a vCenter instance are imported together • Clusters that use vCenter High Availability (VCHA) • vCenter instances that are already connected to an SDDC Manager

* vSphere Virtual Volumes (vVols) framework is deprecated in version 9 and using it is not recommended.

Important:

Some functionalities are not available for imported clusters:

- Adding hosts to imported clusters without the use of the vSphere Client
- Password Management of imported ESX hosts via the Password Management Console of VCF Operations. You can use the workaround in the Broadcom Knowledge Base Article 388859 to create the missing ESXi root credential entries in SDDC Manager to enable password management for these hosts.

Day-N Lifecycle Management and Topology Expansion

After you import a workload domain, operations are divided as follows.

- Lifecycle management for the imported vCenter, NSX Manager, and ESX hosts is orchestrated by VCF Operations.
- Topology expansion is performed directly in the VCF components.
  - Add ESX hosts in vCenter.
  - Add new clusters by using the SDDC Manager workflow or directly in vCenter.
  - Add NSX Edge nodes or Edge clusters in vCenter or NSX Manager.

1. On the VCF Operations top navigation bar, click **Operate** and in the left navigation pane, select **Overview** > **Inventory**.
2. Click **Detailed View**, expand **VCF Instances**, and select the VCF Instance where you are importing the existing vCenter.
3. From the **Add workload domain** drop-down menu, select **Import a vCenter**.

   The **Import a vCenter** wizard opens.
4. On the **General information** page, enter a workload domain name and click **Next**.
5. On the **Specify a vCenter** page, select a vCenter and optionally an NSX Manager instance to import as a workload domain and click **Next**.

   Note:

   Your existing components that use IPv4 and IPv6 can be used to create a workload domain that uses dual-stack (IPv4 and IPv6) networking. The components are imported with their current IP configuration. To use the IPv4 and

IPv6 protocols, SDDC Manager must have dual-stack networking enabled. For more information, see Configure Dual-Stack Networking for SDDC Manager.

Option	Description
Select an existing standalone vCenter from the inventory	Select an existing vCenter from the table. If the vCenter is connected to an NSX Manager instance, activate the toggle and enter the following details: • NSX Manager VIP FQDN **CAUTION:** Do not use capital letters in the NSX Manager FQDN. • NSX Administrator Password • NSX Root Password • NSX Audit Password If your existing NSX Manager instance is version 9.0 or higher and has one or more Edge clusters, the import process adds the discovered Edge node VMs to the VCF inventory, including their user credentials. During the import process, the Edge node credential passwords will be reset. You can retrieve the updated passwords from the VCF credential store. Select the **Enable Edge cluster sync and import NSX Edge node VMs** check box to perform the password reset operation for all the nodes in the existing Edge cluster, as well as for any nodes added later.
Specify an external vCenter	Enter the following details for the external vCenter: • vCenter FQDN **CAUTION:** Do not use capital letters in the vCenter FQDN. • vCenter Root Password • SSO user name • SSO user password If the vCenter is connected to an NSX Manager instance, activate the toggle and enter the following details: • NSX Manager VIP FQDN • NSX Administrator Password • NSX Root Password • NSX Audit Password If your existing NSX Manager instance is version 9.0 or higher and has one or more Edge clusters, the import process adds the discovered Edge node VMs to the VCF inventory, including their user credentials. During the import process, the Edge node credential passwords are reset. You can retrieve the updated passwords from the VCF credential store. Select the **Enable Edge cluster sync and import NSX Edge node VMs** check box to perform the password reset operation for all the nodes in the existing Edge cluster, as well as for any nodes added later.

6. Confirm the certificate thumbprints and click **Next**.
   This includes vCenter, NSX Manager (if you are using an existing instance), and additional thumbprints.

7. If you are importing an existing vCenter instance without an NSX registration, choose whether to create a new or join an existing NSX Manager instance, enter the required details, and click **Next**.
   - **Create a new NSX Manager Instance**

Option	Description
Deployment Size	• **Single NSX Manager Appliance**: Deploys single-node NSX Manager requiring fewer resources. You can choose this option only when importing a vCenter 9.0 and later instance. • **NSX Management Cluster** (recommended): Deploys a three-node NSX Manager cluster to enhance the availability of management and control plane functions, as well as minimize downtime during upgrades.
Appliance Size	• Medium • Large • Extra Large
Appliance 1 FQDN	Enter the FQDN for the first NSX Manager node.
Appliance 2 FQDN	Enter the FQDN for the second NSX Manager node.
Appliance 3 FQDN	Enter the FQDN for the third NSX Manager node. This option is applicable if you chose **NSX Management Cluster** deployment size.
Appliance Cluster FQDN	Enter the NSX Manager cluster FQDN.
NSX overlay networking (This option is no longer available for VCF 9.1.1 and later)*	Choose whether or not to use the ESX management VMkernel for NSX overlay networking. With VCF 9.1.1 and later, by default, the import workflow attempts to configure the cluster for VLAN-backed VPC. Clusters can be configured for TEP-based overlay networking with the SDDC Manager API by using the `BrownfieldOverlaySpec`.

* By default, VCF configures the cluster for VLAN-backed VPC. You can configure a cluster for TEP-based overlay networking with dedicated TEPs by using the API and providing a `BrownfieldOverlaySpec`.

The following table shows the NSX configurations that VCF assigns to a cluster during import when deploying a new NSX instance, based on the vCenter, NSX, and ESX versions and the ESX management VMkernel for overlay networking (`isOverlayOverManagementNetwork`) setting.

vCenter version	NSX version	ESX version	ESX management VMkernel for overlay networking (`isOverlayOverManagementNetwork`)	NSX configuration type	Can configure overlay with dedicated TEPs by using the API
Earlier than 9.0	Earlier than 9.0	Earlier than 9.0	true or false	VLAN	Yes
9.0.x	9.0.x	Earlier than 9.0	true or false	VLAN	Yes
9.0.x	9.0.x	9.0.x	false	VLAN	Yes
9.0.x	9.0.x	9.0.x	true	Overlay over ESX VMkernel networking	Yes
9.1 or later	9.1 or later	Earlier than 9.0	true or false	TEP-less	Yes
9.1 or later	9.1 or later	9.0.x (version mismatch)	true or false	Not applicable. Cluster is skipped due to the ESX / NSX version mismatch.	Not applicable (cluster skipped)
9.1 or later	9.1 or later	9.1 or later (matching versions)	true or false	TEP-less	Yes

Note:

The ESX management VMkernel for overlay networking (`isOverlayOverManagementNetwork`) setting affects the NSX configuration type only when vCenter, NSX, and ESX are all version 9.0.x. In that case, setting it to `true` configures overlay over the ESX VMkernel networking instead of NSX VLAN. For all other supported version combinations, the setting does not change the NSX configuration type.

- **Join existing NSX Manager Instance**
  - Select an existing NSX Manager instance from the list and enter the following details.
    - NSX Administrator Password
    - NSX Root Password
    - NSX Audit Password

  If the shared NSX instance is deployed in an IPv4-only VCF domain, the new workload domain must use IPv4 for communication between the components, even if the workload domain itself is configured for dual-stack networking.

  If you are using an existing NSX Manager instance, only the following configurations are supported. In case host transport nodes are on version earlier than 9.1, new NSX 9.1 features remain inactive until all ESXi hosts in all attached workload domains are upgraded to the matching version.

  In VCF 9.1, you can select the management domain NSX Manager as a shared NSX manager for this workload domain.

Shared NSX Version	Workload Domain Version	State
NSX 9.1	vCenter 9.1	Full features
	vCenter 9.0.x	Pinned state (features limited to vCenter 9.0)
NSX 9.0.x	vCenter 9.0.x	Full features
NSX 4.2.x	vCenter 8.0.x	Legacy mode

8. On the **Prechecks** page, run the prechecks and click **Next**.

   Resolve any issues before proceeding. For large environments, you can export and download the validation report to a `CSV` file.

9. Review your settings and click **Finish**.

Create and Configure a Workload Domain with a Layer 3 Multi-Rack vSAN HCI Cluster

A workload domain with a layer 3 multi-rack vSAN HCI cluster provides scalability, resilience, and optimal resource usage for workloads.

The clusters in such workload domains have the following configuration:

- Layer 3 network fabric between the racks.
- A dedicated NSX host sub-transport node profile for each rack.
- Each rack can be configured as a vSAN fault domain

Deploy the L3 multi-rack vSAN workload domain so ESX hosts in different racks communicate over the L3 network and vSAN spans those racks. You then disable the vSAN automatic policy management, set and assign a default vSAN storage policy to match availability targets, and create vSAN fault domains to map racks and protect against rack-level failure.

For more information on the layer 3 multi-rack cluster design, see the *VMware Cloud Foundation Design Guide*.

Deactivate Auto Policy Management for vSAN ESA Cluster

You cannot use auto policy management in a multi-rack cluster when using vSAN fault domains. This must be disabled if vSAN fault domains are configured for each rack.

1. In a Web browser, log in to the workload domain vCenter at `https://<vcenter_server_fqdn>/ui` as **administrator@vsphere.local**.
2. In the **Hosts and clusters** inventory, navigate to the multi-rack vSAN cluster.
3. On the **Configure** tab for the cluster, select **vSAN** > **Services**.
4. In the **vSAN ESA** pane, click **Edit**.
5. Toggle **Auto-policy Management** off and click **Apply**.

Create and Assign a New Default Storage Policy for vSAN ESA Cluster

By default, the vSAN ESA default storage policy is based on the number of hosts in the cluster during creation. To provide for fault domains that allow rack resiliency in a multi-rack cluster, you must create a default storage policy that is compatible with the number of racks.

1. In the vSphere Client for the workload domain vCenter, from the vSphere Client menu, select **Policies and Profiles**.
2. On the **VM Storage Policies** page, click **Create**.
3. On the **Name and description** page, select the vCenter instance that manages the multi-rack cluster.
4. Type a name and a description for the storage policy and click **Next**.
5. On the **Policy structure** page, select **Enable rules for "vSAN" storage**, and click **Next**.
6. On the **Availability** tab of the **vSAN** page, set **Failures to tolerate** according to the configuration of your environment. You use RAID 5 or RAID 6 (erasure coding) for vSAN ESA.
7. On the **Storage Rules** tab, select **All flash** and click **Next**.
8. On the **Storage compatibility** page, verify that the datastore is compatible and click **Next**.
9. On the **Review and finish** page, review the policy settings and click **Finish**.
10. In the **Storage** inventory of the vSphere Client, navigate to the vSAN datastore.
11. On the **Configure** tab, click **General** and next to the Default Storage Policy pane, click **Edit**.
12. Select the storage policy that you want to assign as default to the vSAN datastore and click **OK**.

Create vSAN Fault Domains

For vSAN multi-rack clusters with more than one ESX node per rack create a vSAN fault domain for each rack for rack resilience.

1. In the vSphere Client for the workload domain vCenter, in the **Hosts and clusters** inventory, navigate to the multi-rack cluster.
2. On the **Configure** tab, select **vSAN** > **Fault Domains**.
3. Click the plus icon.
4. In the **New Fault Domain** dialog box, enter the fault domain name, and select the hosts for the fault domain.
5. Click **Create**.
6. Repeat these steps for the other racks.

During a future cluster expansion, add hosts manually to their respective fault domains.

Delete a Workload Domain

You can delete a workload domain in VCF Operations.

- If remote vSAN datastores are mounted on a cluster in the workload domain, then the workload domain cannot be deleted. To delete such workload domains, you must first migrate any virtual machines from the remote datastore to the local datastore and then unmount the remote vSAN datastores from vCenter.

- If you require access after deleting a workload domain, back up the data. The datastores on the workload domain are destroyed when it is deleted.
- Migrate the virtual machines that you want to keep to another workload domain using vMotion.
- Delete any NSX Edge clusters hosted on the workload domain.

When you delete a workload domain, the clusters within that workload domain are deleted and the hosts are returned to the free pool with a `need cleanup` host state.

Deleting a workload domain also removes the components associated with the workload domain from the management domain. This includes the vCenter instance and the NSX Manager cluster instances.

Note:

If the NSX Manager cluster is shared with any other workload domains, it will not be deleted.

Note:

If the workload domain shares an NSX Manager cluster and VCF Operations for networks latency collection is enabled for the NSX Manager cluster, NSX objects from the domain may be referenced and stop the workload domain deletion. Disable the latency collection and delete the workload domain.

The network pools used by the workload domain are not deleted as part of the workload domain deletion process and must be deleted separately.

CAUTION:
Deleting a workload domain is an irreversible operation. All clusters and virtual machines within the workload domain are deleted and the underlying datastores are destroyed.

It can take up to 20 minutes for a workload domain to be deleted. During this process, you cannot perform any operations on workload domains.

1. Log in to VCF Operations.
2. On the VCF Operations top navigation bar, click **Operate**.
3. In the left pane, select **Inventory** > **Detailed View**.
4. Expand **VCF Instances** and select the VCF Instance that hosts the workload domain you are deleting.
5. Select the workload domain you want to delete and, from the **Actions** drop-down menu, select **Delete domain**.
6. On the dialog box, click **Delete Workload Domain**.

Decommission the hosts that were part of the workload domain, then re-image with ESX and commission them again. See Managing ESX Hosts in VMware Cloud Foundation.

Expand a VCF Domain

You can expand the management domain or a workload domain to add resources to support additional workloads or availability.

To expand a VCF domain, you can:

- Add an ESX host from the global inventory to an SDDC cluster.
  By adding an individual host to an existing VCF domain, you can expand the amount of resources contained within an existing SDDC cluster up to the supported vSphere maximums.
- Add a new SDDC cluster to a VCF domain.
  VCF domains support multiple SDDC clusters. You can add an additional SDDC cluster to an existing VCF domain to provide for increased capacity.
  Starting with VCF 9.1.1 and later, when you add a new SDDC cluster to a VCF domain, the cluster can share a VDS with an existing cluster in the same workload domain.

Post Upgrade Steps for NFS-Based VI Workload Domains

After upgrading VI workload domains that use NFS storage, you must add a static route for hosts to access NFS storage over the NFS gateway. This process must be completed before adding a cluster to the workload domain or adding a host to a cluster in the workload domain.

1. Identify the IP address of the NFS server for the VI workload domain.
2. Identify the network pool associated with the hosts in the cluster and the NFS gateway for the network pool.
   a) Log in to SDDC Manager.
   b) Click **Inventory** > **Workload Domains** and then click the VI workload domain.
   c) Click the **Clusters** tab and then click an NFS-based cluster.
   d) Click the **Hosts** tab and note down the network pool for the hosts.
   e) Click the Info icon next to the network pool name and note down the NFS gateway.
3. Ensure that the NFS server is reachable from the NFS gateway. If a gateway does not exist, create it.
4. Identify the vmknic on each host in the cluster that is configured for NFS traffic.
5. Configure a static route on each host to reach the NFS server from the NFS gateway.

   ```
   esxcli network ip route ipv4 add -g NFS-gateway-IP -n NFS-gateway
   ```

6. Verify that the new route is added to the host using the NFS vmknic.

   ```
   esxcli network ip route ipv4 list
   ```

7. Ensure that the hosts in the NFS cluster can reach the NFS gateway through the NFS vmkernel.
   For example:

   ```
   vmkping -4 -I vmk2 -s 1470 -d -W 5 10.0.22.250
   ```

8. Repeat steps 2 through 7 for each cluster using NFS storage.

Add an ESX Host to an SDDC Cluster in a VCF Domain

You can add an ESX host to an SDDC cluster using the vSphere Client. Adding an ESX host to an SDDC cluster in a VCF domain increases the available resources. You can add multiple hosts at the same time.

- Verify that one or more unassigned hosts are available in the Hosts inventory. See View ESX Host Inventory.
- For information on commissioning hosts, see Commission ESX Hosts.
- Verify that the ESX hosts you want to add is in an active state.
- Verify that the ESX host to be added matches the configuration of the ESX hosts already in the SDDC cluster. This allows the SDDC cluster configuration to remain balanced. If the host to be added does not match the pre-existing hosts in the SDDC cluster, the cluster will be unbalanced and a warning is displayed. The warning does not prevent the expansion and can be dismissed if needed.
- Verify that the ESX host you are adding has the same type of principal storage as the existing ESX hosts in the SDDC cluster. A host using NFS for principal storage will automatically use the same NFS configuration as the other hosts in the SDDC cluster. For a host using VMFS on FC, you must configure zoning, mount the associated volumes, and create the datastore on the host before adding the host to a SDDC cluster. A host using vVols for principal storage will automatically use the same vVols configuration as the other hosts in the SDDC cluster.
- If the SDDC cluster to which you are adding hosts uses a static IP pool for the NSX Host Overlay Network TEPs, that pool must include enough IP addresses for the hosts you are adding.
- Different NIC enumeration is allowed.
- To perform this task in the vSphere Client, you must have access to the management domain vCenter, or your VCF Instance must be configured with VCF SSO and vCenter linking. See Configuring VCF Single Sign-On and Linking vCenter instances in VCF Operations. As an alternative, you can perform this task using the SDDC Manager UI (https://*<sddc-manager-fqdn>*).
- If your environment is on version 9.1.1 or later, and the cluster uses a shared VDS, verify the following:
  – The ESX host you are adding does not exceed the maximum of 16 vSphere Distributed Switch (VDS) memberships per ESX host. An ESX host that is already connected to 16 VDS instances cannot be added to a cluster that uses an additional shared VDS.

  - The cluster is not configured as a vSAN stretched cluster. vSAN stretched clusters do not support shared VDS. If the cluster is a vSAN stretched cluster, the host is added to the cluster but does not use a shared VDS.

Starting with VCF 9.1.1, you can add ESX hosts to a cluster that shares a VDS with another cluster.

When you use the vSphere Client to add an ESX host to an SDDC cluster, it uses the cluster's existing VDS for management, vMotion, vSAN, and host overlay traffic. If the cluster shares a VDS with another cluster in the workload domain, the added host connects to the shared VDS. The host uses the same uplinks and port groups that are configured on the shared VDS.

When you add a host to an SDDC cluster managed with a vSphere Lifecycle Manager (vLCM) image, remediation is scoped to only the newly added hosts. Existing, non-compliant hosts in the cluster are not remediated.

You can add hosts to or remove hosts from multiple different SDDC clusters in parallel. For example, you add three hosts to Cluster A, and while that task is running, you can start a separate task to add (or remove) four hosts to Cluster B. See VMware Configuration Maximums for information about the maximum number of add/remove hosts tasks that you can run in parallel.

1. In the vSphere Client, browse to the vSphere cluster in the inventory.
2. Select the vSphere cluster and click **Actions** > **Add Hosts** > **Add Unassigned Hosts**.
3. Select the host to add to the vSphere cluster and click **Next**.

   You can view both VDS and NSX networking configuration on the cluster. Each VDS has two uplinks in it, and you can assign any free VMNICs on the host to the existing uplinks on the VDS. If the cluster uses a shared VDS, the host connects to the shared VDS uplinks. The uplink assignments on the shared VDS are the same as the uplinks configured for the existing hosts in the cluster.
4. Click **Next**.
5. Review the host, switch configuration, and license details and click **Finish**.

If the cluster uses a shared VDS and you made out-of-band changes to the VDS configuration, run the domain synchronization workflow to reconcile the changes. See Remediate VCF Domain Configuration Drift.

Add an SDDC Cluster to a VCF Domain

If you want to add capacity to a VCF domain, you can use the vSphere Client to add an SDDC cluster to a VCF domain.

- Verify that there are unassigned hosts available in the Hosts inventory. See View ESX Host Inventory.
- For information on commissioning hosts, see Commission ESX Hosts.
- Ensure that the hosts you want to add to the SDDC cluster are in an active state.
- If you are using DHCP for the NSX Host Overlay Network, a DHCP server must be configured and advertising IP addresses on the NSX Host Overlay VLAN of the VCF domain. When NSX creates TEPs for the VCF domain, they are assigned IP addresses from the DHCP server.
- To perform this task in the vSphere Client, you must have access to the management domain vCenter, or your VCF Instance must be configured with VCF SSO and vCenter linking. See Configuring VCF Single Sign-On and Linking vCenter instances in VCF Operations. As an alternative, you can perform this task using the SDDC Manager UI.
- If your environment is on version 9.1.1 and later and you plan to reuse an existing VDS from another cluster:
  - For full VDS reuse, verify that the source cluster and the new cluster are in the same network pool.
  - Verify that the source cluster and the new cluster are in the same data center.
  - Verify that the source cluster and the new cluster are not configured as a vSAN stretched cluster.
  - Verify that the total number of VDS in the vCenter Server does not exceed the vSphere maximum of 128 VDS per vCenter Server.
  - Verify that the ESX hosts you select for the new cluster do not exceed the vSphere maximum of 16 VDS memberships per ESX host.

Starting with version 9.1.1, you can create an SDDC cluster that shares a vSphere Distributed Switch (VDS) with an existing cluster in the same workload domain. Sharing a VDS across clusters eliminates the need to create a dedicated VDS for each cluster and ensures uniform network policy application through a centralized switch configuration. See Sharing a VDS across clusters in a VCF domain.

If a workload domain has multiple clusters, each cluster can use a different type of principal storage, as long as all hosts within an SDDC cluster use the same type.

You can run multiple add cluster tasks at the same time.

Important:

Starting with VMware Cloud Foundation 9.0 and vSphere Foundation 9.0, the vSphere Virtual Volumes capability, also known as vVols, is deprecated and will be removed in a future release of VMware Cloud Foundation and vSphere Foundation. Support for vSphere Virtual Volumes will continue for critical bug fixes only for versions of vSphere 8.x, VMware Cloud Foundation and vSphere Foundation 5.x, and other supported versions until end-of-support for the respective release.

1. In the vSphere Client for the VCF domain vCenter, browse to the VCF instance datacenter in the inventory.
2. Select the datacenter and click **Actions** > **New Cluster** > **Create SDDC Cluster**.

   The **Add cluster** wizard opens.
3. On the **General information** page, enter the name of the cluster and click **Next**.
4. On the **Image** page, configure the settings and click **Next**.

Option	Description
Select an image from the catalog	Select an existing vSphere Lifecycle Manager image from the VCF instance catalog. **Note:** If the cluster image contains a different version of a vendor add-on or component than what is installed on the ESX hosts you add to the cluster, the hosts will be remediated to use the cluster image during cluster creation.
Extract an image from a host	You can use an existing ESX host as a reference host to create a new vSphere Lifecycle Manager image.

5. On the **Storage** page, select the storage type for the SDDC cluster and click **Next**.

Option	Description
vSAN	Select to configure a vSAN-based cluster. Select **Enable vSAN ESA** to configure a vSAN ESA cluster. If you do not select this option, the cluster is configured with vSAN OSA. **Note:** vSAN storage clusters (formerly known as vSAN Max) require vSAN ESA
NFS	Select to configure an NFS-based cluster.
VMFS on FC	Select to configure a Fibre Channel-based cluster
vVol	Select to configure a vVol-based cluster.

   a) Enter the vSAN Storage details and click **Next**.

   This screen only displays if you selected vSAN storage. The vSAN storage options are different for vSAN OSA and vSAN ESA.

Table 988: vSAN OSA Options

Option	Description
vSAN Cluster Type	• **vSAN HCI**: Provides storage and compute resources. • **vSAN Compute Cluster**: Provides compute resources only. Supports two-node configurations without a witness host.
Data-in-transit Encryption	For vSAN HCI, vSAN data-in-transit encryption is a cluster-wide setting. All data and metadata between hosts in the same cluster is encrypted. For vSAN compute cluster, vSAN data-in-transit encryption encrypts all data and metadata between server and client.
Rekey Interval	The rekey interval sets the time duration before vSAN automatically regenerates the host-to-host communication encryption keys. vSAN does not re-encrypt data at rest or reformat disk after the interval expires. The rekey interval is defined in minutes, with the default being 1440 minutes (24 hours).
Failures to Tolerate (vSAN HCI only)	Specify the level of availability you want configured for the cluster. Your selection determines the minimum number of required ESX hosts.
Space efficiency (vSAN HCI only)	Select the check box to enable vSAN deduplication and compression.
Mount Remote Datastore (vSAN Compute Cluster only)	Click **Next** and choose the remote datastore to provide storage for the new cluster. You can mount remote datastores from clusters with different vSAN architectures. A vSAN OSA cluster can mount a vSAN ESA datastore, and a vSAN ESA cluster can mount a vSAN OSA datastore.

Table 989: vSAN ESA Options

Option	Description
vSAN Cluster Type	• **vSAN HCI**: Provides storage and compute resources. • **vSAN Storage**: Provides storage resources only. You can mount a vSAN Storage datastore on other vSAN ESA or vSAN compute clusters.
Network Option (vSAN Storage only)	Select the **vSAN Storage Client Network** checkbox to isolate client or application network traffic from vSAN Storage internal traffic by utilizing dedicated VMkernel ports backed by physical network ports.
Storage Type	Local vSAN datastore. This setting cannot be modified.
Storage Policy	Auto-policy management. This setting cannot be modified. Based on the type of cluster and number of hosts, vSAN creates and assigns a default datastore policy for best capacity utilization after the cluster configuration is completed. Policy details can be viewed in the vSphere Client (**Policies and Profiles** > **VM Storage Policies**).
Data-in-transit Encryption	For vSAN HCI, vSAN data-in-transit encryption is a cluster-wide setting. All data and metadata between hosts in the same cluster is encrypted. For vSAN compute cluster, vSAN data-in-transit encryption encrypts all data and metadata between server and client.

Option	Description
Rekey Interval	The rekey interval sets the time duration before vSAN automatically regenerates the host-to-host communication encryption keys. vSAN does not re-encrypt data at rest or reformat disk after the interval expires. The rekey interval is defined in minutes, with the default being 1440 minutes (24 hours).

b) Enter the NFS Storage details and click **Next**.

This screen only displays if you selected NFS storage.

Option	Description
Datastore Name	Enter a name for the NFS datastore. The datastore name must contain 3-80 characters and cannot contain any leading or trailing whitespace. If you using a previously mounted NFS share, the datastore will be automatically renamed to original datastore name.
Folder	Enter the path to the NFS share. The folder location must contain 3-255 characters and the path cannot contain any whitespace.
NFS Server IP Address	Enter the IP address of the NFS server.

c) Enter the FC Storage details and click **Next**.

This screen only displays if you selected FC storage.

Option	Description
Datastore Name	Enter the name of the VMFS on FC datastore. Datastore name must contain 3-80 characters and cannot contain any leading or trailing whitespace.

d) Enter the vVol Storage details and click **Next**.

This screen only displays if you selected vVols storage.

Option	Description
VASA Protocol Type	Select a VASA protocol type. vVols supports FC, NFS, and iSCSI storage protocol types.
VASA Provider	Select a VASA provider name. The VASA Providers list is determined by the selected VASA protocol type.
Storage Container	Select a storage container. The Storage Containers list is determined by the selected VASA Provider. The list does not display storage containers that are already in use by other clusters.
VASA User	Select a VASA user. The VASA User list is determined by the selected VASA Provider.
Datastore Name	Enter a datastore name. The datastore name must contain 3-80 characters and cannot contain any leading or trailing whitespace.

6. On the **Host Selection** page, select hosts for the SDDC cluster and click **Next**.

   You can use the toggle button to turn **Skip failed hosts during cluster creation** off or on. When this option is off, if you select an unhealthy host, cluster creation fails. When this option is on, if you selected enough healthy hosts to meet the minimum requirements for a new cluster, cluster creation succeeds.

   The hosts must be commissioned with the same storage type that you selected for the cluster.

7. On the **vSphere Distributed Switch** page, create or reuse a vSphere Distributed Switch.

   With VCF 9.1.1 and later, based on the network configuration of the new cluster and the existing VDS, the following sharing modes are available within the same datacenter:

VDS Shared State	Description	Network Pool Configuration
Reuse the existing VDS as is.	Both the existing VDS and all port groups are reused without modification. The new cluster uses the same VDS and port groups as the source cluster.	The new cluster and the source cluster must be on the same network pool.
Reuse the existing VDS and create new port groups.	The existing VDS is reused to maintain policy consistency, but new port groups are created for the new cluster. The new port groups must be distinct from the existing port groups on the reused VDS.	The new cluster can use a different network pool than the source cluster.
Reuse the existing VDS and create a new VDS.	The cluster reuses the existing shared VDS for some network traffic types and creates a separate new VDS for other network traffic types. On the shared VDS, you must create new port groups for the new cluster; reusing existing port groups from the source cluster is not allowed. The new VDS is an additional switch that coexists with the shared VDS, not a replacement for it.	The new cluster can use a different network pool than the source cluster.

Note:

If you want to reuse an existing VDS but need a different Link Aggregation Group (LAG) configuration, use the Create New VDS option and enter the name of the existing VDS. You can then configure the new LAG on the reused VDS. This approach applies the policy reuse mode, where the VDS structure is reused but all port groups are new and distinct.

Create a VDS

1. To use a preconfigured profile, select one of the profiles and click **Next**.

Option	Description
Default	Provides a unified fabric for all traffic types using a single vSphere Distributed Switch.
Storage Traffic Separation	This profile creates two vSphere Distributed Switches with separate physical NICs. One switch is dedicated for storage traffic, while the other is used for all other traffic.
NSX Traffic Separation	This profile creates two vSphere Distributed Switches with separate physical NICs. One switch is dedicated for NSX (Virtual Networking) traffic, while the other is used for all other traffic.
Storage Traffic and NSX Traffic Separation	This profile creates three vSphere Distributed Switches with separate physical NICs. One switch is dedicated for NSX (Virtual Networking) traffic, the second switch is dedicated for storage traffic, while the third is used for all other traffic.

Option	Description
Custom Switch Configuration	Copy from a preconfigured profile or create a new distributed switch configuration. Multiple vSphere Distributed Switches can be configured. Each vSphere Distributed Switch can hold one or more network traffic configurations. Some network traffic types are mandatory. Switch configuration is incomplete until these mandatory traffic types are configured. VM Management, VCF Management, vMotion, vSAN, and NSX network types can be configured only once for a cluster. NSX traffic type can be configured in multiple vSphere Distributed Switches.

2. To create a custom configuration for the distributed switch, click **Create custom configuration**.

Option	Description
Traffic types for the distributed switch	• ESX Management • VM Management • vMotion • vSAN • NSX
Distributed Switch name	Enter a name for the distributed switch.
Uplinks / LAG configuration	VDS Uplinks represent the individual physical network adapters (vmnics) on an ESX host connecting to the physical switch. In this mode, the VDS manages load balancing and failover using software-based algorithms like Route Based on Physical NIC Load (LBT) or Route Based on the Source of the Port ID. The physical switch ports do not need to be configured in a bond or channel; they operate as independent links.
	VDS LAG bundles multiple physical uplinks into a single logical channel using the Link Aggregation Control Protocol (LACP). This requires a corresponding LACP configuration on the physical ToR (Top-of-Rack) switches. The physical switch and VDS negotiate traffic hashing.
Number of uplinks	The number of uplinks dedicated to this distributed switch.
Map uplinks to physical NICs on host	Connects your logical virtual network to your physical network hardware. This mapping pairs a logical uplink, a software object on the distributed switch, with a physical adapter (hardware on the ESXi host).

3. To configure the traffic types for the distributed switch, turn on the toggle for each traffic type and configure the settings.

Option	Description
Distributed port group name	Name of the distributed port group for this traffic type.
Load balancing	*Route based on the source of the port ID*: the distributed switch maps port IDs to specific uplinks based on the load-balancing source of the NSX uplink profile.

Option	Description
	Route Based on Physical NIC Load (LBT) (Default): the distributed switch monitors the saturation of the physical uplinks. If the primary uplinks exceed a set amount of utilization (75% default) for 30 seconds, the distributed switch moves some VM traffic to another active uplink.
	Route Based on Source MAC Hash: the distributed switch uses the VM's MAC address and hashes it to an uplink, which creates even distribution for VMs that might change port IDs but keep MAC addresses.
	Route Based on IP Hash: the distributed switch uses the source and destination IP address of every packet to decide which uplink to use. This allows a single VM to use multiple uplinks simultaneously. **Note:** You must configure LACP (Link Aggregation Control Protocol) on the physical Top-of-Rack switches.
	Use Explicit Failover Order: you define the order in which uplinks are used in case of failure.
Uplink	*Active*: This traffic type uses the uplink immediately during normal operation. In case of multiple active uplinks, traffic of this type is distributed across all active uplinks simultaneously.
	Standby: A standby uplink is connected, but no traffic is sent through it as long as at least one "Active" uplink is healthy. If all active uplinks fail, the distributed switch moves the standby uplink to active status.
	Unused: Traffic of this type never uses this uplink, regardless of the status of the remaining uplinks.

Reuse an Existing VDS

1. To reuse an existing vSphere Distributed switch and reuse its configuration, select **Reuse Existing Cluster VDS**, select the VDS you want to reuse, and click **Next**.

   Note:

   Both the existing VDS and all port groups are reused without modification. The new cluster uses the same VDS and port groups as the source cluster.
2. To reuse an existing vSphere Distributed switch and modify its configuration, select **Reuse Existing Cluster VDS**, select the VDS you want to reuse, and click **Create Custom Switch Configuration**.
   a. In the **Distributed Switches** section, click **Create Distributed Switch**.
   b. On the **Create Distributed Switch** page, select **Use Existing** and from the **VDS** drop-down menu, select the existing VDS.

      Note:

      Reuse Existing Cluster VDS and **Use Existing** are different UI options. **Reuse Existing Cluster VDS** is the initial selection on the vSphere Distributed Switch page that determines whether the cluster reuses an existing VDS. **Use Existing** is a drop-down option on the **Create Distributed Switch** page that appears only when you modify the configuration, and it selects which existing VDS to reference for the custom switch configuration.
   c. Configure the uplink mappings for the shared VDS.

Note:

You can use different uplink tp physical NIC mappings on the shared switch. The uplink names must match the existing VDS uplink names.

Partial reuse of port groups is not supported. All port groups within a VDS must either be shared or distinct.

d. To configure the `Management`, `vMotion`, `vSAN`, and `Public` traffic types for the new cluster, select each type from the **Configure Network Traffic** drop-down menu, configure the settings, and click **Next**.

Option	Description
Load balancing	*`Route based on the source of the port ID`*: the distributed switch maps port IDs to specific uplinks based on the load-balancing source of the NSX uplink profile.
	`Route Based on Physical NIC Load (LBT) (Default)`: the distributed switch monitors the saturation of the physical uplinks. If the primary uplinks exceed a set amount of utilization (75% default) for 30 seconds, the distributed switch moves some VM traffic to another active uplink.
	`Route Based on Source MAC Hash`: the distributed switch uses the VM's MAC address and hashes it to an uplink, which creates even distribution for VMs that might change port IDs but keep MAC addresses.
	`Route Based on IP Hash`: the distributed switch uses the source and destination IP address of every packet to decide which uplink to use. This allows a single VM to use multiple uplinks simultaneously. **Note:** You must configure LACP (Link Aggregation Control Protocol) on the physical Top-of-Rack switches.
	`Use Explicit Failover Order`: you define the order in which uplinks are used in case of failure.
Uplink	*`Active`*: This traffic type uses the uplink immediately during normal operation. In case of multiple active uplinks, traffic of this type is distributed across all active uplinks simultaneously.
	`Standby`: A standby uplink is connected, but no traffic is sent through it as long as at least one "Active" uplink is healthy. If all active uplinks fail, the distributed switch moves the standby uplink to active status.
	`Unused`: Traffic of this type never uses this uplink, regardless of the status of the remaining uplinks.

e. To configure the NSX traffic type for the new cluster, from the **Configure Network Traffic** drop-down menu, select **NSX**, configure the settings, and click **Next**.

Option	Description
Operational Mode	**Select the Apply Default Operation Mode Configured in NSX Manager checkbox to maintain the configured interface mode.**
Transport Zone Type	Select Overlay or VLAN or both.
Transport VLAN	Enter the VLAN ID.
TEP IPv4 Assignment	Choose how IPv4 addresses are assigned to the TEPs. The options are: • **Use DHCP**: IPv4 addresses are assigned from a DHCP server. • **Use IPv4 Pool**: IPv4 addresses are assigned from an IP pool. Specify the IPv4 pool name to be used for TEPs. • **Use VMkernel Adapter**: Assigns the Management VMkernel interface IP address for the TEP.
Transport Zone Name	Enter the VLAN transport zone name
Teaming Policy Uplink Mapping	Map uplinks defined in the selected NSX uplink profile with VDS uplinks. The number of NSX uplinks that are presented for mapping depends on the uplink profile configuration. For example, in the upink-1 (active) row, go to the Physical NICs column, click the edit icon, and type in the name of VDS uplink to complete mapping it with uplink-1 (active). Likewise, complete mapping for the other uplinks.
NSX Uplink Profile Name	Enter the name of the NSX uplink profile

Option	Description
Operational Mode	**Select the Apply Default Operation Mode Configured in NSX Manager checkbox to maintain the configured interface mode.**
NSX Uplink Profile Teaming Policy	Select the mode for the teaming policy: • **Failover Order**: Specify an active uplink along with an optional list of standby uplinks. If the active uplink fails, the next uplink in the standby list replaces the active uplink. No actual load balancing is performed with this option. Standby uplinks and multiple active uplinks are not supported forNSX Edge transport nodes. Also, for anNSX Edge transport node, active uplink used in one profile should not be used in another profile. • **Load Balance Source**: Maps a virtual interface of a VM to an uplink. Traffic sent by this virtual interface will leave the host through this uplink only, and traffic destined to this virtual interface will necessarily enter the virtual switch through this uplink. Select a list of active uplinks. When you configure a transport node, you can pin each interface of the transport node to one active uplink. This configuration allows use of several active uplinks at the same time. No standby uplink is configured in this case. **Note:** To manage VLAN traffic, if you configure a default teaming policy in Load Balance Source mode, then on failure of the first uplink, traffic will not fail over to the second uplink interface. • **Load Balance Source MAC Address**: Select an uplink based on a hash of the source Ethernet.NSX Edgetransport nodes do not support this teaming policy.

8. On the **Review** page, review the details and click **Next**.
9. On the **Validation** page, wait for the validations to complete and click **Finish**.

Shrink a Workload Domain

You can reduce the management domain or a workload domain by removing a host from a vSphere cluster in the workload domain or by deleting a vSphere cluster.

Remove a Host from a vSphere Cluster in a Workload Domain

You can remove a host from a vSphere cluster in the management domain or a workload domain through the Workload Domains page in SDDC Manager UI.

- Delete any workload virtual machines created outside VMware Cloud Foundation or move them to another host.
- You cannot remove a host from a vSphere cluster if that host is hosting an NSX Edge node.
  - Move the NSX Edge node to another host in the same vSphere cluster. You cannot move an NSX Edge node to a host that already hosts another NSX Edge node.

  - If you cannot move the NSX Edge node, delete the NSX Edge node on the host. You cannot delete NSX Edge nodes if doing so would result in an NSX Edge cluster with fewer than two NSX Edge nodes.
- You cannot remove a host from a vSphere cluster if that vSphere cluster was selected for NSX Edge node placement and the NSX Edge node deployment is still in pending state.

Before you remove a host from a vSphere cluster, ensure that you have enough hosts remaining to facilitate the configured vSAN availability. Failure to do so might result in the datastore being marked as read only or in data loss.

Note:

You can add hosts to or remove hosts from multiple different vSphere clusters in parallel. For example, you remove three hosts from Cluster A, and while that task is running, you can start a separate task to remove (or add) four hosts to Cluster B. See VMware Configuration Maximums for information about the maximum number of add/remove hosts tasks that you can run in parallel.

Starting with version 9.1.1, you can remove a host from a cluster that shares a vSphere Distributed Switch (VDS) with another cluster in the workload domain. The shared VDS is not deleted when a host is removed. The host is disconnected from the shared VDS uplinks, but the VDS and its port groups remain available for the remaining clusters.

1. In the navigation pane, click **Inventory** > **Workload Domains**.
2. In the workload domains table, click the name of the workload domain that you want to modify.
3. Click the **Clusters** tab.
4. Click the name of the cluster from which you want to remove a host.
5. Click the **Hosts** tab.
6. Select the host to remove and click **Remove Selected Hosts**.

   An alert appears, asking you to confirm or cancel the action. If the removal results in the number of hosts in the vSphere cluster being less than the minimum number of required hosts, you must click **Force Remove** to remove the host.
7. Click **Remove** to confirm the action.

   The details page for the cluster appears with a message indicating that the host is being removed. When the removal process is complete, the host is removed from the hosts table.

Decommission the host. See Decommission ESX Hosts. After you decommission a host, you must re-image it and commission it before you can use it again.

Delete a vSphere Cluster from a Workload Domain

You can delete a vSphere cluster from the management domain or from a workload domain. vSAN datastores on the ESXi hosts in the deleted cluster are destroyed, while other types of storage are unmounted.

- If remote vSAN datastores are mounted on the cluster, the cluster cannot be deleted. To delete such clusters, you must first migrate any VMs from the remote datastore to the local datastore and then unmount the vSAN remote datastores from vCenter.
- Migrate or back up the VMs and data on the datastore associated with the cluster to another location if you want the ability to restore them later.
- Delete the NSX Edge clusters hosted on the vSphere cluster or shrink the NSX Edge cluster by deleting NSX Edge nodes hosted on the vSphere cluster. You cannot delete NSX Edge nodes if doing so would result in an NSX Edge cluster with fewer than two NSX Edge nodes. For information about deleting an NSX Edge cluster, see KB 78635.

You cannot delete the last cluster in a workload domain. Instead, delete the workload domain.

Starting with VCF 9.1.1, if the cluster you are deleting shares a vSphere Distributed Switch (VDS) with another cluster in the workload domain, the SDDC Manager does not delete the shared VDS. The delete operation removes only the configurations specific to the target cluster, such as port groups and host memberships, while preserving the shared switch infrastructure for the remaining clusters. The VDS is deleted only when the last cluster sharing the VDS is removed.

1. Log in to SDDC Manager at `https://<sddc_manager_fqdn>` with a user assigned the **Administrator** role.
2. In the navigation pane, click **Inventory** > **Workload Domains**.
3. Click the name of the workload domain that contains the vSphere cluster you want to delete.
4. Click the **Clusters** tab.
5. Click the vertical ellipsis (three dots) next to the cluster name and click **Delete Cluster**.



6. Click **Delete Cluster** to confirm that you want to delete the cluster.
   You cannot perform additional workload domain tasks until the Delete Cluster workflow has completed.

Configuration Drift and Out-of-Band Operations in VMware Cloud Foundation

Configuration drift occurs when changes made directly in vCenter or NSX cause the SDDC Manager inventory to become out of sync with the actual environment state. VCF 9.1 provides both a general synchronization workflow for broad inventory reconciliation and targeted out-of-band storage workflow for specific storage changes.

Configuration Drift in VCF

A VCF domain represents a logical unit of application-ready infrastructure that groups ESX hosts managed by a vCenter instance. SDDC Manager maintains an inventory of these domains, including hosts, clusters, distributed switches, network pools, datastores, and NSX configurations.

Configuration drift occurs when the inventory state in SDDC Manager becomes out of sync with the actual state in vCenter and NSX Manager. This happens when administrators make changes directly in vCenter or NSX Manager without using SDDC Manager workflows. These direct changes are called out-of-band (OOB) operations.

When configuration drift occurs, day-N operations such as host add, host remove, stretch, unstretch, and cluster expansion may fail or behave incorrectly because SDDC Manager is working with outdated data.

OOB changes are the primary cause of configuration drift. VCF 9.1 allows administrators to perform the following tasks in vCenter without impacting SDDC Manager:

- Changes to vSphere Distributed Switch (add, remove, teaming policy, MTU, PNICs)
- Change primary datastore
- Changes to datastore
- Manual upgrade in vCenter

The following objects can be modified in NSX:

- Transport Node Profiles
- Uplink Profiles

Additional operations that can cause configuration drift include expanding or reducing workload domains, stretching or unstretching a cluster, and creating or deleting datastores.

How VCF Handles Out-of-Band Changes

VCF 9.1 introduces guardrails that allow certain OOB changes without affecting the configuration state of the environment in VCF Operations. VCF 9.1 resolves configuration drift through two complementary approaches:

- General drift remediation: a synchronization workflow that reconciles the entire VCF domain inventory after OOB changes from vCenter or NSX Manager. This workflow rediscovers NSX Manager nodes, prepares eligible clusters for NSX, and synchronizes hosts, clusters, distributed switches, port groups, network pools, Edge clusters, credentials, and certificates.

- Targeted out-of-band storage operations: a datastore specific workflow for storage changes that the general synchronization workflow does not handle. This workflow addresses the primary datastore reference, OOB stretch and unstretch detection, and vVol to vSAN migration.

Out-of-Band Storage Operations

An OOB storage operation is any storage configuration change that is made directly in vCenter. OOB storage operations include changing the primary datastore, renaming or decommissioning a datastore, and stretching or unstretching a cluster.

When you perform an OOB storage operation, SDDC Manager holds a stale reference to the previous configuration. This creates a split-brain scenario in which the SDDC Manager inventory becomes out of sync with the actual vCenter state. As a result, day-N operations such as add host, remove host, stretch cluster, unstretch cluster, and expand cluster might fail.

VCF 9.1 gathers live data from vCenter and vSAN instead of relying on persisted copies. VCF 9.1 provides tools to manually reconcile the SDDC Manager inventory with the actual state of the environment, where automatic detection is not possible.

VCF 9.1 supports the following OOB storage workflows:

1. Update primary datastore: A new workflow allows administrators to update the primary datastore of a cluster when the original datastore has been changed, renamed, or decommissioned directly in vCenter.
2. OOB stretch and unstretch cluster support: if a cluster is stretched or unstretched directly in vCenter, VCF 9.1 detects the current stretch state and allows subsequent day-N operations from SDDC Manager without requiring manual inventory updates.
3. vVol to vSAN primary datastore migration: due to vVol deprecation in VCF 9.1, you can change the primary datastore type from vVol to any vSAN storage type. This is the only supported cross-type datastore change. This workflow is performed using the Update Primary Datastore workflow.

The following primary datastore changes are not supported in VCF 9.1:

- Cross-type changes other than vVol to vSAN, such as NFS to VMFS or vSAN to NFS.
- vVol to vVol changes.
- vVol to NFS or VMFS changes.

Out-of-Band storage operations are supported even if SDDC Manager is in a straddle state, where SDDC Manager is upgraded to 9.1 but vCenter instances remain on older versions, such as 8.0U3.

The following limitations apply to all OOB storage workflows in VCF 9.1:

- All OOB storage features apply only to clusters created in SDDC Manager. For imported or converted clusters, use the sync operation in VCF Operations.
- VCF 9.1 does not orchestrate workload migration during primary datastore updates. You must migrate workloads to the target datastore before initiating the change.
- Network pool IP changes made OOB on storage VMkernel adapters are not synchronized back to the SDDC Manager network pool.
- Add and remove host operations on clusters with OOB changes must be performed through VCF Operations or SDDC Manager, not directly in vCenter.

Remediate VCF Domain Configuration Drift

If you make out-of-band changes from vCenter or NSX Manager, you can use VCF Operations to sync the SDDC Manager and vCenter inventories.

If the SDDC Manager and vCenter inventories are not synchronized, some workflows may be blocked. You can use this procedure to unblock these tasks.

The synchronization workflow requires you to select a VCF domain. If you must sync multiple workload domains, you run the synchronization workflow multiple times.

When you run a synchronization operation for a VCF domain, SDDC Manager updates its inventory to reflect configuration changes that are made directly in vCenter or NSX Manager and preserves the rest of the domain's existing inventory records. Synchronization covers the following inventory items.

Type	Description
Hosts and clusters	ESX hosts additions and removals.
	Cluster additions, including clusters that use a vSAN stretched cluster configuration.
	ESX hosts that are managed by vCenter but are no longer part of the VCF domain's cluster are removed from the inventory. Credentials managed by SDDC Manager are deleted for those hosts, and their IP addresses are released back to the network pool.
Distributed switches and port groups	vSphere Distributed Switch (VDS) additions and removals.
	Switch properties, including name, maximum transmission unit (MTU), version, network I/O control (NIOC) settings, and Link Aggregation Group (LAG) configuration.
	If the same distributed switch is shared across more than one cluster, synchronization tracks it individually for each cluster it is shared with.
	Port group additions and removals.
	Port group properties, such as name, type, VLAN ID, transport type, and active and standby uplinks. **Note:** Synchronization only discovers and adds port groups that are associated with a recognized infrastructure network type: Management, vMotion, vSAN, vSAN witness or external, NFS, or the port group used by the vCenter appliance network adapter. Port groups that are not associated with one of these traffic types, such as port groups created for general-purpose VM or application network traffic, are not discovered by synchronization and do not appear in the SDDC Manager inventory.
Network pools	Network pool IP ranges, VLAN assignments, MTU values, subnet information, and gateway settings.
	IP addresses that are freed when a host is removed from the VCF domain are returned to the network pool. **Note:** Network pools that are used only by hosts in newly imported clusters are excluded from this synchronization step.
NSX	NSX Manager nodes for the VCF domain are rediscovered. Newly detected nodes are trusted and added to the SDDC Manager inventory.
	NSX Manager nodes that are no longer present in the environment are removed from the SDDC Manager inventory.
	The NSX cluster association for the VCF domain is updated to reflect clusters that are added or removed, including preparing newly discovered clusters for NSX.
	Host Tunnel Endpoint (TEP) IP address assignment mode for NSX-prepared clusters.

Type	Description
NSX Edge clusters	Discovers and imports NSX Edge clusters and Edge node VMs that exist in the domain's NSX Manager but are not yet part of the SDDC Manager inventory.
	Imports the credentials for newly discovered Edge node VMs.
	Resets and rotates the local account passwords for Edge node VMs that SDDC Manager now manages.
	Removes Edge clusters and Edge nodes from the SDDC Manager inventory if they no longer exist in NSX Manager.
	Where supported by the connected NSX version, also discovers and imports NSX Service clusters, in addition to regular Edge clusters.
Credentials and certificates	SSH host keys for newly discovered ESX hosts.
	Certificates for newly discovered management components.
	Local ESX service account credentials are created for newly discovered hosts and removed for hosts that are no longer part of the VCF domain.
Other inventory records **Note:** These VCF domain inventory records are also part of synchronization. In general, these records do not change as a result of synchronization, but they are included to maintain the complete and consistent state of the domain in SDDC Manager inventory.	VCF domain identity and metadata, such as domain name, domain type, and associated tags.
	VMware Avi Load Balancer cluster records associated with the domain.
	Identity broker records associated with the domain.
	VCF component records associated with the domain, including VCF Automation, VCF Operations, VCF Operations for logs.
	SSO domain records.

- During the synchronization workflow, SDDC Manager rediscovers NSX Manager nodes and newly detected nodes are trusted. NSX nodes that are no longer present are removed from the SDDC Manager inventory. All eligible vSphere clusters in the domain that are not yet NSX-configured are automatically prepared for NSX. A cluster is skipped from NSX preparation if:
  - The cluster is already prepared for NSX.
  - The ESXi Host version is incompatible with the NSX version.
- After the synchronization workflow, imported clusters and clusters that are created out-of-band lose their existing network pool configurations.
- The synchronization workflow no longer synchronizes network pool associations for imported clusters. If previous network pool associations for imported clusters exist in the inventory, the sync operation on the domain containing these clusters removes these associations from the SDDC Manager inventory.
- All vSphere clusters in the existing vCenter are in scope for the sync operation. You cannot select a subset of clusters.

1. On the VCF Operations top navigation bar, click **Operate**.
2. In the left pane, select **Inventory** > **Detailed View**.
3. Expand **VCF Instances** and select to the VCF Instance that contains the VCF domain you want to synchronize.
4. Select the domain and, from the **Actions** drop-down menu, select **Sync inventory**.
5. In the **Sync Inventory** dialog box, review and accept the information, and click **Sync inventory**.

   If your existing NSX Manager instance has one or more Edge clusters, the sync process adds the discovered Edge node VMs to the VCF inventory, including their user credentials. During the sync process, the Edge node credential passwords will be reset. You can retrieve the updated passwords from the VCF credential store.

   Select the **Enable Edge cluster sync and import NSX Edge node VMs** check box to perform the password reset operation for all the nodes in the existing Edge cluster, and for any nodes added later.

Select the Primary Datastore of a Cluster in SDDC Manager

You can select a new primary datastore for a cluster in SDDC Manager after the original datastore has been changed or decommissioned directly in vCenter. Supported scenarios include changing from a vVol datastore to a vSAN datastore, and selecting a different datastore of the same type, such as when moving to a datastore with larger capacity or recovering from the loss of the original datastore. Without this update workflow, SDDC Manager holds a stale reference, which causes day-N workflows such as adding a workload domain to fail.

- Verify that the cluster is created in SDD Manager. For imported or converted clusters, use the sync operation.
- Verify that the target datastore is a shared datastore available on the cluster in vCenter.
- If you are changing from a vVol datastore, verify that the target vSAN datastore is already configured on the cluster in vCenter.
- If you are changing from a vVol datastore, migrate all workloads to the target datastore before initiating this change. VCF 9.1 does not perform workload migration.

Due to vVol deprecation in VCF 9.1, you can change the primary datastore type from vVol to any vSAN storage type, including vSAN OSA, vSAN ESA, vSAN Remote, and vSAN Storage. This is the only supported cross-type datastore change. SDDC Manager handles the associated network pool reconfiguration automatically.

For the vVol-to-vSAN datastore change, the following apply:

- Network pool IP changes made OOB on storage VMkernel adapters are not synchronized back to the SDDC Manager network pool.
- The original network pool is not removed even if it becomes empty. This behavior is by design to prevent unintended side effects.
- VCF does not orchestrate workload migration. You must migrate workloads to the target datastore before initiating the change.

1. Log in to SDDC Manager with a user assigned the **Administrator** role.
2. In the left pane, navigate to **Inventory** > **Workload Domains**.
3. Click the VCF domain.
4. On the VCF domain page, click the **Clusters** tab.
5. Click the ellipsis drop-down menu next to the cluster with the updated datastore and select **Update Primary Datastore**.

   The **Update Primary Datastore** wizard opens.
6. On the **Datastore Selection** page, select the datastore you want to configure as the primary and click **Next**.
7. On the **Validation** page, wait for the validation to complete and click **Finish**.

Add a VASA Provider for vVols Storage in Workload Domains

To use vVols storage for workload domains, you must add a VASA provider.

Before you add the VASA provider details in VCF Operations , ensure that you have set up the VASA provider. For information about setting up a VASA provider, see the vSphere Storage documentation.

> **Important:**
>
> Starting with VMware Cloud Foundation 9.0 and vSphere Foundation 9.0, the vSphere Virtual Volumes capability, also known as vVols, is deprecated and will be removed in a future release of VMware Cloud Foundation and vSphere Foundation. Support for vSphere Virtual Volumes will continue for critical bug fixes only for versions of vSphere 8.x, VMware Cloud Foundation and vSphere Foundation 5.x, and other supported versions until end-of-support for the respective release.

1. Log in to VCF Operations.
2. In the left pane, navigate to **Administration** > **SDDC Manager**.
3. On the **SDDC Manager** page, click the **Storage Settings** tab.
4. Click **Add VASA Provider**.
5. Enter a name for the VASA provider.
6. Enter a URL for the VASA provider.
7. Enter VASA provider user credentials.
8. Enter VASA provider storage container details.
9. Click **Save**.

Note:

You can add additional VASA user credentials and storage containers.

Stretching vSAN Clusters in VMware Cloud Foundation

You can stretch a vSAN cluster in a workload domain across two availability zones within a region. Both availability zones must contain an equal number of hosts to ensure failover in case any of the availability zones goes down.

VMware Cloud Foundation 9.0 supports stretching the following types of vSAN clusters using the SDDC Manager API:

- vSAN ESA/OSA
- vSAN compute
- vSAN storage (formerly known as vSAN Max)

In addition to stretching existing clusters, you can use the SDDC Manager API to create a stretched vSAN compute cluster. The following procedures provide the prerequisites and steps for stretching existing clusters, as well as creating a stretched vSAN compute cluster. You can use the examples as the basis for performing these tasks in your own environment. For more information about all the available parameters see the SDDC Manager API Reference Guide.

vSAN Compute Only clusters support a two-node configuration (one host per availability zone). A witness host is not required for this specific topology.

The default management domain vSphere cluster must be stretched before a workload domain cluster can be stretched. This ensures that the NSX control plane and management VMs (vCenter, NSX management cluster, and SDDC Manager) remain accessible if the stretched cluster in the primary availability zone goes down.

VM-Host affinity rules for availability zone placement that you create manually in vCenter do not apply to VCF management services VMs. VCF management services use Kubernetes Cluster API Provider vSphere (CAPV) node anti-affinity for host placement. vSphere Distributed Resource Scheduler (DRS) rules are not required and are not honored for these VMs.

With VCF 9.1 introduces support for the following remote datastore architectures in a mixed mode:

Cluster Type	vSAN OSA Remote Datastore	vSAN ESA Remote Datastore	vSAN Storage Remote Datastore
vSAN OSA Client	Not Applicapble	Supported	Supported
vSAN ESA Client	Supported	Not Applicable	Supported
vSAN Compute Only	Supported	Supported	Supported

Note:
You cannot stretch a cluster in the following conditions:

- The cluster shares a vSAN Storage Policy with any other clusters.
- The cluster includes DPU-backed hosts.
- The cluster includes hosts from different subnets or networks (L3) within an individual availability zone.

- You cannot stretch a vSAN ESA or vSAN Max cluster if Global Deduplication is enabled.

Some use cases for stretching a cluster are described below.

- Planned maintenance
  You can perform a planned maintenance on an availability zone without any downtime and then migrate the applications after the maintenance is completed.
- Automated recovery
  Stretching a cluster automatically initiates VM restart and recovery, and has a low recovery time for the majority of unplanned failures.
- Disaster avoidance
  With a stretched cluster, you can prevent service outages before an impending disaster.

For design details about stretched clusters, see Stretched vSphere Cluster Model.

Deploy and Configure vSAN Witness Host

vSAN HCI and vSAN storage stretched clusters require a witness host deployed in a vSAN witness zone, which must be different from the location of both availability zones. vSAN compute stretched cluster do not require a witness host.

You deploy the vSAN witness host using an appliance instead of using a dedicated physical ESX host as a witness host. The witness host does not run virtual machines and must run the same version of ESX as the ESX hosts in the stretched cluster. It must also meet latency and Round Trip Time (RTT) requirements.
There are separate vSAN witness appliances for vSAN OSA and vSAN ESA. You must deploy the witness appliance that matches the cluster type that you are stretching.

Cluster Type	vSAN Witness Appliance
vSAN ESA	vSAN ESA appliance
vSAN OSA	vSAN OSA appliance
vSAN storage	vSAN ESA appliance
vSAN compute	n/a

Deploy vSAN Witness Host

You deploy the vSAN witness host for a stretched cluster at a site which is isolated from the existing availability zones to prevent propagation of failure or outage in the data center.

Download the VMware vSAN Witness Appliance `.ova` file from the Broadcom Support Portal.

- For stretching a vSAN OSA cluster, download the appliance for vSAN OSA.
- For stretching a vSAN ESA cluster, download the appliance for vSAN ESA.
- For stretching a vSAN storage cluster, download the appliance for vSAN ESA.
- For stretching a vSAN compute cluster or creating a vSAN compute stretch cluster, no vSAN witness appliance is required.

1. In a web browser, log in to vCenter at `https://vcenter_fqdn/ui`.
2. Select **Menu** > **Hosts and Clusters**.
3. In the inventory panel, expand **vCenter** > **Datacenter**.
4. Right-click the cluster and select **Deploy OVF template**.
5. On the **Select an OVF template** page, select **Local file**, click **Upload files**, browse to the location of the vSAN witness host `OVA` file, and click **Next**.
6. On the **Select a name and folder** page, enter a name for the virtual machine and click **Next**.
7. On the **Select a compute resource** page, select a compute resource, and click **Next**.
8. On the **Review details** page, review the settings and click **Next**.
9. On the **License agreements** page, accept the license agreement and click **Next**.
10. On the **Configuration** page, select **Medium** and click **Next**.
11. On the **Select storage** page, select a datastore and click **Next.**
12. On the **Select networks** page, select a portgroup for the witness and management network, and click **Next**.
13. On the **Customize template** page, enter the root password for the witness, select a network for vSAN Traffic, enter the details for each source network, and click **Next**.

    If you do not specify management and secondary network details, DHCP is used.
14. On the **Ready to complete** page, click **Finish** and wait for the process to complete.
15. Power on the vSAN witness host.
    a) In the inventory panel, navigate to **vCenter** > **Datacenter** > **Cluster**.
    b) Right-click the vSAN witness VM and from the **Actions** menu, select **Power** > **Power on**.

Register vSAN Witness Host

Before you can configure the vSAN witness host, you must register it with vCenter.

1. Use the vSphere Client to log in to the vCenter containing the cluster that you want to stretch.
2. In the vSphere Client, navigate to the datacenter.
3. Right-click the datacenter and select **Add Host**.

   Important:
   You must add the vSAN witness host to the datacenter. Do not add it to a folder or cluster..
4. On the **Name and location** page, enter the Fully Qualified Domain Name (FQDN) of the vSAN witness host and click **Next**.

   Note:
   Do not use the IP address.
5. On the **Connection settings** page, enter the root credentials for the witness host and click **Next**.
6. On the **Host summary** page, review the summary of the host details and click **Next**.
7. On the **Host lifecycle** page, select an option and specify a vSphere Lifecycle Manager image.
8. On the **Assign license** page, click **Next**.
9. Review the summary and click **Finish**.

Configure NTP on the Witness Host

To prevent time synchronization issues, configure the NTP service on the vSAN witness host.

1. In the inventory panel of the vSphere Client, select **vCenter** > **Datacenter**.
2. Select the vSAN witness host and click the **Configure** tab.
3. Configure the NTP client on the vSAN witness host.
   a) In the **System** section, click **Time configuration** and click the **Add Service** button.
   b) Select **Use Network Time Protocol (enable NTP client)**.
   c) Configure the following settings and click **OK**.

Setting	Value
NTP Servers	NTP server address
Start NTP Service	Selected
NTP Service Startup Policy	Start and stop with host

Configure the VMkernel Adapters on the vSAN Witness Host

To enable vSAN data network communication between the availability zones, configure the witness network on the vSAN witness host.

1. In the inventory panel of the vSphere Client, select **vCenter** > **Datacenter**.
2. Select the vSAN witness host and click the **Configure** tab.
3. Remove the dedicated witness traffic VMkernel adapter on the vSAN Witness host.
   a) In the **Networking** section, click **VMkernel adapters**.
   b) Select the kernel adapter **vmk1** with `secondaryPg` as **Network label** and click **Remove**.
   c) On the **Remove VMkernel adapter** dialog box, click **Remove**
4. Remove the secondary switch from the vSAN witness host.
   a) In the left pane, select **Networking** > **Virtual switches**.
   b) Expand the **Standard switch: secondary switch** section.
   c) Click the vertical ellipsis and from the drop-down menu, select **Remove**.
   d) On the **Remove standard switch** dialog box, click **Yes**.
5. Remove the virtual machine network port group on the vSAN witness host.
   a) Expand the **Standard switch: vSwitch0** section.
   b) In the **VM Network** pane, click the vertical ellipsis and from the drop-down menu, select **Remove**.
   c) On the **Remove port group** dialog box, click **Yes**.
6. Enable witness traffic on the VMkernel adapter for the management network of the vSAN witness host.
   a) On the **VMkernel adapters** page, select the **vmk0** adapter and click **Edit**.
   b) In the **vmk0 - edit settings** dialog box, click **Port properties**, select the **vSAN** check box, and click **OK**.

Stretch a vSAN ESA or OSA Cluster Using the SDDC Manager API

You can stretch a vSAN cluster (ESA or OSA) in the management domain or workload domain using a JSON specification and the SDDC Manager API.

- See Stretched vSAN Storage Model for design requirements and recommendations.
- Verify that vCenter is operational.
- Verify that you have completed the Planning and Preparation Workbook with the management domain or workload domain deployment option included. See Planning and Preparation.
- Verify that your environment meets the requirements listed in the Prerequisite Checklist sheet in the Planning and Preparation Workbook. See Planning and Preparation.
- Create a network pool for availability zone 2.
- Commission vSAN ESA or OSA hosts for availability zone 2. See Commission ESX Hosts.
- Ensure that you have enough hosts such that there is an equal number of hosts on each availability zone. This is to ensure that there are sufficient resources in case an availability zone goes down completely.

- Deploy and configure a vSAN witness host. See Deploy and Configure vSAN Witness Host.
- If you are stretching a cluster in a workload domain, the default management vSphere cluster must already be stretched.

  Note:
  You cannot stretch a cluster in the following cases:

  - The cluster shares a vSAN Storage Policy with any other clusters.
  - The cluster includes DPU-backed hosts.
  - The cluster includes hosts within an availability zone that have different subnets or networks (L3).
  - The cluster has vSphere Supervisor enabled.

When you stretch a cluster, SDDC Manager modifies the site disaster tolerance setting for storage policy associated with datastore of that cluster from **None - standard cluster** to **Site mirroring - stretched cluster**. This affects all VMs using default datastore policy in that cluster. If you do not want to change the site disaster tolerance setting for specific VMs, apply a different storage policy to those VMs before stretching the cluster.

1. Create a JSON specification file in a text editor.

   The following example is for an environment with a single vSphere Distributed Switch. If you have multiple vSphere Distributed Switches, see the SDDC Manager API Reference Guide for details about creating a JSON specification.

   Note:

   The ESX hosts that you are adding to availability zone 2 must use the same vmnic to vSphere Distributed Switch mapping as the existing hosts in availability zone 1.

   a) Use the sample JSON specification for standard (non-LACP) uplink configurations.

```
{
  "clusterStretchSpec": {
    "deployWithoutLicenseKeys": true,
    "hostSpecs": [
      {
        "id": "<ESX host 1 ID>",
        "hostname": "sfo02-m01-r01-esx01.sfo.rainpole.io",
        "hostNetworkSpec": {
          "networkProfileName": "sfo02-m01-r01-network-profile01",
          "vmNics": [
            {
              "id": "vmnic0",
              "vdsName": "sfo-m01-cl01-vds01",
              "uplink": "uplink1"
            },
            {
              "id": "vmnic1",
              "vdsName": "sfo-m01-cl01-vds01",
              "uplink": "uplink2"
            }
          ]
        }
      },
      {
        "id": "<ESX host 2 ID>",
        "hostname": "sfo02-m01-r01-esx02.sfo.rainpole.io",
        "hostNetworkSpec": {
          "networkProfileName": "sfo02-m01-r01-network-profile01",
          "vmNics": [
            {
              "id": "vmnic0",
              "vdsName": "sfo-m01-cl01-vds01",
```

```
            "uplink": "uplink1"
          },
          {
            "id": "vmnic1",
            "vdsName": "sfo-m01-cl01-vds01",
            "uplink": "uplink2"
          }
        ]
      }
    },
    {
      "id": "<ESX host 3 ID>",
      "hostname": "sfo02-m01-r01-esx03.sfo.rainpole.io",
      "hostNetworkSpec": {
        "networkProfileName": "sfo02-m01-r01-network-profile01",
        "vmNics": [
          {
            "id": "vmnic0",
            "vdsName": "sfo-m01-cl01-vds01",
            "uplink": "uplink1"
          },
          {
            "id": "vmnic1",
            "vdsName": "sfo-m01-cl01-vds01",
            "uplink": "uplink2"
          }
        ]
      }
    },
    {
      "id": "<ESX host 4 ID>",
      "hostname": "sfo02-m01-r01-esx04.sfo.rainpole.io",
      "hostNetworkSpec": {
        "networkProfileName": "sfo02-m01-r01-network-profile01",
        "vmNics": [
          {
            "id": "vmnic0",
            "vdsName": "sfo-m01-cl01-vds01",
            "uplink": "uplink1"
          },
          {
            "id": "vmnic1",
            "vdsName": "sfo-m01-cl01-vds01",
            "uplink": "uplink2"
          }
        ]
      }
    }
  ],
  "networkSpec": {
    "networkProfiles": [
      {
        "isDefault": true,
        "name": "sfo02-m01-r01-network-profile01",
        "nsxtHostSwitchConfigs": [
          {
            "ipAddressPoolName": "sfo02-m01-r01-ip-pool01-host",
```

```
            "uplinkProfileName": "sfo02-m01-r01-uplink-profile01",
            "vdsName": "sfo-m01-cl01-vds01",
            "vdsUplinkToNsxUplink": [
              {
                "nsxUplinkName": "uplink1",
                "vdsUplinkName": "uplink1"
              },
              {
                "nsxUplinkName": "uplink2",
                "vdsUplinkName": "uplink2"
              }
            ]
          }
        ]
      }
    ],
    "nsxClusterSpec": {
      "ipAddressPoolsSpec": [
        {
          "name": "sfo02-m01-r01-ip-pool01-host",
          "subnets": [
            {
              "cidr": "10.12.14.0/24",
              "gateway": "10.12.14.1",
              "ipAddressPoolRanges": [
                {
                  "start": "10.12.14.101",
                  "end": "10.12.14.132"
                }
              ]
            }
          ]
        }
      ],
      "uplinkProfiles": [
        {
          "name": "sfo02-m01-r01-uplink-profile01",
          "transportVlan": 1214,
          "teamings": [
            {
              "name": "DEFAULT",
              "policy": "LOADBALANCE_SRCID",
              "standByUplinks": [],
              "activeUplinks": [
                "uplink1",
                "uplink2"
              ]
            }
          ]
        }
      ]
    }
  },
  "isEdgeClusterConfiguredForMultiAZ": true,
  "witnessSpec": {
    "fqdn": "sfo-m01-cl01-vsw01.sfo.rainpole.io",
    "vsanCidr": "10.21.10.0/24",
```

```
      "vsanIp": "10.21.10.218"
    },
    "witnessTrafficSharedWithVsanTraffic": false
  }
}
```

Note:

The `networkProfileName` field references a sub-Transport Node Profile (sub-TNP), not a new standalone Transport Node Profile. SDDC Manager automatically creates this sub-TNP during the stretch operation. The referenced profile must have `isDefault` set to *`false`*. Do not create the profile manually before submitting the API request.

b) Use the sample JSON specification for LACP uplink configurations.

```
{
  "clusterStretchSpec": {
    "deployWithoutLicenseKeys": true,
    "hostSpecs": [
      {
        "id": "<ESX-HOST-UUID-1>",
        "hostname": "<FQDN-HOST-1>",
        "hostNetworkSpec": {
          "networkProfileName": "<NSX-Sub-Cluster-Profile-AZ2>",
          "vmNics": [
            {
              "id": "vmnic0",
              "vdsName": "<vDS-Name-AZ1>",
              "uplink": "LAG-0"
            },
            {
              "id": "vmnic1",
              "vdsName": "<vDS-Name-AZ1>",
              "uplink": "LAG-1"
            }
          ]
        }
      },
      {
        "id": "<ESX-HOST-UUID-2>",
        "hostname": "<FQDN-HOST-2>",
        "hostNetworkSpec": {
          "networkProfileName": "<NSX-Sub-Cluster-Profile-AZ2>",
          "vmNics": [
            {
              "id": "vmnic0",
              "vdsName": "<vDS-Name-AZ1>",
              "uplink": "LAG-0"
            },
            {
              "id": "vmnic1",
              "vdsName": "<vDS-Name-AZ1>",
              "uplink": "LAG-1"
            }
          ]
        }
      },
      {
        "id": "<ESX-HOST-UUID-3>",
        "hostname": "<FQDN-HOST-3>",
```

```
      "hostNetworkSpec": {
        "networkProfileName": "<NSX-Sub-Cluster-Profile-AZ2>",
        "vmNics": [
          {
            "id": "vmnic0",
            "vdsName": "<vDS-Name-AZ1>",
            "uplink": "LAG-0"
          },
          {
            "id": "vmnic1",
            "vdsName": "<vDS-Name-AZ1>",
            "uplink": "LAG-1"
          }
        ]
      }
    },
    {
      "id": "<ESX-HOST-UUID-4>",
      "hostname": "<FQDN-HOST-4>",
      "hostNetworkSpec": {
        "networkProfileName": "<NSX-Sub-Cluster-Profile-AZ2>",
        "vmNics": [
          {
            "id": "vmnic0",
            "vdsName": "<vDS-Name-AZ1>",
            "uplink": "LAG-0"
          },
          {
            "id": "vmnic1",
            "vdsName": "<vDS-Name-AZ1>",
            "uplink": "LAG-1"
          }
        ]
      }
    }
  ],
  "networkSpec": {
    "networkProfiles": [
      {
        "isDefault": false,
        "name": "<NSX-Sub-Cluster-Profile-AZ2>",
        "nsxtHostSwitchConfigs": [
          {
            "ipAddressPoolName": "<TEP-Pool-Name-AZ2>",
            "uplinkProfileName": "<Uplink-Profile-Name-AZ2>",
            "vdsName": "<vDS-Name-AZ1>",
            "vdsUplinkToNsxUplink": [
              {
                "nsxUplinkName": "LAG",
                "vdsUplinkName": "LAG"
              }
            ]
          }
        ]
      }
    ]
  },
```

```
    "nsxClusterSpec": {
      "ipAddressPoolsSpec": [
        {
          "name": "<TEP-Pool-Name-AZ2>",
          "subnets": [
            {
              "cidr": "<TEP-CIDR-AZ2>",
              "gateway": "<TEP-Gateway-AZ2>",
              "ipAddressPoolRanges": [
                {
                  "start": "<TEP-IP-Start>",
                  "end": "<TEP-IP-End>"
                }
              ]
            }
          ]
        }
      ],
      "uplinkProfiles": [
        {
          "name": "<Uplink-Profile-Name-AZ2>",
          "transportVlan": "<Transport-VLAN-ID>",
          "teamings": [
            {
              "name": "DEFAULT",
              "policy": "FAILOVER_ORDER",
              "standByUplinks": [],
              "activeUplinks": [
                "LAG"
              ]
            }
          ]
        }
      ]
    },
    "isEdgeClusterConfiguredForMultiAZ": false,
    "witnessSpec": {
      "fqdn": "<Witness-Host-FQDN>",
      "vsanCidr": "<vSAN-CIDR-Witness>",
      "vsanIp": "<vSAN-IP-Witness>"
    },
    "witnessTrafficSharedWithVsanTraffic": false
  }
}
```

Table 990: API Field Relationships for LACP Configurations

Field	Location in API Specification	Requirement
`networkProfileName`	`hosts[].hostNetworkSpec`	Must match `networkProfiles[].name` exactly. References the automatically created sub-TNP for the new availability zone.
`vmNics[].uplink`	`hosts[].hostNetworkSpec.vmNics[]`	For LACP, use the LAG name (*`LAG-1`*). For standard uplinks, use *`uplink1`*.
`uplinkProfiles[].name`	`nsxClusterSpec.uplinkProfiles[]`	Must match `nsxtHostSwitchConfigs[].uplinkProfileName` exactly.

`nsxtHostSwitchConfigs[].uplinkProfileName`	`nsxClusterSpec.networkProfiles[].nsxtHostSwitchConfigs[]`	Must match an existing entry in `nsxClusterSpec.uplinkProfiles[].name`.
`vdsUplinkToNsxUplink[].vdsUplinkName`	`nsxClusterSpec.networkProfiles[].nsxtHostSwitchConfigs[]`	For LACP, use the LAG name (*`LAG-1`*). For standard uplinks, use *`uplink1`*.
`teamings[].activeUplinks[]`	`nsxClusterSpec.uplinkProfiles[].teamings[]`	For LACP, use the LAG name (*`LAG-1`*).
`networkProfiles[].isDefault`	`nsxClusterSpec.networkProfiles[]`	Must be set to *`false`*.

Note:

Replace the example values in the JSON file with the correct values for your environment.

`isEdgeClusterConfiguredForMultiAZ` should be set to `"true"` if the cluster hosts an NSX Edge cluster and `"false"` if the cluster does not host an NSX Edge cluster.

2. Log in to VCF Operations.
3. In the navigation pane, click **Developer Center** > **APIs & SDKs**.
4. Click the **API Explorer** link on the SDDC Manager API.

   Click the VCF Instance which contains the cluster you want to stretch.
5. Retrieve and replace the unique IDs for each ESX host in the JSON specification file.
   a) Navigate to **Hosts** > **GET /v1/hosts**.
   b) In the **Status** text box, enter `UNASSIGNED_USEABLE` and click **Execute**.
   c) In the **Response** section, click **PageOfHost**, copy the `id` element of each host, and replace the respective value in the JSON specification file.

ESX Host	Value
ESX Host 1	*`ESX host 1 ID`*
ESX Host 2	*`ESX host 2 ID`*
ESX Host 3	*`ESX host 3 ID`*
ESX Host 4	*`ESX host 4 ID`*

6. Retrieve the unique ID for the cluster you want to stretch.
   a) Navigate to **Clusters** > **GET /v1/clusters**.
   b) Click **Execute**.
   c) In the **Response** section, click **PageOfCluster**, copy the `id` element of the cluster.
   You will need the cluster ID for subsequent steps.
7. Validate the JSON specification file.
   a) Navigate to **Clusters** > **POST /v1/clusters/{id}/validations** .
   b) In the **Value** text box, enter the unique ID for the cluster that you retrieved in step 5.
   c) In the **clusterUpdateSpec** text box, type

```
{
"clusterUpdateSpec":
}
```

   d) Paste the JSON specification.
   For example:

```
{
"clusterUpdateSpec": {
"clusterStretchSpec": {
"hostSpecs": [
{
"hostname": "sfo02-w01-esx01.sfo.rainpole.io",
```

```
"hostNetworkSpec": {
"networkProfileName": "sfo-w01-az2-nsx-np01",
"vmNics": [
{
"id": "vmnic0",
"uplink": "uplink1",
"vdsName": "sfo-w01-cl01-vds01"
},
{
"id": "vmnic1",
"uplink": "uplink2",
"vdsName": "sfo-w01-cl01-vds01"
}
]
},
"id": "<ESX host 1 ID>"
},
{
"hostname": "sfo02-w01-esx02.sfo.rainpole.io",
"hostNetworkSpec": {
"networkProfileName": "sfo-w01-az2-nsx-np01",
"vmNics": [
{
"id": "vmnic0",
"uplink": "uplink1",
"vdsName": "sfo-w01-cl01-vds01"
},
{
"id": "vmnic1",
"uplink": "uplink2",
"vdsName": "sfo-w01-cl01-vds01"
}
]
},
"id": "<ESX host 2 ID>"
},
{
"hostname": "sfo02-w01-esx03.sfo.rainpole.io",
"hostNetworkSpec": {
"networkProfileName": "sfo-w01-az2-nsx-np01",
"vmNics": [
{
"id": "vmnic0",
"uplink": "uplink1",
"vdsName": "sfo-w01-cl01-vds01"
},
{
"id": "vmnic1",
"uplink": "uplink2",
"vdsName": "sfo-w01-cl01-vds01"
}
]
},
"id": "<ESX host 3 ID>"
},
{
"hostname": "sfo02-w01-esx04.sfo.rainpole.io",
```

```
"hostNetworkSpec": {
"networkProfileName": "sfo-w01-az2-nsx-np01",
"vmNics": [
{
"id": "vmnic0",
"uplink": "uplink1",
"vdsName": "sfo-w01-cl01-vds01"
},
{
"id": "vmnic1",
"uplink": "uplink2",
"vdsName": "sfo-w01-cl01-vds01"
}
]
},
"id": "<ESX host 4 ID>"
}
],
"isEdgeClusterConfiguredForMultiAZ": "true",
"networkSpec": {
"networkProfiles": [
{
"isDefault": false,
"name": "sfo-w01-az2-nsx-np01",
"nsxtHostSwitchConfigs": [
{
"ipAddressPoolName": "sfo-w01-az2-host-ip-pool01",
"uplinkProfileName": "sfo-w01-az2-host-uplink-profile01",
"vdsName": "sfo-w01-cl01-vds01",
"vdsUplinkToNsxUplink": [
 {
  "nsxUplinkName": "uplink-1",
  "vdsUplinkName": "uplink1"
 },
 {
  "nsxUplinkName": "uplink-2",
  "vdsUplinkName": "uplink2"
 }
]
}
]
}
],
"nsxClusterSpec": {
"ipAddressPoolsSpec": [
{
"description": "WLD01 AZ2 Host TEP Pool",
"name": "sfo-w01-az2-host-ip-pool01",
"subnets": [
{
 "cidr": "172.16.44.0/24",
 "gateway": "172.16.44.253",
 "ipAddressPoolRanges": [
  {
   "end": "172.16.44.200",
   "start": "172.16.44.10"
  }
```

```
 ]
}
]
}
],
"uplinkProfiles": [
{
"name": "sfo-w01-az2-host-uplink-profile01",
"teamings": [
{
 "activeUplinks": [
  "uplink-1",
  "uplink-2"
 ],
 "name": "DEFAULT",
 "policy": "LOADBALANCE_SRCID",
 "standByUplinks": []
}
],
"transportVlan": 1644
}
]
}
},
"witnessSpec": {
"fqdn": "sfo-w01-cl01-vsw01.sfo.rainpole.io",
"vsanCidr": "172.17.11.0/24",
"vsanIp": "172.17.11.219"
},
"witnessTrafficSharedWithVsanTraffic": false
}
}
```

Note:

Replace the example values in the JSON file with the correct values for your environment.

`isEdgeClusterConfiguredForMultiAZ` should be set to `"true"` if the cluster hosts an NSX Edge cluster and `"false"` if the cluster does not host an NSX Edge cluster.

e) Click **Execute**.
f) In the confirmation dialog box, click **Continue**.
g) In the **Response** section, expand the **result** section and verify that the response is `SUCCEEDED`.

8. Stretch the cluster with the JSON specification.
   a) Navigate to **Clusters** > **PATCH /v1/clusters/{id}**.
   b) Paste the unique `ID` of the cluster in the **Value** text-box.
   c) In the **clusterUpdateSpec** text box, paste the JSON specification.
   For example:

```
{
"clusterStretchSpec": {
"deployWithoutLicenseKeys": true,
"hostSpecs": [
{
"hostname": "sfo02-w01-esx01.sfo.rainpole.io",
"hostNetworkSpec": {
"networkProfileName": "sfo-w01-az2-nsx-np01",
"vmNics": [
{
```

```
"id": "vmnic0",
"uplink": "uplink1",
"vdsName": "sfo-w01-cl01-vds01"
},
{
"id": "vmnic1",
"uplink": "uplink2",
"vdsName": "sfo-w01-cl01-vds01"
}
]
},
"id": "<ESX host 1 ID>"
},
{
"hostname": "sfo02-w01-esx02.sfo.rainpole.io",
"hostNetworkSpec": {
"networkProfileName": "sfo-w01-az2-nsx-np01",
"vmNics": [
{
"id": "vmnic0",
"uplink": "uplink1",
"vdsName": "sfo-w01-cl01-vds01"
},
{
"id": "vmnic1",
"uplink": "uplink2",
"vdsName": "sfo-w01-cl01-vds01"
}
]
},
"id": "<ESX host 2 ID>"
},
{
"hostname": "sfo02-w01-esx03.sfo.rainpole.io",
"hostNetworkSpec": {
"networkProfileName": "sfo-w01-az2-nsx-np01",
"vmNics": [
{
"id": "vmnic0",
"uplink": "uplink1",
"vdsName": "sfo-w01-cl01-vds01"
},
{
"id": "vmnic1",
"uplink": "uplink2",
"vdsName": "sfo-w01-cl01-vds01"
}
]
},
"id": "<ESX host 3 ID>"
},
{
"hostname": "sfo02-w01-esx04.sfo.rainpole.io",
"hostNetworkSpec": {
"networkProfileName": "sfo-w01-az2-nsx-np01",
"vmNics": [
{
```

```
"id": "vmnic0",
"uplink": "uplink1",
"vdsName": "sfo-w01-cl01-vds01"
},
{
"id": "vmnic1",
"uplink": "uplink2",
"vdsName": "sfo-w01-cl01-vds01"
}
]
},
"id": "<ESX host 4 ID>"
}
],
"isEdgeClusterConfiguredForMultiAZ": "true",
"networkSpec": {
"networkProfiles": [
{
"isDefault": false,
"name": "sfo-w01-az2-nsx-np01",
"nsxtHostSwitchConfigs": [
{
"ipAddressPoolName": "sfo-w01-az2-host-ip-pool01",
"uplinkProfileName": "sfo-w01-az2-host-uplink-profile01",
"vdsName": "sfo-w01-cl01-vds01",
"vdsUplinkToNsxUplink": [
 {
  "nsxUplinkName": "uplink-1",
  "vdsUplinkName": "uplink1"
 },
 {
  "nsxUplinkName": "uplink-2",
  "vdsUplinkName": "uplink2"
 }
]
}
]
}
],
"nsxClusterSpec": {
"ipAddressPoolsSpec": [
{
"description": "WLD01 AZ2 Host TEP Pool",
"name": "sfo-w01-az2-host-ip-pool01",
"subnets": [
{
 "cidr": "172.16.44.0/24",
 "gateway": "172.16.44.253",
 "ipAddressPoolRanges": [
  {
   "end": "172.16.44.200",
   "start": "172.16.44.10"
  }
 ]
}
]
}
```

```
],
"uplinkProfiles": [
{
"name": "sfo-w01-az2-host-uplink-profile01",
"teamings": [
{
 "activeUplinks": [
  "uplink-1",
  "uplink-2"
 ],
 "name": "DEFAULT",
 "policy": "LOADBALANCE_SRCID",
 "standByUplinks": []
}
],
"transportVlan": 1644
}
]
}
},
"witnessSpec": {
"fqdn": "sfo-w01-cl01-vsw01.sfo.rainpole.io",
"vsanCidr": "172.17.11.0/24",
"vsanIp": "172.17.11.219"
},
"witnessTrafficSharedWithVsanTraffic": false
}
}
```

Note:

Replace the example values in the JSON file with the correct values for your environment.

`isEdgeClusterConfiguredForMultiAZ` should be set to `"true"` if the cluster hosts an NSX Edge cluster and `"false"` if the cluster does not host an NSX Edge cluster.

d) Click **Execute**.

e) On the confirmation dialog box, click **Continue**.

You can monitor the stretch cluster creation in **Fleet Management** > **Tasks**.

Create a vSAN Compute Stretch Cluster Using the SDDC Manager API

You can create a vSAN compute stretch cluster using the POST /v1/clusters API.

- See Stretched vSAN Compute Cluster Storage Model for design requirements and recommendations.
- A minimum of four ESX hosts commissioned with the vSAN type vSAN Compute Cluster. See Commission ESX Hosts.
- ESX, vCenter, and SDDC Manager must all be version 9.0 or later.
- An existing remote datastore. It can be a vSAN ESA, vSAN OSA, or vSAN storage datastore.
- The server cluster providing the remote datastore must be stretched. See Stretch a vSAN Storage Cluster Using the SDDC Manager API.

No vSAN witness host is required for a stretched vSAN compute cluster.

If you want to stretch a vSAN compute cluster that you already created, see Stretch a vSAN Compute Cluster Using the SDDC Manager API.

This procedure provides two different sample JSON specification files for the different topologies:

Topology	Description
Symmetric	This reflects a topology that has multiple routes to and from the server cluster, with **all routes** having the same bandwidth and latency characteristics. This is not typical in stretched cluster deployments but may arise if the vSAN compute cluster is deployed at different sites unrelated to where the stretched vSAN storage cluster resides.
Asymmetric	This typically reflects a topology that has multiple routes to and from the server cluster, with **one route** having optimal bandwidth and latency characteristics. This is typical in stretched cluster deployments where a client stretched cluster needs to access storage on a server stretched cluster would ideally access the object data in the same site, known as **inter-site-connectivity**. In this case you specify the site affinity between server and client sites in the JSON specification file. For example, if your environment looks like this: • AZ1 (Server): East Coast • AZ2 (Server): West Coast • AZ1 (Client): East Coast • AZ2 (Client): West Coast Then you would use an asymmetric topology with the following site affinity: • AZ1 (Client) → AZ1 (Server) • AZ2 (Client) → AZ2 (Server)

1. Create a JSON specification file in a text editor.

   Symmetric vSAN remote datastore topology:

```
{
  "domainId": "<domain ID>",
  "computeSpec": {
    "clusterSpecs": [
      {
        "name": "sfo-w01-cl02",
        "hostSpecs": [
          {
            "id": "ESX host 1 ID",
            "hostname": "sfo01-w01-r01-esx04.sfo.rainpole.io",
            "azName": "sfo-w01-r01-c01_primary-az-faultdomain",
            "hostNetworkSpec": {
              "networkProfileName": "sfo01-w01-r01-network-profile01",
              "vmNics": [
                {
                  "id": "vmnic0",
                  "vdsName": "sfo-w01-cl02-vds01",
                  "uplink": "uplink1"
                },
                {
                  "id": "vmnic1",
                  "vdsName": "sfo-w01-cl02-vds01",
                  "uplink": "uplink2"
                }
              ]
            }
```

```
  },
  {
    "id": "ESX host 2 ID",
    "hostname": "sfo01-w01-r01-esx05.sfo.rainpole.io",
    "azName": "sfo-w01-r01-c01_primary-az-faultdomain",
    "hostNetworkSpec": {
      "networkProfileName": "sfo01-w01-r01-network-profile01",
      "vmNics": [
        {
          "id": "vmnic0",
          "vdsName": "sfo-w01-cl02-vds01",
          "uplink": "uplink1"
        },
        {
          "id": "vmnic1",
          "vdsName": "sfo-w01-cl02-vds01",
          "uplink": "uplink2"
        }
      ]
    }
  },
  {
    "id": "ESX host 3 ID",
    "hostname": "sfo02-w01-r01-esx04.sfo.rainpole.io",
    "azName": "sfo-w01-r01-c01_secondary-az-faultdomain",
    "hostNetworkSpec": {
      "networkProfileName": "sfo02-w01-r01-network-profile01",
      "vmNics": [
        {
          "id": "vmnic0",
          "vdsName": "sfo-w01-cl02-vds01",
          "uplink": "uplink1"
        },
        {
          "id": "vmnic1",
          "vdsName": "sfo-w01-cl02-vds01",
          "uplink": "uplink2"
        }
      ]
    }
  },
  {
    "id": "ESX host 4 ID",
    "hostname": "sfo02-w01-r01-esx05.sfo.rainpole.io",
    "azName": "sfo-w01-r01-c01_secondary-az-faultdomain",
    "hostNetworkSpec": {
      "networkProfileName": "sfo02-w01-r01-network-profile01",
      "vmNics": [
        {
          "id": "vmnic0",
          "vdsName": "sfo-w01-cl02-vds01",
          "uplink": "uplink1"
        },
        {
          "id": "vmnic1",
          "vdsName": "sfo-w01-cl02-vds01",
          "uplink": "uplink2"
```

```
        }
      ]
    }
  }
],
"datastoreSpec": {
  "vsanRemoteDatastoreClusterSpec": {
    "isStretched": "true",
    "vsanRemoteDatastoreSpec": [
      {
        "datastoreUuid": "<-- ENTER UUID OF REMOTE DATASTORE HERE -->",
        "networkTopology": "Symmetric"
      }
    ]
  }
},
"networkSpec": {
  "vdsSpecs": [
    {
      "name": "sfo-w01-cl02-vds01",
      "mtu": "9000",
      "portGroupSpecs": [
        {
          "name": "sfo-w01-cl02-pg-esxi-mgmt",
          "transportType": "MANAGEMENT",
          "standByUplinks": [],
          "teamingPolicy": "loadbalance_loadbased",
          "activeUplinks": [
            "uplink1",
            "uplink2"
          ]
        },
        {
          "name": "sfo-w01-cl02-pg-vmotion",
          "transportType": "VMOTION",
          "standByUplinks": [],
          "teamingPolicy": "loadbalance_loadbased",
          "activeUplinks": [
            "uplink1",
            "uplink2"
          ]
        },
        {
          "name": "sfo-w01-cl02-pg-vsan",
          "transportType": "VSAN",
          "standByUplinks": [],
          "teamingPolicy": "loadbalance_loadbased",
          "activeUplinks": [
            "uplink1",
            "uplink2"
          ]
        }
      ],
      "nsxtSwitchConfig": {
        "hostSwitchOperationalMode": "STANDARD",
        "transportZones": [
          {
```

```
            "name": "vlan-tz-sfo-w01-nsx01",
            "transportType": "VLAN"
          },
          {
            "name": "overlay-tz-sfo-w01-nsx01",
            "transportType": "OVERLAY"
          }
        ]
      }
    }
  ],
  "nsxClusterSpec": {
    "nsxTClusterSpec": {
      "ipAddressPoolsSpec": [
        {
          "name": "sfo01-w01-r01-ip-pool01-host",
          "subnets": [
            {
              "cidr": "10.13.14.0/24",
              "gateway": "10.13.14.1",
              "ipAddressPoolRanges": [
                {
                  "start": "10.13.14.101",
                  "end": "10.13.14.132"
                }
              ]
            }
          ]
        },
        {
          "name": "sfo02-w01-r01-ip-pool01-host",
          "subnets": [
            {
              "cidr": "10.14.14.0/24",
              "gateway": "10.14.14.1",
              "ipAddressPoolRanges": [
                {
                  "start": "10.14.14.101",
                  "end": "10.14.14.132"
                }
              ]
            }
          ]
        }
      ],
      "uplinkProfiles": [
        {
          "name": "sfo01-w01-r01-uplink-profile01",
          "transportVlan": 1314,
          "teamings": [
            {
              "name": "DEFAULT",
              "policy": "LOADBALANCE_SRCID",
              "standByUplinks": [],
              "activeUplinks": [
                "uplink1",
                "uplink2"
```

```
            ]
          }
        ]
      },
      {
        "name": "sfo02-w01-r01-uplink-profile01",
        "transportVlan": 1414,
        "teamings": [
          {
            "name": "DEFAULT",
            "policy": "LOADBALANCE_SRCID",
            "standByUplinks": [],
            "activeUplinks": [
              "uplink1",
              "uplink2"
            ]
          }
        ]
      }
    ]
  }
},
"networkProfiles": [
  {
    "isDefault": true,
    "name": "sfo01-w01-r01-network-profile01",
    "nsxtHostSwitchConfigs": [
      {
        "ipAddressPoolName": "sfo01-w01-r01-ip-pool01-host",
        "uplinkProfileName": "sfo01-w01-r01-uplink-profile01",
        "vdsName": "sfo-w01-cl02-vds01",
        "vdsUplinkToNsxUplink": [
          {
            "nsxUplinkName": "uplink1",
            "vdsUplinkName": "uplink1"
          },
          {
            "nsxUplinkName": "uplink2",
            "vdsUplinkName": "uplink2"
          }
        ]
      }
    ]
  },
  {
    "isDefault": false,
    "name": "sfo02-w01-r01-network-profile01",
    "nsxtHostSwitchConfigs": [
      {
        "ipAddressPoolName": "sfo02-w01-r01-ip-pool01-host",
        "uplinkProfileName": "sfo02-w01-r01-uplink-profile01",
        "vdsName": "sfo-w01-cl02-vds01",
        "vdsUplinkToNsxUplink": [
          {
            "nsxUplinkName": "uplink1",
            "vdsUplinkName": "uplink1"
          },
```

```
                    {
                      "nsxUplinkName": "uplink2",
                      "vdsUplinkName": "uplink2"
                    }
                  ]
                }
              ]
            }
          ]
        },
        "clusterImageId": "<personality ID>"
      }
    ]
  },
  "deployWithoutLicenseKeys": "true"
}
```

Asymmetric vSAN remote datastore topology:

```
{
  "domainId": "<domain ID>",
  "computeSpec": {
    "clusterSpecs": [
      {
        "name": "sfo-w01-cl02",
        "hostSpecs": [
          {
            "id": "ESX host 1 ID",
            "hostname": "sfo01-w01-r01-esx04.sfo.rainpole.io",
            "azName": "sfo-w01-r01-c01_primary-az-faultdomain",
            "hostNetworkSpec": {
              "networkProfileName": "sfo01-w01-r01-network-profile01",
              "vmNics": [
                {
                  "id": "vmnic0",
                  "vdsName": "sfo-w01-cl02-vds01",
                  "uplink": "uplink1"
                },
                {
                  "id": "vmnic1",
                  "vdsName": "sfo-w01-cl02-vds01",
                  "uplink": "uplink2"
                }
              ]
            }
          },
          {
            "id": "ESX host 2 ID",
            "hostname": "sfo01-w01-r01-esx05.sfo.rainpole.io",
            "azName": "sfo-w01-r01-c01_primary-az-faultdomain",
            "hostNetworkSpec": {
              "networkProfileName": "sfo01-w01-r01-network-profile01",
              "vmNics": [
                {
                  "id": "vmnic0",
                  "vdsName": "sfo-w01-cl02-vds01",
                  "uplink": "uplink1"
                },
                {
```

```
            "id": "vmnic1",
            "vdsName": "sfo-w01-cl02-vds01",
            "uplink": "uplink2"
          }
        ]
      }
    },
    {
      "id": "ESX host 3 ID",
      "hostname": "sfo02-w01-r01-esx04.sfo.rainpole.io",
      "azName": "sfo-w01-r01-c01_secondary-az-faultdomain",
      "hostNetworkSpec": {
        "networkProfileName": "sfo02-w01-r01-network-profile01",
        "vmNics": [
          {
            "id": "vmnic0",
            "vdsName": "sfo-w01-cl02-vds01",
            "uplink": "uplink1"
          },
          {
            "id": "vmnic1",
            "vdsName": "sfo-w01-cl02-vds01",
            "uplink": "uplink2"
          }
        ]
      }
    },
    {
      "id": "ESX host 4 ID",
      "hostname": "sfo02-w01-r01-esx05.sfo.rainpole.io",
      "azName": "sfo-w01-r01-c01_secondary-az-faultdomain",
      "hostNetworkSpec": {
        "networkProfileName": "sfo02-w01-r01-network-profile01",
        "vmNics": [
          {
            "id": "vmnic0",
            "vdsName": "sfo-w01-cl02-vds01",
            "uplink": "uplink1"
          },
          {
            "id": "vmnic1",
            "vdsName": "sfo-w01-cl02-vds01",
            "uplink": "uplink2"
          }
        ]
      }
    }
  ],
  "datastoreSpec": {
    "vsanRemoteDatastoreClusterSpec": {
      "vsanRemoteDatastoreSpec": [
        {
          "datastoreUuid": "<-- ENTER UUID OF REMOTE DATASTORE HERE -->",
          "networkTopology": "Asymmetric",
          "siteAffinity": [
            {
              "serverSite": "sfo-w01-c01_primary-az-faultdomain",
```

```
            "clientSite": "sfo-w01-c01_primary-az-faultdomain"
          },
          {
            "serverSite": "sfo-w01-c01_secondary-az-faultdomain",
            "clientSite": "sfo-w01-c01_secondary-az-faultdomain"
          }
        ]
      }
    ]
  }
},
"networkSpec": {
  "vdsSpecs": [
    {
      "name": "sfo-w01-cl02-vds01",
      "mtu": "9000",
      "portGroupSpecs": [
        {
          "name": "sfo-w01-cl02-pg-esxi-mgmt",
          "transportType": "MANAGEMENT",
          "standByUplinks": [],
          "teamingPolicy": "loadbalance_loadbased",
          "activeUplinks": [
            "uplink1",
            "uplink2"
          ]
        },
        {
          "name": "sfo-w01-cl02-pg-vmotion",
          "transportType": "VMOTION",
          "standByUplinks": [],
          "teamingPolicy": "loadbalance_loadbased",
          "activeUplinks": [
            "uplink1",
            "uplink2"
          ]
        },
        {
          "name": "sfo-w01-cl02-pg-vsan",
          "transportType": "VSAN",
          "standByUplinks": [],
          "teamingPolicy": "loadbalance_loadbased",
          "activeUplinks": [
            "uplink1",
            "uplink2"
          ]
        }
      ],
      "nsxtSwitchConfig": {
        "hostSwitchOperationalMode": "STANDARD",
        "transportZones": [
          {
            "name": "vlan-tz-sfo-w01-nsx01",
            "transportType": "VLAN"
          },
          {
            "name": "overlay-tz-sfo-w01-nsx01",
```

```
            "transportType": "OVERLAY"
          }
        ]
      }
    }
  ],
  "nsxClusterSpec": {
    "nsxTClusterSpec": {
      "ipAddressPoolsSpec": [
        {
          "name": "sfo01-w01-r01-ip-pool01-host",
          "subnets": [
            {
              "cidr": "10.13.14.0/24",
              "gateway": "10.13.14.1",
              "ipAddressPoolRanges": [
                {
                  "start": "10.13.14.101",
                  "end": "10.13.14.132"
                }
              ]
            }
          ]
        },
        {
          "name": "sfo02-w01-r01-ip-pool01-host",
          "subnets": [
            {
              "cidr": "10.14.14.0/24",
              "gateway": "10.14.14.1",
              "ipAddressPoolRanges": [
                {
                  "start": "10.14.14.101",
                  "end": "10.14.14.132"
                }
              ]
            }
          ]
        }
      ],
      "uplinkProfiles": [
        {
          "name": "sfo01-w01-r01-uplink-profile01",
          "transportVlan": 1314,
          "teamings": [
            {
              "name": "DEFAULT",
              "policy": "LOADBALANCE_SRCID",
              "standByUplinks": [],
              "activeUplinks": [
                "uplink1",
                "uplink2"
              ]
            }
          ]
        },
        {
```

```
        "name": "sfo02-w01-r01-uplink-profile01",
        "transportVlan": 1414,
        "teamings": [
          {
            "name": "DEFAULT",
            "policy": "LOADBALANCE_SRCID",
            "standByUplinks": [],
            "activeUplinks": [
              "uplink1",
              "uplink2"
            ]
          }
        ]
      }
    ]
  }
},
"networkProfiles": [
  {
    "isDefault": true,
    "name": "sfo01-w01-r01-network-profile01",
    "nsxtHostSwitchConfigs": [
      {
        "ipAddressPoolName": "sfo01-w01-r01-ip-pool01-host",
        "uplinkProfileName": "sfo01-w01-r01-uplink-profile01",
        "vdsName": "sfo-w01-cl02-vds01",
        "vdsUplinkToNsxUplink": [
          {
            "nsxUplinkName": "uplink1",
            "vdsUplinkName": "uplink1"
          },
          {
            "nsxUplinkName": "uplink2",
            "vdsUplinkName": "uplink2"
          }
        ]
      }
    ]
  },
  {
    "isDefault": false,
    "name": "sfo02-w01-r01-network-profile01",
    "nsxtHostSwitchConfigs": [
      {
        "ipAddressPoolName": "sfo02-w01-r01-ip-pool01-host",
        "uplinkProfileName": "sfo02-w01-r01-uplink-profile01",
        "vdsName": "sfo-w01-cl02-vds01",
        "vdsUplinkToNsxUplink": [
          {
            "nsxUplinkName": "uplink1",
            "vdsUplinkName": "uplink1"
          },
          {
            "nsxUplinkName": "uplink2",
            "vdsUplinkName": "uplink2"
          }
        ]
```

```
                    }
                  ]
                }
              ]
            },
            "clusterImageId": "<personality ID>"
          }
        ]
      },
      "deployWithoutLicenseKeys": "true"
    }
```

Note:
Replace the example values in the JSON file with the correct values for your environment.

2. Log in to VCF Operations.
3. In the navigation pane, click **Developer Center** > **APIs & SDKs**.
4. Click the **API Explorer** link on the SDDC Manager API.
5. Click the VCF Instance in which you want to create the stretched cluster.
6. Retrieve the unique IDs for each ESX host that will be part of the stretched cluster and add the IDs to the JSON specification file.
   a) Navigate to **Hosts** > **GET /v1/hosts**.
   b) In the **Status** text box, enter `UNASSIGNED_USEABLE` and click **Execute**.
   c) In the **Response** section, click **PageOfHost**, copy the `id` element of each host, and replace the respective value in the JSON specification file.

ESX Host	Value
ESX Host 1	*ESX host 1 ID*
ESX Host 2	*ESX host 2 ID*
ESX Host 3	*ESX host 3 ID*
ESX Host 4	*ESX host 4 ID*

7. Retrieve the ID of the workload domain in which you want to create the cluster and add the ID to the JSON specification file.
   a) Navigate to **Domains** > **GET /v1/domains**.
   b) Click **Execute**.
   c) In the **Response** section, click the domain name, copy the `id` element, and replace the *<domain ID>* value in the JSON specification file.
8. Retrieve the vSphere Lifecycle Manager image ID to use for the cluster and add the ID to the JSON specification file.
   a) Navigate to **Personalities** > **GET /v1/personalities**.
   b) Click **Execute**.
   c) In the **Response** section, click the personality representing the image you want to use, copy the `personalityId` element, and replace the *<personality ID>* value in the JSON specification file.
9. Retrieve the datastore Uuid for the server cluster providing the remote datastore and add it to the JSON specification file.
   a) Navigate to **Clusters** > **GET /v1/clusters**.
   b) Click **Execute**.
   c) In the **Response** section, click **PageOfCluster**, and copy the `id` element of the server cluster providing the remote datastore.
   d) Navigate to **Clusters** > **GET /v1/clusters{id}/datastores**.
   e) In the **Value** text box, enter the unique ID for the server cluster and click **Execute**.
   f) In the **Response** section, click the datastore, copy the `id` element, and replace the *<datastore ID>* value in the JSON specification file.
10. Navigate to **Clusters** > **POST /v1/clusters**.
11. Paste the content from your JSON specification file in the **Value** text box for the `body` parameter and click **Execute**.
12. On the confirmation dialog box, click **Continue**.

    You can monitor the stretch cluster creation in **Fleet Management** > **Tasks**.

Stretch a vSAN Compute Cluster Using the SDDC Manager API

You can stretch an existing vSAN compute cluster using the PATCH /v1/clusters/{id} API.

- See Stretched vSAN Compute Cluster Storage Model for design requirements and recommendations.
- A minimum of three ESX hosts commissioned with the vSAN type vSAN Compute Cluster. See Commission ESX Hosts.
- An existing vSAN compute cluster with at least three ESX hosts.
- ESX, vCenter, and SDDC Manager must all be version 9.0 or later.
- The server cluster providing the remote datastore must be stretched. See Stretch a vSAN Storage Cluster Using the SDDC Manager API.
- Unmount all datastores from the vSAN compute cluster, except for the primary datastore.
- Unregister all VMs running on the primary datastore.

No vSAN witness host is required for a stretched vSAN compute cluster.

If you want to create a new stretched vSAN compute cluster, see Create a vSAN Compute Stretch Cluster Using the SDDC Manager API.

1. Create a JSON specification file in a text editor.

```
{
  "clusterStretchSpec": {
    "deployWithoutLicenseKeys": true,
    "hostSpecs": [
      {
        "id": "<ESX host 1 ID>",
        "hostname": "sfo02-w01-r01-esx01.sfo.rainpole.io",
        "hostNetworkSpec": {
```

```
        "networkProfileName": "sfo02-w01-r01-network-profile01",
        "vmNics": [
          {
            "id": "vmnic0",
            "vdsName": "sfo-w01-cl01-vds01",
            "uplink": "uplink1"
          },
          {
            "id": "vmnic1",
            "vdsName": "sfo-w01-cl01-vds01",
            "uplink": "uplink2"
          }
        ]
      }
    },
    {
      "id": "<ESX host 2 ID>",
      "hostname": "sfo02-w01-r01-esx02.sfo.rainpole.io",
      "hostNetworkSpec": {
        "networkProfileName": "sfo02-w01-r01-network-profile01",
        "vmNics": [
          {
            "id": "vmnic0",
            "vdsName": "sfo-w01-cl01-vds01",
            "uplink": "uplink1"
          },
          {
            "id": "vmnic1",
            "vdsName": "sfo-w01-cl01-vds01",
            "uplink": "uplink2"
          }
        ]
      }
    },
    {
      "id": "<ESX host 3 ID>",
      "hostname": "sfo02-w01-r01-esx03.sfo.rainpole.io",
      "hostNetworkSpec": {
        "networkProfileName": "sfo02-w01-r01-network-profile01",
        "vmNics": [
          {
            "id": "vmnic0",
            "vdsName": "sfo-w01-cl01-vds01",
            "uplink": "uplink1"
          },
          {
            "id": "vmnic1",
            "vdsName": "sfo-w01-cl01-vds01",
            "uplink": "uplink2"
          }
        ]
      }
    }
  ],
  "networkSpec": {
    "networkProfiles": [
      {
```

```
      "isDefault": true,
      "name": "sfo02-w01-r01-network-profile01",
      "nsxtHostSwitchConfigs": [
        {
          "ipAddressPoolName": "sfo02-w01-r01-ip-pool01-host",
          "uplinkProfileName": "sfo02-w01-r01-uplink-profile01",
          "vdsName": "sfo-w01-cl01-vds01",
          "vdsUplinkToNsxUplink": [
            {
              "nsxUplinkName": "uplink1",
              "vdsUplinkName": "uplink1"
            },
            {
              "nsxUplinkName": "uplink2",
              "vdsUplinkName": "uplink2"
            }
          ]
        }
      ]
    }
  ],
  "nsxClusterSpec": {
    "ipAddressPoolsSpec": {
      "name": "sfo02-w01-r01-ip-pool01-host",
      "subnets": [
        {
          "cidr": "10.14.14.0/24",
          "gateway": "10.14.14.1",
          "ipAddressPoolRanges": [
            {
              "start": "10.14.14.101",
              "end": "10.14.14.132"
            }
          ]
        }
      ]
    },
    "uplinkProfiles": [
      {
        "name": "sfo02-w01-r01-uplink-profile01",
        "transportVlan": 1414,
        "teamings": [
          {
            "name": "DEFAULT",
            "policy": "LOADBALANCE_SRCID",
            "standByUplinks": [],
            "activeUplinks": [
              "uplink1",
              "uplink2"
            ]
          }
        ]
      }
    ]
  }
},
"isEdgeClusterConfiguredForMultiAZ": true
```

```
    }
  }
```

Note:
Replace the example values in the JSON file with the correct values for your environment.

2. Log in to VCF Operations.
3. In the navigation pane, click **Developer Center** > **APIs & SDKs**.
4. Click the **API Explorer** link on the SDDC Manager API.
5. Click the VCF Instance which contains the cluster you want to stretch.
6. Retrieve the unique IDs for each ESX host that will be part of the stretched cluster and add the IDs to the JSON specification file.
   a) Navigate to **Hosts** > **GET /v1/hosts**.
   b) In the **Status** text box, enter `UNASSIGNED_USEABLE` and click **Execute**.
   c) In the **Response** section, click **PageOfHost**, copy the `id` element of each host, and replace the respective value in the JSON specification file.

ESX Host	Value
ESX Host 1	*ESX host 1 ID*
ESX Host 2	*ESX host 2 ID*
ESX Host 3	*ESX host 3 ID*

7. Retrieve the unique ID for the cluster you want to stretch.
   a) Navigate to **Clusters** > **GET /v1/clusters**.
   b) Click **Execute**.
   c) In the **Response** section, click **PageOfCluster**, copy the `id` element of the cluster.
   You will need the cluster ID for the subsequent step.
8. Stretch the cluster with the JSON specification.
   a) Navigate to **Clusters** > **PATCH /v1/clusters/{id}**.
   b) Paste the unique `ID` of the cluster in the **Value** text-box.
   c) In the **clusterUpdateSpec** text box, paste the JSON specification.
   d) Click **Execute**.
   e) On the confirmation dialog box, click **Continue**.
   You can monitor the stretch cluster creation in **Fleet Management** > **Tasks**.

Stretch a vSAN Storage Cluster Using the SDDC Manager API

You can stretch an vSAN storage cluster using the PATCH /v1/clusters/{id} API. You must stretch a vSAN storage server cluster before you can stretch any of its vSAN compute client clusters.

- See Stretched vSAN Storage Cluster Storage Model for design requirements and recommendations.
- Commission a minimum of three ESX hosts with the vSAN type vSAN Storage (formerly known as vSAN Max). See Commission ESX Hosts.
- An existing vSAN storage cluster with a minimum of three ESX hosts.
- Deploy and configure a vSAN ESA witness host. See Deploy and Configure vSAN Witness Host.
- Unmount any vSAN compute cluster datastores that are mounted to the vSAN storage cluster.

1. Create a JSON specification file in a text editor.

   The following example is for an environment with two vSphere Distributed Switches. This is typcial in a vSAN storage cluster where vSAN storage traffic separation is configured with a dedicated vDS just for the vSAN storage traffic and all other traffic including the vSAN client traffic is on a separate vDS.

Note:

The ESX hosts that you are adding to availability zone 2 must use the same vmnic to vSphere Distributed Switch mapping as the existing hosts in availability zone 1.

```
{
  "clusterStretchSpec": {
    "hostSpecs": [
      {
        "id": "<ESX host 1 ID>",
        "hostname": "sfo02-w01-r01-esx01.sfo.rainpole.io",
        "hostNetworkSpec": {
          "networkProfileName": "sfo02-w01-r01-network-profile01",
          "vmNics": [
            {
              "id": "vmnic0",
              "vdsName": "sfo-w01-cl01-vds01",
              "uplink": "uplink1"
            },
            {
              "id": "vmnic1",
              "vdsName": "sfo-w01-cl01-vds01",
              "uplink": "uplink2"
            },
            {
              "id": "vmnic2",
              "vdsName": "sfo-w01-cl01-vds02",
              "uplink": "uplink1"
            },
            {
              "id": "vmnic3",
              "vdsName": "sfo-w01-cl01-vds02",
              "uplink": "uplink2"
            }
          ]
        }
      },
      {
        "id": "<ESX host 2 ID>",
        "hostname": "sfo02-w01-r01-esx02.sfo.rainpole.io",
        "hostNetworkSpec": {
          "networkProfileName": "sfo02-w01-r01-network-profile01",
          "vmNics": [
            {
              "id": "vmnic0",
              "vdsName": "sfo-w01-cl01-vds01",
              "uplink": "uplink1"
            },
            {
              "id": "vmnic1",
              "vdsName": "sfo-w01-cl01-vds01",
              "uplink": "uplink2"
            },
            {
              "id": "vmnic2",
              "vdsName": "sfo-w01-cl01-vds02",
              "uplink": "uplink1"
```

```
          },
          {
            "id": "vmnic3",
            "vdsName": "sfo-w01-cl01-vds02",
            "uplink": "uplink2"
          }
        ]
      }
    },
    {
      "id": "<ESX host 3 ID>",
      "hostname": "sfo02-w01-r01-esx03.sfo.rainpole.io",
      "hostNetworkSpec": {
        "networkProfileName": "sfo02-w01-r01-network-profile01",
        "vmNics": [
          {
            "id": "vmnic0",
            "vdsName": "sfo-w01-cl01-vds01",
            "uplink": "uplink1"
          },
          {
            "id": "vmnic1",
            "vdsName": "sfo-w01-cl01-vds01",
            "uplink": "uplink2"
          },
          {
            "id": "vmnic2",
            "vdsName": "sfo-w01-cl01-vds02",
            "uplink": "uplink1"
          },
          {
            "id": "vmnic3",
            "vdsName": "sfo-w01-cl01-vds02",
            "uplink": "uplink2"
          }
        ]
      }
    }
  ],
  "networkSpec": {
    "networkProfiles": [
      {
        "isDefault": true,
        "name": "sfo-w01-r01-network-profile-01",
        "nsxtHostSwitchConfigs": [
          {
            "ipAddressPoolName": "sfo02-w01-r01-ip-pool01-host",
            "uplinkProfileName": "sfo02-w01-r01-uplink-profile01",
            "vdsName": "sfo-w01-cl01-vds01",
            "vdsUplinkToNsxUplink": [
              {
                "nsxUplinkName": "uplink1",
                "vdsUplinkName": "uplink1"
              },
              {
                "nsxUplinkName": "uplink2",
                "vdsUplinkName": "uplink2"
```

```
                }
              ]
            }
          ]
        }
      ],
      "nsxClusterSpec": {
        "ipAddressPoolsSpec": [
          {
            "name": "sfo02-w01-r01-ip-pool01-host",
            "subnets": [
              {
                "cidr": "10.14.14.0/24",
                "gateway": "10.14.14.1",
                "ipAddressPoolRanges": [
                  {
                    "start": "10.14.14.101",
                    "end": "10.14.14.132"
                  }
                ]
              }
            ]
          }
        ],
        "uplinkProfiles": [
          {
            "name": "sfo02-w01-r01-uplink-profile-01",
            "transportVlan": 1414,
            "teamings": [
              {
                "name": "DEFAULT",
                "policy": "LOADBALANCE_SRCID",
                "standByUplinks": [],
                "activeUplinks": [
                  "uplink1",
                  "uplink2"
                ]
              }
            ]
          }
        ]
      }
    },
    "isEdgeClusterConfiguredForMultiAZ": false,
    "witnessSpec": {
      "fqdn": "sfo-w01-cl01-vsw01.sfo.rainpole.io",
      "vsanCidr": "10.21.10.0/24",
      "vsanIp": "10.21.10.219"
    },
    "witnessTrafficSharedWithVsanTraffic": false,
    "deployWithoutLicenseKeys": true
  }
}
```

Note:
Replace the example values in the JSON file with the correct values for your environment.

2. Log in to VCF Operations.
3. In the navigation pane, click **Developer Center** > **APIs & SDKs**.
4. Click the **API Explorer** link on the SDDC Manager API.
5. Click the VCF Instance which contains the cluster you want to stretch.
6. Retrieve the unique IDs for each ESX host that will be part of the stretched cluster and add the IDs to the JSON specification file.
   a) Navigate to **Hosts** > **GET /v1/hosts**.
   b) In the **Status** text box, enter `UNASSIGNED_USEABLE` and click **Execute**.
   c) In the **Response** section, click **PageOfHost**, copy the `id` element of each host, and replace the respective value in the JSON specification file.

ESX Host	Value
ESX Host 1	*ESX host 1 ID*
ESX Host 2	*ESX host 2 ID*
ESX Host 3	*ESX host 3 ID*

7. Retrieve the unique ID for the cluster you want to stretch.
   a) Navigate to **Clusters** > **GET /v1/clusters**.
   b) Click **Execute**.
   c) In the **Response** section, click **PageOfCluster**, copy the `id` element of the cluster.
   You will need the cluster ID for the subsequent step.
8. Stretch the cluster with the JSON specification.
   a) Navigate to **Clusters** > **PATCH /v1/clusters/{id}**.
   b) Paste the unique `ID` of the cluster in the **Value** text-box.
   c) In the **clusterUpdateSpec** text box, paste the JSON specification.
   d) Click **Execute**.
   e) On the confirmation dialog box, click **Continue**.
   You can monitor the stretch cluster creation in **Fleet Management** > **Tasks**.

Expand a Stretched Cluster in VMware Cloud Foundation

You can expand a stretched cluster by adding ESX hosts. It is recommended that you add the same number of hosts to both availability zones for symmetry and cluster balance.

1. Commission the additional hosts to the global inventory.
   See Commission ESX Hosts.

   Note:
   Make sure to select the correct network pool for the availability zone to which you are adding each host.

2. Get the UIDs of the hosts you commissioned.
   a) In the navigation pane, click **Developer Center** > **API Explorer**.
   b) Under APIs for managing hosts, click **GET /v1/hosts**.
   c) Click **Execute**.
   d) Click **Download** to download the JSON file.
   e) Open the JSON file and copy the the UIDs of the hosts.
3. Get the ID of the cluster you are expanding.
   a) In the API Explorer, navigate to APIs for managing clusters and click `GET /v1/clusters`.
   b) Click **Execute**.
   c) Click **Download** to download the JSON file.
   d) Open the JSON file and copy the the cluster ID for the cluster you are expanding.
4. Get the primary and secondary availability zone names from vCenter.
   a) In a web browser, log in to the vCenter at `https://vcenter_fqdn/ui`.
   b) Select **Menu** > **Hosts and Clusters**.
   c) In the inventory panel, expand **vCenter** > **Datacenter**.
   d) Select the cluster and then click the **Configure** tab.
   e) Under **vSAN**, select **Fault Domains**.
   f) Note the primary and secondary availability zone names.
5. Prepare the JSON request body.
   a) Click **Patch /v1/clusters/id**.
   b) Under **ClusterUpdateSpec** field, click **Cluster Update Data ClusterUpdateSpec{ ... }**.
   c) Click **Download** to download the JSON file.
   d) Edit the downloaded JSON file so that it contains only the expand section similar to the example below. In the `azName` field, type the primary and secondary names you had retrieved in step 4.

      Note:
      The ESXi hosts that you are adding must use the same vmnic to vSphere Distributed Switch mapping as the existing hosts in the stretched cluster. For example: If existing hosts map vmnic0 and vmnic1 to vSphere Dstributed Switch 1 and vmnic2 and vmnic3 to vSphere Distributed Switch 2, then the hosts you are adding must map the same vmincs to the same vSphere Distributed Switches.

```
{
  "clusterExpansionSpec": {
     "hostSpecs": [ {
        "id": "ESX host 1 ID",
        "azName":"primary/secondary",
        "hostNetworkSpec": {
         "vmNics": [{
          "id": "vmnic0",
          "vdsName": "<vSphere Distributed Switch 1>"
         },
         {
          "id": "vmnic1",
          "vdsName": "<vSphere Distributed Switch 2>"
         }
        ],
        "deployWithoutLicenseKeys": "true"
       }
      }, {
        "id": "ESX host 2 ID",
        "azName":"primary/secondary",
        "hostNetworkSpec": {
         "vmNics": [{
          "id": "vmnic0",
```

```
            "vdsName": "<vSphere Distributed Switch 1>"
           },
           {
            "id": "vmnic1",
            "vdsName": "<vSphere Distributed Switch 2>"
           }
          ],
          "deployWithoutLicenseKeys": "true"
        }
      }, {
          "id": "ESX host 3 ID",
          "azName":"primary/secondary",
          "hostNetworkSpec": {
           "vmNics": [{
            "id": "vmnic0",
            "vdsName": "<vSphere Distributed Switch 1>"
           },
           {
            "id": "vmnic1",
            "vdsName": "<vSphere Distributed Switch 2>"
           }
          ],
          "deployWithoutLicenseKeys": "true"
        }
      }, {
          "id": "ESX host 4 ID",
          "azName":"primary/secondary",
          "hostNetworkSpec": {
           "vmNics": [{
            "id": "vmnic0",
            "vdsName": "<vSphere Distributed Switch 1>"
           },
           {
            "id": "vmnic1",
            "vdsName": "<vSphere Distributed Switch 2>"
           }
          ],
          "deployWithoutLicenseKeys": "true"
        }
      } ]
  }
}
```

6. Run the expand cluster API.
   a) For the **ClusterUpdateSpec** field, update the cluster ID (you retrieved this in step 3) and JSON file with the payload you prepared in step 5.
   b) Click **Execute**.
   c) Monitor the task until it is completed.
7. If required, SSH in to each newly added host and add a static route to the vSAN network of the witness host. Also add static routes in the witness if it could not reach the vSAN network of the newly added hosts.
8. Update the value of **Host failure cluster tolerates** to the number of hosts in AZ1 after cluster expansion.
   a) Log in to the management vCenter instance.
   b) Select **Cluster** and click the **Configure** tab.
   c) Under **Services**, click **vSphere Availability** and then click **Edit**.
   d) On the **Admission Control** page of the **Edit Cluster Settings** dialog box, set host failures cluster tolerates to the number of hosts in availability zone 1 and click **OK**.

Unstretch a Cluster in VMware Cloud Foundation

This procedure describes how to unstretch a vSAN cluster in VMware Cloud Foundation.

- Consolidate or delete all virtual machine snapshots on the vSAN cluster before unstretching the cluster.
- If you are unstretching a vSAN storage cluster (formerly known as vSAN Max), unmount any stretched vSAN compute clusters to which the vSAN storage cluster is providing a remote datastore.

1. Get the ID of the cluster you are unstretching.
   a) In the VCF Operations console, click**Developer Center** > **APIs & SDKs**.
   b) Click the **API Explorer** link for the SDDC Manager API.
   c) In the API Explorer, navigate to APIs for managing clusters and click `GET /v1/clusters`.
   d) Click **Execute**.
   e) Click **Download** to download the JSON file.
   f) Open the JSON file and copy the cluster ID for `SDDC-Cluster1`.
2. Prepare the JSON request body.
   a) Click **Patch /v1/clusters/id**.
   b) Under **ClusterUpdateSpec** field, click **Cluster Update Data ClusterUpdateSpec{ ... }**.
   c) Click **Download** to download the JSON file.
   d) Edit the downloaded JSON file so that it contains only the unstretch information similar to the example below.

      Note:

      The *azToRemove* variable is required for compute only clusters which are stretched through SDDC Manager version 9.1 or higher.

      ```
      { "clusterUnstretchSpec": {"azToRemove": <availability_zone>} }
      ```
3. Run the unstretch cluster API.
   a) For the **ClusterUpdateSpec** field, update the cluster UID (you retrieved this in step 1) and unstretch JSON file with the payload you prepared in step 2.
   b) Click **Execute**.

      The unstretch cluster task is displayed in the SDDC Manager task panel.
   c) Monitor the unstretch cluster task till it is completed.

      All hosts from AZ2 are removed from the unstretched cluster and the cluster is converted to standard vSAN cluster.

Replace a Failed Host in a Stretched Cluster

If an ESX host or ESX host component in a stretched cluster fails, it is recommended that you replace the host with a new host.

- Image the replacement host with the same ESX version as the other hosts in the cluster.

- Check the health of the cluster.

1. Get the IDs of the hosts to be removed.
   a) In the VCF Operations console, click**Developer Center** > **APIs & SDKs**.
   b) Click the **API Explorer** link for the SDDC Manager API.
   c) Under APIs for managing hosts, click **GET /v1/hosts**.
   d) Click **Execute**.
   e) Click **Download** to download the JSON file.
   f) Open the JSON file and copy the the ID of the host to be removed.
2. Get the ID of the cluster from where the host is to be removed.
   a) In the API Explorer, navigate to APIs for managing clusters and click `GET /v1/clusters`.
   b) Click **Execute**.
   c) Click **Download** to download the JSON file.
   d) Open the JSON file and copy the the cluster ID.
3. Prepare the JSON request body.
   a) Click **Patch /v1/clusters/id**.
   b) Under **ClusterUpdateSpec**, click **Cluster Update Data ClusterUpdateSpec{ ... }**.
   c) Click **Download** to download the JSON file.
   d) Edit the JSON file so that it contains only the compact section similar to the example below.

```
{
  "clusterCompactionSpec": {
     "hosts": [ {
        "id": "ESX host 1 ID"
     }, {
        "id": "ESX host 2 ID"
     }, {
        "id": "ESX host 3 ID"
     } ]
  }
}
```

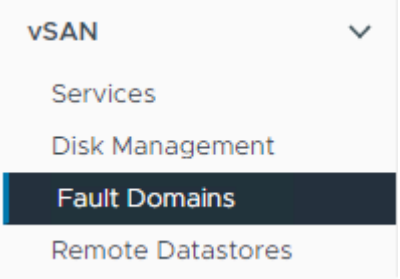
4. Run the compact cluster API.
   a) In the `id` field, replace the values with the host IDs you retrieved in step 1.
   b) Click **Execute**.
   c) Monitor the task till it is completed.
5. Decommission the host to be removed.
   See Decommission ESX Hosts.
6. Commission the replacement host to the same network pool as the removed host.
   See Commission ESX Hosts.
7. Expand the cluster to add the commissioned host to the cluster. See Expand a Stretched Cluster in VMware Cloud Foundation.
8. If required, SSH in to each newly added host and add a static route to the vSAN network of the witness host.

Change the vSAN Witness Host in a Stretched Cluster

You can replace or change the vSAN witness host for a stretched cluster using the vSphere Client without affecting lifecycle operations.

Verify that the vSAN witness host is not in use by another cluster, has a VMkernel configured for vSAN traffic, and has no vSAN partitions on its disks.

1. In a web browser, log in to vCenter at `https://vcenter_fqdn/ui`.
2. Select **Menu** > **Hosts and Clusters**.
3. In the inventory panel, expand **vCenter** > **Datacenter**.
4. Select the stretched cluster and click **Configure**.
5. Under vSAN, click **Fault Domains**.



6. Click the **Change** button.
7. Select a new host to use as the vSAN witness host and click **Next**.
8. Claim disks on the new witness host and click **Next**.
9. Review the configuration and click **Finish**.

Configuring NSX Federation for the VCF Management Components in VCF

With NSX Federation, you can federate the management domain NSX or a workload domain NSX across VCF instances. You configure NSX Federation for the VCF management components that connect to an NSX overlay in the management domain between two regions. The steps to configure NSX Federation can apply to a workload domain as well.

For information about the maximum number of federated NSX Managers and other NSX federation maximums, see VMware Configuration Maximums.

Note:

Workload domains that share an NSX Manager are considered a single location.

You perform some of the tasks on the first NSX instance while others you perform on the second federated NSX instance.

NSX Instance	Tasks to be Performed
First Instance	1. Create the NSX Global Manager Clusters for NSX Federation 2. Prepare the First Instance NSX Local Manager for NSX Federation 3. Enable NSX Federation in the First VCF Instance
Enable high availability for NSX Federation Control Plane in an additional instance	1. Create the NSX Global Manager Clusters for NSX Federation 2. Set the Standby NSX Global Manager
Additional Instance	1. Deploy NSX Edge Cluster to Additional Instance 2. Add an Additional Instance Location to NSX Global Manager 3. Stretching Fleet level management network between VMware Cloud Foundation Instances: a. Connect the Additional VCF Instance to the Cross-Instance Tier-0 Gateway b. Add the Additional VCF Instance as a Location to the Cross-Instance Tier-1 Gateway

Create the NSX Global Manager Clusters for NSX Federation

An NSX Federation environment contains an active and a standby NSX Global Manager and one or more NSX Local Manager instance or NSX Local Manager clusters. The standby NSX Global Manager appliance provides high availability and disaster recovery.

You perform the following tasks to create the NSX Global Manager cluster at each location.

- Set the Active NSX Global Manager in the Management Domain
- Set the Standby NSX Global Manager

Deploy the first NSX Global Manager Appliance

You deploy three NSX Global Manager nodes in the management domain.

1. Download the NSX OVF file from the Broadcom Support Portal.
2. In a web browser, log in to the vSphere Client at `https://vcenter_fqdn/ui`.
3. Right-click the default cluster in the management domain and select **Deploy OVF template**.
4. Select **Local file**, click **Upload files**, and navigate to the OVA file.
5. Click **Next**.
6. Enter a name and a location for the NSX Manager VM, and click **Next**.
   The name you enter appears in the vSphere and vCenter inventory.
7. Select the compute resource on which to deploy the NSX Manager appliance page and click **Next**.
8. Review the OVF template details and click **Next**.
9. Accept the license agreement and click **Next**.
10. Specify the deployment configuration size and click **Next.**
    You can refer to VMware Configuration Maximums to ensure that you choose the correct size for the scale or your environment.
11. Specify storage for the configuration and disk files.
    - Select the virtual disk format.
    - Select the VM storage policy.
    - Specify the datastore to store the NSX Manager appliance files.
    - Click **Next**.

    Note:
    The virtual disk format is determined by the selected VM storage policy when using a vSAN datastore.
12. Select the management network as the destination network and click **Next**.
13. In the **Application section**, enter the system root, CLI admin, and audit passwords for the NSX Manager.
    Your passwords must comply with the password strength restrictions.
    - At least 12 characters
    - At least one lower-case letter
    - At least one upper-case letter
    - At least one digit
    - At least one special character
    - At least five different characters

14. In the **Optional parameters** section, leave the password fields blank.
15. In the **Network Properties** section, enter the hostname of the NSX Manager.
16. For **Rolename**, select the **NSX Global Manager** role.
17. Enter the default gateway, management network IPv4, and management network netmask.
18. In the **DNS** section, enter the DNS Server list and Domain Search list.
19. In the **Services Configuration** section, enter the NTP Server list and enable SSH.
20. Verify that all your custom OVF template specification is accurate and click **Finish** to initiate the deployment.
21. After the deployment is complete, power on the NSX Global Manager node.
    Right-click the NSX Global Manager VM and, from the **Actions** menu, select **Power** > **Power on**.
22. In a web browser, log in to NSX Global Manager at `https://gm_node1_fqdn/`.
23. Accept the end-user license agreement and click **Continue**.
24. Join the VMware Customer Experience Program and click **Save**.
25. Repeat steps 4 to 22 to deploy two additional NSX Global Manager nodes.

Cluster the NSX Global Manager Appliances

Form a cluster from the three NSX Global Manager nodes you deployed in the management domain.

1. Log in to the first NSX Global Manager appliance by using an Secure Shell client as the `admin` account.
2. Run the following command to retrieve the NSX Global Manager cluster ID.
   ```
   get cluster config | find Id:
   ```
3. Run the following command to retrieve the thumbprint of the NSX Global Manager API certificate.
   ```
   get certificate api thumbprint
   ```
4. Log in to the second NSX Global Manager appliance by using an Secure Shell client as the `admin` account.
5. Add the second appliance to the cluster.
   ```
   join first_node_IP cluster-id cluster_ID username admin password nsx_admin_password
   thumbprint api_thumbprint
   ```
6. After the warning message that data will be lost on the appliance, enter `Yes`.
   The cluster creation and stabilization process might take up to 15 minutes.
7. Verify that the status for every cluster service group is UP before making any other cluster changes.
   ```
   get cluster status
   ```
8. Repeat the steps to join the third node to the cluster.
9. Verify the cluster status on the web interface.
   a) Log in to the NSX Global Manager web interface and select **Configuration** > **Global Manager Appliances**.
   b) Verify that the **Cluster status** is green that the cluster node is **Available**.

Create an Anti-Affinity Rule for the NSX Global Manager Cluster

Create an anti-affinity rule to ensure that the NSX Global Manager nodes run on different ESX hosts. If an ESX host is unavailable, the NSX Global Manager nodes on the other hosts continue to provide support for the NSX management and control planes.

1. In a web browser, log in to the management domain vCenter at `https://vcenter_fqdn/ui`.
2. Select **Menu** > **Hosts and Clusters**.
3. In the inventory, expand **vCenter** > **Datacenter**.
4. Select the vSphere cluster where the NSX Global Manager nodes are running and click the **Configure** tab.
5. Select **VM/Host rules** and click **Add**.
6. Enter the rule details.

Option	Description
Name	Type a name for the rule.
Enable rule	Select this option.
Type	Select **Separate Virtual Machines**.
Members	Click **Add**, select the three NSX Global Manager nodes, and click **OK**.

7. Click **OK** in the Create VM/Host rule dialog box.

Assign a Virtual IP Address to the NSX Global Manager Cluster

To provide fault tolerance and high availability to NSX Global Manager nodes, assign a virtual IP address (VIP) to the NSX Global Manager cluster.

1. In a web browser, log in to the NSX Global Manager node at `https://gm_node_1-fqdn/`.
2. Navigate to**System** > **Global Manager Appliances**.
3. Click **Set Virtual IP** and enter the VIP address for the cluster.

   Ensure that VIP is part of the same subnet as the other management nodes.
4. Click **Save**.
5. Verify that the VIP is working correctly by logging in to the NSX Global Manager using the FQDN associated with the virtual IP address assigned to the cluster.

Prepare the First Instance NSX Local Manager for NSX Federation

To prepare for NSX Federation, you create an IP pool in the NSX Local Manager. The NSX Global Manager assigns IP addresses from this pool to the Edge nodes for remote tunnel end point (RTEP) interfaces. You also set the global fabric MTU to match the end-to-end MTU between instances.

You perform this procedure at each location to prepare the RTEP networking.

1. In a web browser, log in to NSX Global Manager cluster for the management domain at `https://lm_vip_fqdn/`.
2. On the main navigation bar, click **Networking**.
3. Create an IP pool for RTEP in Local Manager
   a) In the navigation pane, select **IP Address Pools** and click **Add IP address pool**.
   b) Enter a name.
   c) Under **Subnets**, click **Set**.
   d) In the **Set Subnets** dialog box, click **Add subnet** > **IP Ranges**.
   e) Configure the **CIDR** and **Gateway** and click **Add**.
   f) Click **Add** and then click **Apply**.
   g) Click **Save**.
4. Configure MTU for RTEP.
   a) On the main navigation bar, click **System**.
   b) Select **Fabric** > **Settings**.
   c) Under **Global Fabric Settings**, Click **Edit** for Remote Tunnel Endpoint.
   d) Enter maximum MTU supported between the network fabric between locations

Enable NSX Federation in the First VCF Instance

To enable NSX Federation in VCF, set the NSX Global Manager as active and add the existing NSX Local Manager in the management domain as a location to the NSX Global Manager.

Set the Active NSX Global Manager in the Management Domain

Activate the NSX Global Manager in the management domain.

1. In a web browser, log in to NSX Global Manager cluster for the management domain at `https://gm_vip_fqdn/`.
2. Click **System** and then select **Location Manager**.
3. Click **Make Active** and enter a name for the active NSX Global Manager.
4. Click **Save**.

Add the First Instance Location to NSX Global Manager

Add the NSX Manager cluster in the first instance in the management domain as a location to the NSX Global Manager.

> **Important:**
>
> Before adding a location to NSX Global manager ensure no unsupported features are configured on the NSX Local Manager.
>
> If the edge cluster was deployed as a Centralized Transit Gateway, the following unsupported objects must first be reconfigured or deleted:
>
> - Edge cluster in the VPC Connectivity profile
> - External IP blocks in the VPC Connectivity profile
> - Private Transit Gateway IP block in VPC Connectivity profile
> - External Connection in the VPC Connectivity profile
> - External Connection removed from the Transit Gateway
> - External Connection deleted

1. Obtain the certificate thumbprint of the NSX Local Manager cluster.
   a) Log in to vSphere Client and open the web console for one of the NSX Managers.
   b) Log in as the **Admin** user.
   c) To enable SSH on the NSX Manager, run the command.

```
start service ssh
```

d) Use a Secure Shell (SSH) client and to log in NSX Manager as the **Admin** user.
e) Retrieve the NSX Local Manager cluster VIP thumbprint by running the command.

```
get certificate cluster thumbprint
```

f) To deactivate SSH, run the command.

```
stop service ssh
```

2. Add NSX Manager as a location to the NSX Global Manager.
   a) Log in to the active NSX Global Manager at `https://active_gm_vip_fqdn/`.
   b) Navigate to**System** > **Location Manager** and click **Add On-Prem Location**.
   c) In the **Add New Location** dialog box, configure the settings.

Option	Description
Location Name	Enter a name for the location.
FQDN/IP	Enter the FQDN or IP address of the NSX Manager cluster VIP. Do not enter an individual NSX Manager FQDN or IP.
Username and Password	Provide the **admin** credentials for the NSX Manager at the location.
SHA-256 Thumbprint	Add the thumbprint you retrieved.
Check Compatibility	Click **Check Compatibility** to ensure that the location can be added. This checks that the NSX version is compatible.

   d) Click **Save**.
3. Configure networking on the NSX Local Manager nodes.
   a) On the **Location Manager** page, in the **Locations** section, click **Networking** for the location you are adding.
   b) Click **Configure**.

      The **Local Manager Edge cluster** view opens.
   c) Expand the edge cluster and for the first edge node click the ellipsis icon and select **Edit Remote Tunnel Endpoint**
   d) In the **Remote Tunnel Endpoint Configuration** pane, configure the settings.

Option	Value
Host Switch	*nsxDefaultHostSwitch* or *overlay-vlan-switch-xxx*
Teaming Policy Name	Select **Use Default**.
RTEP VLAN	Enter the VLAN for the host.
IP Pool for all Nodes	Select the IP pool.

   e) Repeat for the remaining NSX edges in the NSX edge cluster.
   f) Click **Save**.

Import the First Instance NSX Local Manager Configuration

You import segments, tier-0 gateways, and tier-1 gateways from the NSX Local Manager to the NSX Global Manager.

1. Log in to the NSX Global Manager at **https://*active_gm_vip_fqdn*/**.
2. On the **System** tab, select the **Location Manager** pane.
3. Under **Locations**, select **Actions** and click **Import to GM**.
4. Verify that you have a recent backup and click **Proceed to import**.
5. In the Preparing for import dialog box, click **Next** and then click **Import**.

   Wait for a confirmation that the import is successful.

Set the Standby NSX Global Manager

You provide high availability of the active NSX Global Manager by configuring the NSX Global Manager in the second instance as standby to the active cluster. In case of failure of the cluster in the first instance, you can use the cluster in the second instance to provide networking capabilities.

Create the standby NSX Global Manager cluster. See Create the NSX Global Manager Clusters for NSX Federation.

1. Obtain the certificate thumbprint of the Standby NSX Global Manager cluster.
   a) Log in to vSphere Client and open the web console for one of the NSX Managers and log in as the **Admin** user.
   b) To enable SSH on the NSX Manager, run the command.
      ```
      start service ssh
      ```
   c) Use a Secure Shell (SSH) client and to log in NSX Manager with the **Admin** user.
   d) Retrieve the NSX Global Manager cluster thumbprint by running the command.
      ```
      get certificate cluster thumbprint
      ```
   e) To deactivate SSH, run the command.
      ```
      stop service ssh
      ```
2. Add an NSX Global Manager instance.
   a) Log in to the active NSX Global Manager at `https://active_gm_vip_fqdn/`.
   b) Navigate to**System** > **Location Manager** and click **Add Standby**.
   c) In the **Add New Location** dialog box, configure the settings.

Option	Description
Location Name	Enter a name for the location.
FQDN/IP	Enter the FQDN or IP address of the NSX Global Manager cluster thumbprin.
Username and Password	Provide the **admin** credentials for the NSX Manager at the location.
SHA-256 Thumbprint	Add the thumbprint you retrieved.
Check Compatibility	Click **Check Compatibility** to ensure that the location can be added. This checks that the NSX version is compatible.

   d) Click **Save**.

Stretch the VCF Fleet Management Segment Between VCF Instances

You prepare the environment and stretch the network between both locations.

The management domain NSX Manager instance has a Tier-0 gateway, Tier-1 gateway, and one segment created during the deployment of the VCF fleet level components. When the NSX Manager instance is federated, the Tier-1 gateway and the Tier-0 gateways are imported to become global objects managed by the NSX Global Manager for cross-instance use. See VCF Management Dedicated VLAN and NSX Stretched Overlay Segment Network Model.

Both gateways are configured to have locations for both instances. You can manage the ingress and egress routing for the cross-instance segment when you move them between VMware Cloud Foundation instances.

> **Note:**
> Cross-instance segments cannot have overlapping IP addresses/ranges.

Deploy an NSX Edge Cluster in the Additional VCF Instance

You deploy an NSX edge cluster in the additional VCF Instance that is used by the cross-instance Tier-1 and Tier-0 getaways.

You deploy can NSX edge cluster by using NSX Manager UI or SDDC Manager API without a Tier-1 or Tier-0 gateway. The Tier-1 and Tier-0 gateways of the first location are then stretched to this location.

This procedure covers the creation of the NSX edge cluster by using the SDDC Manager API. To use the NSX UI, see Create an NSX Edge Cluster and Add Edge Nodes.

Note:

You cannot create a Centralized Transit Gateway and later import it to NSX Global Manager as some objects created block the import.

1. Create a JSON file for the deployment of the NSX edge cluster.

```
{
  "edgeClusterName": "lax-m01-ec01",
  "edgeClusterProfileType": "CUSTOM",
  "edgeClusterProfileSpec": {
    "bfdAllowedHop": 255,
    "bfdDeclareDeadMultiple": 3,
    "bfdProbeInterval": 1000,
    "edgeClusterProfileName": "lax-m01-ecp01",
    "standbyRelocationThreshold": 30
  },
  "edgeClusterType": "NSX-T",
  "edgeRootPassword": "<password>",
  "edgeAdminPassword": "<password>",
  "edgeAuditPassword": "<password>",
  "edgeFormFactor": "MEDIUM",
  "mtu": 8900,
  "edgeNodeSpecs": [
    {
      "edgeNodeName": "lax-m01-r01-en01.lax.rainpole.io",
      "managementIP": "10.21.10.75/24",
      "managementGateway": "10.21.10.1",
      "edgeTepGateway": "10.21.19.1",
      "edgeTep1IP": "10.21.19.2/24",
      "edgeTep2IP": "10.21.19.3/24",
      "edgeTepVlan": 2119,
      "clusterId": "25c01ec7-04a9-4d71-bcec-bab1be23e018",
      "interRackCluster": "false",
      "vmManagementPortgroupName": "m01-cl01-vds01-pg-vm-mgmt",
      "vmManagementPortgroupVlan": 2110
    },
    {
      "edgeNodeName": "lax-m01-r01-en02.lax.rainpole.io",
      "managementIP": "10.21.10.76/24",
      "managementGateway": "10.21.10.1",
      "edgeTepGateway": "10.21.19.1",
      "edgeTep1IP": "10.21.19.4/24",
      "edgeTep2IP": "10.21.19.5/24",
      "edgeTepVlan": 2119,
      "clusterId": "25c01ec7-04a9-4d71-bcec-bab1be23e018",
      "interRackCluster": "false",
      "vmManagementPortgroupName": "m01-cl01-vds01-pg-vm-mgmt",
      "vmManagementPortgroupVlan": 2110
    }
  ]
}
```

2. Log in to the SDDC Manager UI and navigate to **Developer Center**.
3. Validate your JSON file.

```
POST /v1/edge-clusters/validations
```

4. Deploy the edge cluster by using the validated json file.

   ```
   POST /v1/edge-clusters
   ```

Add an Additional Instance Location to NSX Global Manager

Add the NSX Manager in the additional instance in the management domain as a location to the NSX Global Manager.

1. Obtain the certificate thumbprint of the NSX Local Manager cluster.
   a) Log in to vSphere Client and open the web console for one of the NSX Managers.
   b) Log in as the **Admin** user.
   c) To enable SSH on the NSX Manager, run the command.

      ```
      start service ssh
      ```

   d) Use a Secure Shell (SSH) client and to log in NSX Manager as the **Admin** user.
   e) Retrieve the NSX Manager cluster VIP thumbprint.

      ```
      get certificate cluster thumbprint
      ```

   f) Stop the SSH service on the NSX Manager.

      ```
      stop service ssh
      ```

2. Add NSX Manager as a location to the NSX Global Manager.
   a) In a Web browser, log in to the active NSX Global Manager at `https://active_gm_vip_fqdn/`.
   b) Navigate to **System** > **Location Manager** and click **Add On-Prem Location**.
   c) In the **Add New Location** dialog box, enter the location details.

Option	Description
Location Name	Enter a name for the location.
FQDN/IP	Enter the FQDN or IP address of the NSX Manager cluster VIP. Do not enter an individual NSX Manager FQDN or IP.
Username and Password	Provide the **admin** user credentials for the NSX Manager at the location.
SHA-256 Thumbprint	Add the thumbprint you retrieved.
Check Compatibility	Click **Check Compatibility** to ensure that the location can be added. This checks that the NSX version is compatible.

   d) Click **Save**

3. Configure networking on the NSX Local Manager nodes.
   a) On the **Location Manager** page, in the **Locations** section, click **Networking** for the location you are adding.
   b) Click **Configure**.

      The **Local Manager Edge cluster** view opens.
   c) Expand the edge cluster and for the first edge node click the ellipsis and select **Edit Remote Tunnel Endpoint**
   d) In the **Remote Tunnel Endpoint Configuration** pane, configure the settings.

Option	Value
Host Switch	*nsxDefaultHostSwitch* or *overlay-vlan-switch-xxx*
Teaming Policy Name	Select **Use Default**.
RTEP VLAN	Enter the VLAN for the host.
IP Pool for all Nodes	Select the IP pool.

   e) Click **Save**.

      Note:

      After the new location is added, you are prompted to import objects and policies in this location. Do not import objects at this stage, there is no requirement to import any objects in the additional location.

Prepare the Additional VCF Instance NSX Local Manager for NSX Federation

To prepare for NSX Federation, you create an IP pool in the NSX Local Manager. The NSX Global Manager assigns IP addresses from this pool to the NSX Edge nodes for remote tunnel end point (RTEP) interfaces. You also set the global fabric MTU to match the end-to-end MTU between instances.

1. In a web browser, log in to NSX Global Manager cluster for the management domain at `https://lm_vip_fqdn/`.
2. On the main navigation bar, click **Networking**.
3. Create an IP pool for RTEP in Local Manager
   a) In the navigation pane, select **IP Address Pools** and click **Add IP address pool**.
   b) Enter a name.
   c) Under **Subnets**, click **Set**.
   d) In the **Set Subnets** dialog box, click **Add subnet** > **IP Ranges**.
   e) Configure the **CIDR** and **Gateway** and click **Add**.
   f) Click **Add** and then click **Apply**.
   g) Click **Save**.
4. Configure MTU for RTEP.
   a) On the main navigation bar, click **System**.
   b) Select **Fabric** > **Settings**.
   c) Under **Global Fabric Settings**, Click **Edit** for Remote Tunnel Endpoint.
   d) Enter maximum MTU supported between the network fabric between locations.

      To avoid issues, the MTU in the second instance should match the MTU in the first instance.

Add Segments for the Cross-Instance Tier-0 Gateway

You add global segments as uplinks for cross-instance Tier-0 gateway in the additional VCF Instance.

The NSX Edge uplinks require a VLAN transport zone for the VLAN segments connected to the Tier-0 gateway interfaces. Take note of the VLAN transport zone configured for the uplink traffic.

1. In a web browser, log in to the NSX Global Manager cluster for the management domain at `https://gm_vip_fqdn/`.
2. On the main navigation bar, click **Networking.**
3. Create NSX segments for the Tier-0 uplinks.
   a) In the navigation pane, select **Segments** and click **Add Segment**.
   b) Enter a name.
   c) Under **Locations** select the additional location.
   d) Under **Transport Zone** select the VLAN transport zone used for the edge uplinks.
   e) Under **Uplink Teaming Policy** select the assigned teaming policy to the first uplink
   f) Enter the **VLAN** for Edge Uplink 1
   g) Repeat these steps for the Edge Uplink 2 using the following settings.

Option	Description
VLAN	Edge Upink 2 VLAN
Uplink Teaming Policy	Teaming Policy assigned to Second Uplink

Connect the Additional VCF Instance to the Cross-Instance Tier-0 Gateway

You turn the standard Tier-0 gateway into a cross-instance Tier-0 gateway by connecting the additional VCF instances to it. You configure uplink interfaces, BGP, and route redistribution for the additional VCF instances.

The Tier-0 gateway is configured as the primary in both locations to allow for an automated failover of network traffic. Traffic egress is through the primary site and is controlled by the Tier-1 gateway configuration.

1. In a web browser, log in to the NSX Global Manager cluster at `https://active_gm_vip_fqdn/`.
2. Add the VCF instance as a location on the Tier-0 gateway.
   a) On the NSX Manager main navigation bar, click **Networking**.
   b) In the navigation pane, select **Tier-0 Gateways**.
   c) On the Tier-0 Gateway page, click the vertical eclipses for the *cross-instance_tier0* gateway and click **Edit**.
   d) Click **Add Location** and enter the required information.

Setting	Value
Location	Select the location name of the instance being added.
Edge Cluster	Select the Edge cluster name of the instance being added.

   e) Click **Save**.
3. Set interfaces for the instance on the tier-0 gateway.
   a) Expand **Interfaces** and click **Set**.
   b) Click **Add interface**.
   c) Enter a name for the interface and select the instance location.
   d) Set the type to **External** and enter the IP address for the interface.
   e) Select the segment that the interface is connected to and the Edge node corresponding to the instance.
   f) Set the MTU.
   g) Repeat these steps to create three additional interfaces.
4. Configure BGP neighbors.
   a) Expand BGP and under BGP Neighbors, click **Set**.

      You can enable BFD if the network supports and is configured for BFD.
   a) Click **Add BGP neighbor**
   b) Enter the **IP address** for the neighbor and select the instance **location**.
   c) Enter the **remote AS** and **source addresses** for the neighbor.
   d) Click **Timers & Password** and set the **Hold Down Time** to 12 and **Keep Alive Time** to 4.
   e) Enter the BGP neighbor password, click **Save**, and then click **Close**.
   f) Repeat these steps to add another BGP neighbor.
5. Configure Route Re-Distribution
   a) Expand **Route Re-Distribution** and next to the location you are adding, click **Set**.
   b) In the **Set Route Re-distribution** dialog box, click **Add Route-Redistribution**.
   c) Enter *default* as name and, under **Route re-distribution**, click **Set**.
   d) In the **Set Route Re-distribution** dialog box, select all listed sources and click **Apply**.
   e) Click **Add** to finish editing the default route redistribution and click **Apply**.
   f) Click **Save**
6. Click **Close editing**.

Add the Additional VCF Instance as a Location to the Cross-Instance Tier-1 Gateway

Add each additional instance as a location on the cross-instance Tier-1 gateway to enable cross-instance workloads.

1. In a web browser, log in to the NSX Global Manager cluster at `https://active_gm_vip_fqdn/`.
2. On the NSX Manager main navigation bar, click **Networking**.
3. In the navigation pane, select **Tier-1 Gateways**.
4. On the Tier-1 Gateway page, click the vertical eclipses for the *cross-instance_tier1* gateway and click **Edit**.
5. Click **Add Location** and enter the following values.

Setting	Value
Location	Select the location of this instance.
Edge Cluster	Select the NSX Edge cluster of this instance.
Mode	Set to **Secondary**.

6. Click **Save** and click **Close Editing**.

Managing Network Connectivity in *vCenter*

Starting with VCF 9.0, a simplified workflow is introduced to set up network connectivity and perform network configuration.

The new NSX-based workflow for network connectivity is available both in vCenter and NSX for you to have a uniform experience of performing network configurations from either of the components. The new workflow applies many default settings that are recommended by Broadcom for simplifying the deployment process. The workflow covers various scenarios that are related to deployment parameters, transport zones, and uplink profile. They help reduce the number of user inputs that are required to perform network configuration. The workflow also provides a topology visualization to correlate your inputs in the user interface. Contextual visualization helps you get a view of the setup before completing the tasks in the workflow.

Based on the kind of network connectivity you have selected, you can create an edge cluster with a centralized gateway or a distributed gateway with VLAN. For more information, see Workload Networking Detailed Designs and NSX Edge Cluster Detailed Design. To configure a distributed Virtual Extensible Local Area Network (VXLAN) connection, use NSX UI. For more information, see Add a Distributed VXLAN Connection.

Configure Centralized Network Connectivity with Edge Clusters

Create this connection if you have an environment that requires static or dynamic routing with physical fabric. A centralized connection is the one where the traffic is routed through a NSX gateway on an edge node. You can use this task to configure a centralized connectivity with edge cluster deployment from vCenter. For more information on network connectivity using NSX, see Setting up Centralized Connectivity with Edge Clusters.

If you want to deploy edges on a vSphere cluster, ensure that the following requirements are met. VCF Installer and VCF Operations fulfill various prerequisites by default.

- The vSphere cluster must have ESX hosts with vSphere Distributed Switch uplinks that are configured.
- If you are using DHCP to allocate MGMT or TEP IP addresses, then create and configure a DHCP scope accordingly.
- The Compute Manager must be registered with NSX Manager.
- Plan for Edge TEP VLAN. The same VLAN as the Host TEP can be reused. For more information on creating TEPs, see Create an IP Pool for Host Tunnel Endpoint IP Addresses.
- Edges can be deployed on a host-group within a vSphere cluster or on a vSphere cluster. All hosts in the host-group or the vSphere cluster must have identical access to management network, gateway uplinks, host, and Edge TEP networks.
- The vSphere cluster must have a Transport Node Profile (TNP) attached to it. For more information on applying a TNP, see Prepare Cluster Hosts as Transport Nodes by Using TNP.
- **Activate NSX on DVPG** option must be turned on for the host cluster on which the edge has to be deployed to use the same VLAN for edge and host overlay. To turn it on, go to **System > Fabric > Hosts** page for cluster in NSX Manager.
- Edge node reserves 100% CPU shares and memory. vSphere cluster and resource pool must have enough resources to deploy at least two edge nodes.

- NSX Edge VM Resource Requirements:
  - **Small**: 4-GB memory, 2 vCPU, 200-GB disk space. The NSX Edge Small VM appliance size is suitable for lab and proof-of-concept deployments.
  - **Medium**: 8-GB memory, 4 vCPU, 200-GB disk space. The NSX Edge Medium appliance size is suitable for production environments with load balancing.
  - **Large**: 32-GB memory, 8 vCPU, 200-GB disk space. The NSX Edge Large appliance size is suitable for production environments with load balancing.
  - **XLarge**: 64-GB memory, 16 vCPU, 200-GB disk space. The NSX Edge Extra Large appliance size is suitable for production environments with load balancing.
- Management network for NSX Edge must have connectivity to NSX Manager.
- DNS entries for NSX Edge components should be populated in your managed DNS server.
- If dynamic routing is desired, set up 2 BGP Peers on a ToR switch with an interface IP, local ASN, and BGP password. For more information about configuring BGP, see Configure BGP.
- If you are using Border Gateway Protocol (BGP), reserve a local ASN to use for the NSX Edge cluster's Tier-0 Interfaces.
- If static routing is desired, you can configure HA VIP for active/standby Tier-0 Gateway and can use VIP as the Next Hop IP on Top-of-Rack (ToR) switch.
- When setting up the centralized connectivity by using the simplified workflow in vCenter, you must add a minimum of two NSX Edge nodes to a NSX Edge cluster. You can add up to 10 edges to an edge cluster. Once a NSX Edge cluster is successfully created, you see the NSX Edge cluster list on subsequent visits to the **Network Connectivity** page.
- For more information on planning resources, see Planning and Preparation.

If you have upgraded to VCF 9.0, you can use this workflow to manage edge clusters created in the previous version. Also, you can also use this workflow to set up a Tier-0 gateway for an edge cluster, add or delete an edge, and reset passwords. Based on these configurations, you can start creating centralized gateway connectivity with edge cluster deployment.

1. In the *vSphere Client*, navigate to the vCenter for the Workload Domain and select **Networks > Transit Gateways**.
2. Click **Setup Network Connectivity**.
3. In the **Gateway Type** page, select **Centralized Connection**.
4. Click **Next**.
5. Review and complete the **Networking Prerequisites**.
6. Click **Continue**.
7. Configure an Edge Cluster.
   a) Enter a name for the edge cluster.
   b) Enter the value for **Tunnel Endpoint MTU**. The maximum packet size that is allowed over a tunnel is already selected from the **Global Fabric Settings**.

      Note:

      The Edge TEP MTU is the same across the host MTU within the same overlay Transport Zone, so you must set the Global Fabric MTU accordingly.
8. Define the settings for edge nodes.
   a) Select an appropriate size for edge cluster in **Edge Form Factor.**
   b) A minimum of two Edge Nodes are required to deploy an Edge Cluster. Once you have added a node, you can add another edge node by cloning the previously added edge node. Cloning is recommended only when the cloned

Edge is part of the same vSphere cluster with identical networks (edge management, pNIC mapping, VLAN for Edge TEP).

9. Manage passwords.
   a) If the **Manage Edge Password with VCF** option is turned on, then VCF generates the password to manage the Edge nodes. If the **Manage Edge Password with VCF** is turned off, then you can manually set the password.
   b) Enter the CLI and Root Administrator passwords to manage the Edge Nodes.

      Your passwords must comply with the password strength restrictions.

      - At least 12 characters
      - At least one lower-case letter
      - At least one upper-case letter
      - At least one digit
      - At least one special character
      - At least five different characters

Option	Description
CLI Username	Enter the name for the CLI user.
CLI password	Enter the password for the CLI user.
CLI Confirm Password	Enter the password again for confirmation.
System Root password	Enter the password for the root admin user.
System Root Confirm Password	Enter the password again for confirmation.

10. To add a NSX Edge Node to a NSX Edge Cluster, click **Add** to add an **Edge Node** in the **Edge Cluster** page.

Option	Description
Edge Node Name (FQDN)	Enter the FQDN for the NSX Edge node. Each node must have a unique FQDN.
vSphere Cluster	Select a vSphere cluster to host the NSX Edge node. You can select a standard vSphere cluster or a stretched vSphere cluster. All the NSX Edge nodes in an NSX Edge cluster must be hosted on vSphere clusters of the same type. If the vSphere cluster you select already hosts management virtual machines that are connected to the host Management port group, the **VM Management Port group VLAN** settings are not available.
Resource Pool (Optional)	Select the resource pool for the Edge node. **Note:** A resource pool must be created before running the workflow.
Host Group Affinity	Enter **Yes** if you want the NSX Edge to be pinned to a host group affinity. Edge host affinity enables the Edge to be pinned to a given host or group of hosts. Best Effort Restart (BER) policy is applied to the Edge VMs that ensures a graceful shutdown of the Edge VM when the host enters maintenance mode.
Host Group	If you have selected the **Host Group Affinity** as **Yes**, then select a host group to which the Edge node must be pinned. **Note:** A host group must be created before running the workflow.

Option	Description
Data Store	Select a data store where the Edge node virtual machine is placed.
Management IP	
IP Allocation	Select DHCP or Static.
Port Group	Select the port group for the node. • For management workload, VM management port group is created. If a different port group is required, then create the port group before performing this operation. • For a VI workload, VM management port group is not created. Create a port group before performing this operation.
Management IP CIDR	Enter a CIDR for Management IP if you have selected **Static** option for IP allocation. Each node must have a unique management IP.
Default Gateway	Enter the default gateway address if you selected the Static option for IP allocation.
Uplinks	If **Use the host overlay network configuration from the selected vSphere cluster** checkbox is selected, the existing overlay network configuration (such as VLAN ID and TEP IP assignment) is automatically fetched from the hosts in the selected vSphere cluster. This option is enabled if the selected cluster/host is prepared and **NSX enabled on DVPG** setting is **Yes**. If this checkbox is not selected, manually enter the information for uplink configuration.
Edge Node Uplink Mapping	ESX is added to at least one VDS by which ESX pNICs get mapped to uplinks on VDS. Map the Edge NIC to these ESX pNICs. The pNICS are also auto-selected when there are two pNICs on a VDS. In this case, if you choose the first pNIC as an active uplink for the first interface (fp-eth0) the other pNIC is automatically selected as a standby uplink. For the second interface (fp-eth1), exactly the reverse pNICs are selected as active and standby uplinks. If there are multiple pNICs or pNICs connected to multiple VDS, pNICs are not auto-selected and you have to manually map them.
TEP VLAN	Select the VLAN ID for Tunnel Endpoints (TEP). This field is auto-populated if the **Use the host overlay network configuration from the selected vSphere cluster** checkbox is selected.
IP Allocation (TEP)	Select the IP allocation method for TEP. Define a vLAN ID for the Host TEP network. You can select from the following options. Based on the IP allocation method, provide the other required details. • DHCP - DHCP Server provides IP addresses to the hosts. The external DHCP server allocates the TEP IP. • IP Pool - Select an IP Pool for allocations. • Static IP List - Provide values IPv4 Static List, Static IPv4 Gateway, and IPv4 Subnet Mask. This field is auto-populated if the **Use the host overlay network configuration from the selected vSphere cluster** checkbox is selected.
IP Pool	Select the IP pool from the drop-down menu.

11. Under **Uplinks**, select **Use the host overlay network configuration from the selected vSphere Cluster**. When you enable this check-box, it automatically gets the existing overlay network configuration from the hosts in the selected vSphere cluster.
12. Click **Apply**
13. In the **Edge Cluster** page, use **Clone** to clone the Edge Node.
14. Configure the settings that cannot be cloned.

Option	Description
Edge Node Name (FQDN)	Enter the FQDN for the NSX Edge node. Each node must have a unique FQDN.

15. Click **Apply**.
16. In the **Workload Domain Connectivity** page, configure the parameters to enable external connectivity.
17. To skip gateway and routing configurations for now and configure them later, click **Skip.**

    In this case, the system hides **Gateway and Routing** fields and the edges are deployed without the gateway and routing configurations. You can configure these settings later from vCenter. Navigate to the vCenter instance for the management or workload domain and select **Networks > Network Connectivity**. On the **Edge Cluster** page, click **Set** under **Gateway**.

Option	Description
Gateway Name	Enter a name for the gateway.
High Availability Mode	Select the HA mode for the gateway.
Routing Configuration	
Gateway Routing Type	Select from BGP or Static options.
Local Autonomous System	Enter the autonomous system number if the gateway routing type is set to BGP.

18. Set **Gateway Uplinks**. For each edge node, configure your gateway uplink to establish an uplink connectivity between the gateway and Top of the Rack router. You can only configure up to two Edge Nodes with Gateway uplinks if HA mode is set to Active Standby.

Option	Description
Gateway Interface VLAN	Enter the gateway interface vLAN. This field is displayed when **Gateway Routing Type** is set to **Static**.
Gateway Interface CIDR	Enter the gateway IP address. This field is displayed when **Gateway Routing Type** is set to **Static**.
BGP Peer IP	Enter first uplink BGP peer IP to establish a connection between a Tier-0 gateway and a router in your physical infrastructure. This field is displayed when **Gateway Routing Type** is set to **BGP**.
MTU	Enter the MTU for the NSX Edge cluster. The MTU value can be between 1500-9000. This field is displayed when **Gateway Routing Type** is set to **BGP**.
BGP Peer Password	Enter the BGP password. This field is displayed when **Gateway Routing Type** is set to **BGP**.

Option	Description
Confirm Password	Confirm the BGP password. This field is displayed when **Gateway Routing Type** is set to **BGP**.

19. To enable networking inside and outside VPC, IP blocks must be pre-provisioned where a transit gateway is defined. Under VPC configuration, set the parameters.

    To edit this configuration in the *vSphere Client*, navigate to the vCenter instance for the management or workload domain and select **Networks > Network Connectivity**.

Option	Description
External IP Blocks	Enter external IP blocks that can be used to create public subnets in the VPC. The entered IP blocks allow outside connectivity to VPC workloads either on public subnets or through external IPs.
Private - Transit Gateway IP Blocks	Enter private IP blocks to create private subnets in the VPC. The entered IP blocks are available for inter-VPC communication and it is not advertised by the Transit Gateway to the outside datacenter.

20. Review your configurations under **Edge Cluster Details** and **Workload Domain Connectivity**.
21. Click **Deploy**.

Add an NSX Edge Cluster

Learn how to add an edge cluster. Having a multi-node cluster of NSX Edges helps ensure here is at least one NSX Edge is always available.

- Ensure that there is at least one NSX Edge node which is ready and it must not be part of any existing cluster.
- Optionally, create an NSX Edge cluster profile for high availability (HA). You can also use the default NSX Edge cluster profile.

To create a tier-0 logical router or a tier-1 router, with stateful services such as NAT and load balancer, you must associate it with an NSX Edge cluster.

Therefore, even if you have only one NSX Edge, it must belong to an NSX Edge cluster to be useful. An NSX Edge transport node can be added to only one NSX Edge cluster. An NSX Edge cluster can be used to back multiple logical routers.

After you have created the NSX Edge cluster, you can edit it to add more NSX Edges.

You can also configure failure domains for NSX edge nodes. A failure domain is a logical grouping of NSX Edge nodes within an NSX Edge Cluster. If there is a failure affecting multiple NSX Edge node, failure domains compliment the auto placement algorithm and guarantee service availability. For more information about configuring a failure domain, see Configure Failure Domains.

1. On the vSphere Client home page, navigate to **Networking> Configure.**
2. On the **Configure** tab, expand **Networking** and select **Edge Clusters**.
3. Click **Add Cluster**.
4. In the **Edge Cluster** page, enter the information.

Option	Description
Edge Cluster Name	Enter a name for the NSX Edge cluster.
Tunnel Endpoint MTU	The Edge TEP MTU is the same across the host MTU within the same overlay Transport Zone, so you must set the Global Fabric MTU accordingly.
NSX Edge Nodes	

Edge Form Factor	Select an appropriate form factor for the Edge cluster. • **Small**: 4-GB memory, 2 vCPU, 200-GB disk space. The NSX Edge Small VM appliance size is suitable for lab and proof-of-concept deployments. • **Medium**: 8-GB memory, 4 vCPU, 200-GB disk space. The NSX Edge Medium appliance size is suitable for production environments with load balancing. • **Large**: 32-GB memory, 8 vCPU, 200-GB disk space. The NSX Edge Large appliance size is suitable for production environments with load balancing. • **XLarge**: 64-GB memory, 16 vCPU, 200-GB disk space. The NSX Edge Extra Large appliance size is suitable for production environments with load balancing.
Minimum 2 Edge Nodes are required to deploy an Edge Cluster	For details on configuring an Edge Node, see Add an NSX Edge Node to an NSX Edge Cluster. Once you have added a node, you can add another edge node by cloning the previously added edge node.
Auto-generate passwords with VCF to manage Edge nodes	If the **Auto- generate passwords with VCF to manage Edge nodes** is turned on, then the password is automatically generated. If the **Manage Edge Password with VCF** is turned off, then you can manually set the password. **Note:** To view or reset the password after edge deployment, see Reset password.

5. Add Edge Node credentials (optional).

   Your passwords must comply with the password strength restrictions.

   - At least 12 characters
   - At least one lower-case letter
   - At least one upper-case letter
   - At least one digit
   - At least one special character
   - At least five different characters

   Option

Option	Description
CLI Username	Enter the name for the CLI user.
CLI password	Enter the password for the CLI user.
CLI Confirm Password	Enter the password again for confirmation.

6. Add root credentials.

   Your passwords must comply with the password strength restrictions.

   - At least 12 characters
   - At least one lower-case letter
   - At least one upper-case letter
   - At least one digit
   - At least one special character

- At least five different characters

Option	Description
System Root password	Enter the password for the root admin user.
System Root Confirm Password	Enter the password again for confirmation.

Add an NSX Edge Node to an NSX Edge Cluster

Learn how to add a NSX Edge Node to an Edge Cluster through vCenter. It provides a simplified workflow, centralized management, reduced complexity, and improved operational efficiency for virtual private cloud environments.

When setting up the centralized connectivity by using the simplified workflow in vCenter, you must add a minimum of two Edge nodes to an Edge cluster. You can add up to 10 edges to an edge cluster. Once an Edge cluster is successfully created, you see an Edge cluster list on subsequent visits to the **Network Connectivity** page.

1. In the *vSphere Client*, navigate to the vCenter for the management or workload domain and select **Networking> Configure.**
2. On the **Configure** tab, expand **Networking** and select **Edge Clusters**.
3. Click **Add Cluster**.
4. On the **Edge Cluster** page, click **Add Node** to add an **Edge Node**.

Option	Description
Edge Node Name (FQDN)	Enter the FQDN for the NSX Edge node. Each node must have a unique FQDN.
vSphere Cluster	Select a vSphere cluster to host the NSX Edge node. You can select a standard vSphere cluster or a stretched vSphere cluster. All the NSX Edge nodes in an NSX Edge cluster must be hosted on vSphere clusters of the same type. If the vSphere cluster you select already hosts management virtual machines that are connected to the host Management port group, the **VM Management Port group VLAN** settings are not available.
Resource Pool (Optional)	Select the resource pool for the Edge node. **Note:** A resource pool must be created before running the workflow.
Host Group Affinity	Enter **Yes** if you want the Edge to be pinned to a host group affinity. Edge host affinity enables the Edge to be pinned to a given host or group of hosts. Best Effort Restart (BER) policy is applied to the Edge VMs that ensures a graceful shutdown of the Edge VM when the host enters maintenance mode.
Host Group	If you have selected the **Host Group Affinity** as **Yes**, then select a host group to which the Edge node must be pinned. **Note:** A host group must be created before running the workflow.
Data Store	Select a data store where the Edge node virtual machine is placed.
Management IP	
IP Allocation	Select DHCP or Static.

Option	Description
Port Group	Select the port group for the node. • For management workload, VM management port group is created. If a different port group is required, then create the port group before performing this operation. • For a VI workload, VM management port group is not created. Create a port group before performing this operation.
Management IP CIDR	Enter a CIDR for Management IP if you have selected **Static** option for IP allocation. Each node must have a unique management IP.
Default Gateway	Enter the default gateway address if you selected the Static option for IP allocation.
Uplinks	If **Use the host overlay network configuration from the selected vSphere cluster** checkbox is selected, the existing overlay network configuration (such as VLAN ID and TEP IP assignment) is automatically fetched from the hosts in the selected vSphere cluster. This option is enabled if the selected cluster/host is prepared and **NSX enabled on DVPG** setting is **Yes**. If this checkbox is not selected, manually enter the information for uplink configuration.
Edge Node Uplink Mapping	ESX is added to at least one VDS by which ESX pNICs get mapped to uplinks on VDS. Map the Edge NIC to these ESX pNICs. The pNICS are also auto-selected when there are two pNICs on a VDS. In this case, if you choose the first pNIC as an active uplink for the first interface (fp-eth0) the other pNIC is automatically selected as a standby uplink. For the second interface (fp-eth1), exactly the reverse pNICs are selected as active and standby uplinks. If there are multiple pNICs or pNICs connected to multiple VDS, pNICs are not auto-selected and you have to manually map them.
TEP VLAN	Select the VLAN ID for Tunnel Endpoints (TEP). This field is auto-populated if the **Use the host overlay network configuration from the selected vSphere cluster** checkbox is selected.
IP Allocation (TEP)	Select the IP allocation method for TEP. Define a vLAN ID for the Host TEP network. You can select from the following options. Based on the IP allocation method, provide the other required details. • DHCP - DHCP Server provides IP addresses to the hosts. The external DHCP server allocates the TEP IP. • IP Pool - Select an IP Pool for allocations. • Static IP List - Provide values IPv4 Static List, Static IPv4 Gateway, and IPv4 Subnet Mask. This field is auto-populated if the **Use the host overlay network configuration from the selected vSphere cluster** checkbox is selected.
IP Pool	Select the IP pool from the drop-down menu.
VLAN MTU Check	Click **Run Check** to perform checks on the distributed switch or host switches that are associated with available transport nodes in the cluster or host group.

5. Under **Uplinks**, select **Use the host overlay network configuration from the selected vSphere Cluster**. When you enable the checkbox, it automatically gets the existing overlay network configuration from the hosts in the selected vSphere cluster.
6. Click **Apply.**
7. In the **Edge Cluster** page, use **Clone** to clone the Edge Node.
8. Configure the settings that cannot be cloned. Enter the FQDN for the NSX Edge node. Each node must have a unique FQDN.
9. Click **Apply**.
10. In the **Workload Domain Connectivity** page, configure the parameters to enable external connectivity.
    a) To skip gateway and routing configurations for now and configure them later, click **Skip.**

       In this case, the system hides the **Gateway and Routing** fields and the edges are deployed without the gateway and routing configurations. You can configure these settings later from vCenter. Navigate to the vCenter instance for the management or workload domain and select **Networks > Network Connectivity**. On the **Edge Cluster** page, click **Set** under **Gateway**.

Option	Description
Gateway Name	Enter the gateway name.
High Availability Mode	Select the HA mode for the gateway.
Routing Configuration	
Gateway Routing Type	Select from BGP or Static options.
Local Autonomous System	Enter the autonomous system number if the gateway routing type is set to BGP.

    b) Set **Gateway Uplinks**. For each edge node, configure your gateway uplink to establish an uplink connectivity between the gateway and Top of the Rack router. You can only configure up to two Edge Nodes with Gateway uplinks if HA mode is set to **Active Standby**.

Option	Description
Gateway Interface VLAN	Enter the gateway interface vLAN. This field is displayed when **Gateway Routing Type** is set to **Static**.
Gateway Interface CIDR	Enter the gateway IP address. This field is displayed when **Gateway Routing Type** is set to **Static**.
BGP Peer IP	Enter first uplink BGP peer IP to establish a connection between a Tier-0 gateway and a router in your physical infrastructure. This field is displayed when **Gateway Routing Type** is set to **BGP**.
MTU	Enter the MTU for the NSX Edge cluster. The MTU value can be between 1500-9000. This field is displayed when **Gateway Routing Type** is set to **BGP**.
BGP Peer Password	Enter the BGP password. This field is displayed when **Gateway Routing Type** is set to **BGP**.
Confirm Password	Confirm the BGP password. This field is displayed when **Gateway Routing Type** is set to **BGP**.

    c) To enable networking inside and outside VPC, IP blocks must be pre-provisioned where a transit gateway is defined. Under **VPC configuration**, set the parameters. To edit this configuration in the *vSphere Client*,

navigate to the vCenter instance for the management or workload domain and select **Networks > Network Connectivity**.

Option	Description
External IP Blocks	Enter external IP blocks that can be used to create public subnets in the VPC. The entered IP blocks allow outside connectivity to VPC workloads either on public subnets or through external IPs.
Private - Transit Gateway IP Blocks	Enter private IP blocks to create private subnets in the VPC. The entered IP blocks are available for inter-VPC communication and are not advertised by the Transit Gateway to the outside datacenter.

11. Review your configurations under **Edge Cluster Details** and **Workload Domain Connectivity**.
12. Click **Deploy**.

Configure Distributed Network Connectivity

Distributed connection allows a lightweight distributed connection to the data-center fabric without the need to deploy Edge nodes or configure dynamic routing. In addition to on-demand networking and distributed routing, it provides distributed services—specifically External IPs (1:1 NAT) and distributed DHCP. It can be complemented by the Virtual Network Appliance Cluster which offers extended NAT support and Load Balancer hence allowing to deploy the Supervisor Cluster.

- Dedicated VLANs and subnets must be available for the Distributed Transit Gateway connection to the ToR. The same VLAN must be configured on all the ToR switches where the ESX hosts are connected which offers VPCs. All the hosts running VPC workloads must have access to these VLANs. The Transit Gateway can be scoped to specific vSphere clusters by defining and associating a specific Network Span.
- Reserve external IP blocks for VPC connectivity. External IP blocks must be the same as or a subset of dedicated VLAN subnets. For more information, see Create a Virtual Private Cloud in vCenter.

A distributed connection is the one where a Transit Gateway connects to a VLAN provisioned in the fabric. This happens directly on the host which implies the VLAN must be available on the *ESX* hosts (on physical uplinks that are used by the **NSX** VMkernel). The workflow is available in both **NSX** and vCenter. This section describes how to use the workflow from vCenter to configure a distributed connectivity. To set up a distributed gateway connectivity for the first time from vCenter, perform the following steps:

1. In the *vSphere Client*, navigate to the vCenter for the Workload Domain and select **Networks > Network Connectivity**.
2. Click **Setup Network Connectivity**.
3. In the **Transit Gateway** page, select **Distributed Connectivity**.

   Select gateway connectivity for the created VPCs that align with the infrastructure requirements and networking preferences.
4. Complete the **Networking prerequisites**. Click **Continue**.
5. Enter the information on **External Network Connectivity Details** page.

Option	Description
VLAN ID	Enter the VLAN ID to which the traffic is to be routed.
Gateway CIDR IPv4 Address	Enter the gateway address through which the traffic should be routed to the external network.
VPC Configuration	
VPC External IPv4 Blocks	Enter IP blocks for VPC.
Private - TGW IPv4 Blocks	Enter IP blocks for Private - TGW IPv4.

6. Enter the information for VPC services. The **Virtual Networking Appliance Cluster (VNAC)** configures the VPS connectivity profile.
7. Enter a name for the cluster and select the form factor from the **Node Size** drop-down menu.

   The default size is medium.
8. Click **Add Node** and enter the information.

Option	Description
Node Name (FQDN)	Enter the FQDN for the NSX Edge node. Each node must have a unique FQDN.
vSphere Cluster	Select a vSphere cluster to host the NSX Edge node. You can select a standard vSphere cluster or a stretched vSphere cluster. All the NSX Edge nodes in an NSX Edge cluster must be hosted on vSphere clusters of the same type. If the vSphere cluster you select already hosts management virtual machines that are connected to the host Management port group, the **VM Management Port group VLAN** settings are not available.
Resource Pool (Optional)	Select the resource pool for the Edge node. **Note:** A resource pool must be created before running the workflow.
Host Group Affinity	Enter Yes if you want the Edge to be pinned to a host group affinity. Edge host affinity enables the Edge to be pinned to a given host or group of hosts. Best Effort Restart (BER) policy is applied to the Edge VMs that ensures a graceful shutdown of the Edge VM when the host enters maintenance mode.
Host Group	If you have selected the **Host Group Affinity** as **Yes**, then select a host group to which the Edge node must be pinned. **Note:** A host group must be created before running the workflow.
Data Store	Select a data store where the Edge node virtual machine is placed.
Management Network Settings	
IP Address Type	• IPv4 Only • IPv4 & IPv6 • IPv6 Only
IP Assignment	Select DHCP or Static.
Port Group	Select the port group for the node. • For management workload, VM management port group is created. If a different port group is required, then create the port group before performing this operation. • For a VI workload, VM management port group is not created. Create a port group before performing this operation.
Management IP CIDR	Enter a CIDR for Management IP if you have selected **Static** option for IP allocation. Each node must have a unique management IP.
Default Gateway	Enter the default gateway address if you selected the Static option for IP allocation.

9. On the **Review** page, review the details and click **Deploy**.

Create a Virtual Networking Appliance Cluster

Edge-based service cluster is a simplified wrapper to deploy cluster of edge nodes that are referred as Virtual Network Appliance (VNA). VNA enables stateful services for VPC workloads that are connected with Distributed Transit Gateway. To deliver services through the VNA node, a generic traffic redirection support is added on DTGW that will redirect qualifying traffic to the VNA nodes. The traffic redirection framework is generic and can support service nodes such as Avi Load Balancer. Like NSX Edge Nodes, service node lifecycle management that is available in NSX and vCenter UI shares the edge system health metrics to VCF operations and advanced network metrics to VCF network operations.

1. In the *vSphere Client* home page, select the **Networking** tab and click **Configure.**
2. On the **Configure** tab, expand **Networking** and select **VNA Clusters**.
3. Click **Add Cluster**.
4. Enter a name for the cluster and select the node size from the **Form Factor** drop-down menu. The default size is medium.
5. Click **Add Node** and enter the information.

Option	Description
Node Name (FQDN)	Enter the FQDN for the VNA node. Each node must have a unique FQDN.
vSphere Cluster	Select a vSphere cluster to host the VNA node. You can select a standard vSphere cluster or a stretched vSphere cluster. All the VNA nodes in a VNA cluster must be hosted on the same type of vSphere cluster.
Resource Pool (Optional)	Select the resource pool for the VNA node. **Note:** A resource pool must be created before running the workflow.
Data Store	Select a data store where the Edge node virtual machine is placed.
Management Network Settings	
IP Address Type	• IPv4 Only • IPv4 & IPv6 • IPv6 Only
IP Assignment	Select DHCP or Static.
Port Group	Select the port group for the node. • For management workload, VM management port group is created. If a different port group is required, then create the port group before performing this operation. • For a VI workload, VM management port group is not created. Create a port group before performing this operation.
Management CIDR	Enter a CIDR for Management IP if you have selected **Static** option for IP allocation. Each node must have a unique management IP.
Default Gateway	Enter the default gateway address if you selected the Static option for IP allocation.

6. Click **Apply**.

Managing Virtual Private Clouds in *vCenter*

Learn how to manage Virtual Private Clouds (VPC) in *vCenter*.

A virtual private cloud (VPC) is a pool of shared network resources that are allocated within a cloud environment, providing isolation between the different organizations using the resources. A subnet is a network with an IP CIDR in VPC and a workload is connected to a subnet.

A hierarchical folder-based approach is used to represent NSX projects and VPC in the *vSphere Client* that facilitates separate management by its various users under the Virtual Managed Folder. With this framework, NSX DVPGs were systematically organized in their respective folders. This streamlined the structural hierarchy between NSX and the *vCenter* inventory.

VPCs, subnets, and various VPC services were configured only from NSX. To connect the workloads, administrators had to navigate to *vCenter*. VCF offers a simplified networking solution. VPCs and Subnets are primarily NSX objects that are integrated in *vCenter*. The administrator manages the VPC and subnet in *vCenter*. This implementation allows the VC admin to create, read, update, and delete the NSX VPC from the vSphere Client. You can also navigate from and to NSX objects within the *vCenter*. For more information on VPCs in NSX, see Virtual Private Cloud in NSX.

VCF provides a consistent networking consumption model that is based on Virtual Private Clouds (VPCs) and Transit Gateways (TGW). This model is natively multi-tenant and can be used not only from NSX, but also from VCF (vCenter, VCF Automation, and monitored from VCF Operations). It aims to provide a standardized, simple, and powerful Cloud networking experience to the network administrators and cloud users from VCF. This cloud networking consumption is built using the following objects:

- **External Connections**: Defines outside connectivity for a Transit Gateway to the datacenter. An external connection can be of the following types:
  - Centralized
  - Distributed
- **Transit Gateway (TGW)**: Gateway inter-connecting VPCs and connecting to an external datacenter through external connections.
- **Virtual Private Cloud (VPC)**: On-demand logical networks connected together that can be either advertised (public subnets) or exposed through networking services like External IPs or NAT. For simplicity, VPCs in VCF use these profiles:
  - VPC Connectivity Profile: Defines outside connectivity configuration for VPCs such as the Transit Gateway they are connected to, the IP Blocks available for consumption, and if a default outbound NAT is configured
  - VPC Service Profile: Defines the common configurations for services available in the VPC such as DHCP, DHCP Relay, or different subnet profiles.

External Connections

External Connection defines how the Transit Gateway (TGW) interconnects the VPCs to the network outside NSX and defines the services that are available on it. Those connections are created from Default (/infra) and define the connection to the external environment.

The following types of External Connections are configurable from the vCenter server for different requirements. The workflow is also available in VMware NSX. For more information about setting the connectivity through VMware NSX., see *Setting up Network Connectivity* in the NSX section. This section describes how to use the workflow from vCenter.

You can select one based on the features you require for your network.

- **Centralized connection:** Connects the transit gateway through a Tier-0 gateway provisioned from NSX. The Tier-0 gateway is deployed on NSX Edge nodes and provides interconnection to the physical fabric. This connection type requires you to deploy NSX Edge nodes and manage their lifecycle.

  Centralized connections allow stateful services and dynamic routing with the physical fabric (BGP) at Tier-0 level. They also allow NAT configuration at the transit gateway level. Create a centralized connection if you have an environment that requires full-scale NSX networking and security services (such as DHCP, NAT, and gateway firewall for VPC and transit gateway) and other stateful centralized network services.
- **Distributed VLAN connection:** Connects the transit gateway directly to VLANs provisioned in the fabric. The connection happens directly on the ESX hosts, which mandates the availability of VLANs on the ESX.

  A distributed VLAN connection allows lightweight connectivity to the data-center fabric without needing to deploy NSX Edge nodes or configure dynamic routing. Stateful services are supported with this connection type using Virtual Network Appliances (VNAs).

A Transit Gateway can be created with a centralized connectivity or it can be created with a distributed connectivity. A TGW can also have multiple external connections to support complex routing, such as separating Internet and Intranet paths. All external connections for a TGW must be of the same type, that is, either all centralized or all distributed. A TGW cannot have a mixed type of connections. Note that a VPC in a project can connect to only one TGW.

In both cases, creating an external connection requires certain pre-requisites to be fulfilled. For a Centralized connection, it requires a Tier-0 gateway deployed with NSX Edge nodes and external routing properly configured. For a Distributed connection, it implies that the VLANs used for outside connections are available on ESX hosts on the same uplinks (physical NICs) as NSX overlay VMkernel adapter.

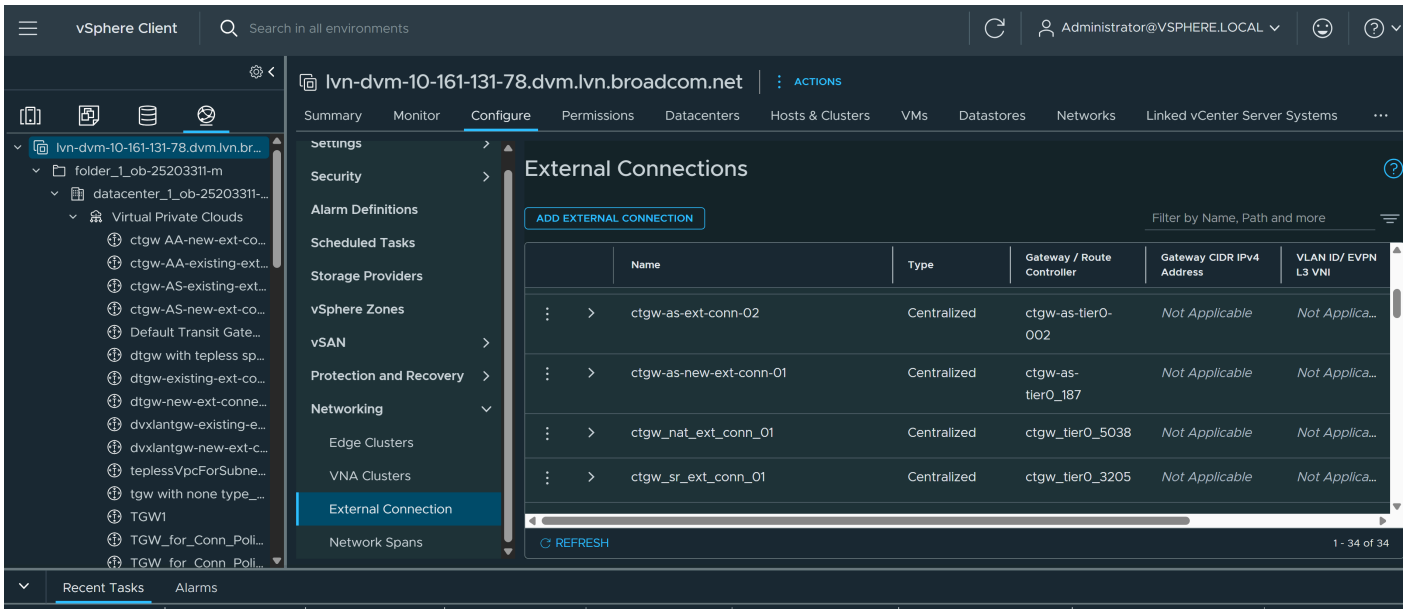
Add a Centralized Connection

You can add a centralized connection to connect the transit gateway through a Tier-0 Gateway.

Tier-0 must be deployed with Edge nodes and external routing should be properly configured.

1. In the *vSphere Client* home page, select the **Networking** tab and click **Configure.**
2. On the **Configure** tab, expand **Networking** and select **External Connection**.

   Figure 396: External Connection



3. Add **External Connection**. Enter the information.

Option	Description
Name	Enter a name for the external connection.
Type	Click to select the type of connection you want to create. The centralized connection is selected by default.
Tier-0 Gateway	Select the Tier-0 or Tier-0 VRF gateway through which you want to route the traffic to external networks.
Advanced Settings	
Remote Networks	Remote networks to be reachable through this connection for the VPCs. The default gateway is 0.0.0.0/0. If you do not specify this field, then only routes of specific networks are programmed else the default route will be programmed on the transit gateway to route traffic to the connection.

Option	Description
External IP Blocks	Select the external IP blocks from the drop-down list. IP subnets from the selected external IP blocks are advertised as static routes to the tier-0 or tier-0 VRF gateway. If you do not specify this field, IP subnets from all the external IP blocks are advertised.
Private - Transit Gateway IP Blocks	If you turn on this toggle, then the private transit gateway IP blocks are advertised through the external connection. The default behavior is to advertise only external IP blocks (all external IP blocks unless you mention the specific ones to be advertised in the **External IP Blocks** field). If you turn on this toggle, you must also add the transit gateway route advertisement rule, "TGW_PRIVATE", to the transit gateway attachment to get the private transit gateway IP blocks advertised.
Provider Outbound SNAT	If this toggle is on, SNAT is enabled for all the outbound traffic irrespective of the subnet type. Source IP that does not belong to the selected External IP Blocks are NATed at the source. a provider can control the outgoing traffic irrespective of the subnet type. Any other NAT rule that is created on a transit gateway uplink has a higher priority than the provider outbound SNAT rule. You can control all outbound traffic by turning on this toggle that works in conjunction with **External IP Blocks** and **SNAT IP Block** fields.
SNAT IP Block	Specify the IP ranges or IP CIDRs to be used for the provider outbound SNAT translated IP. You must also specify this IP block in the **External IP Blocks** field if the **Provider Outbound SNAT** toggle is on.

4. Click **Save**.

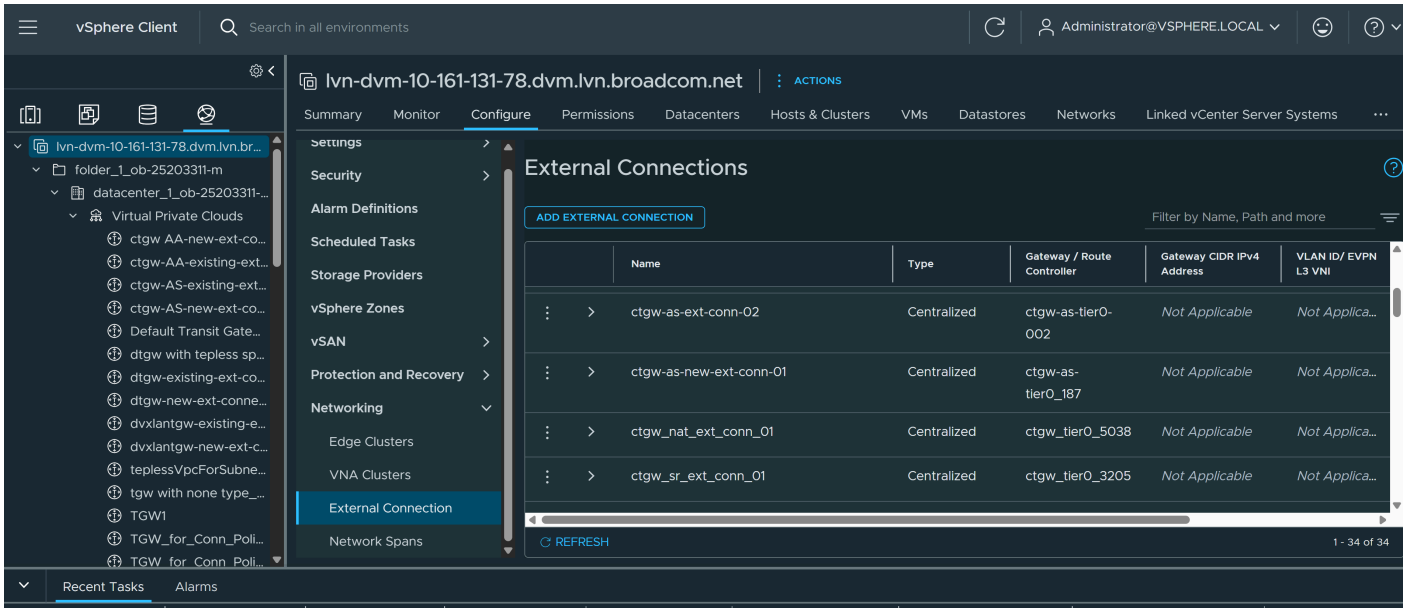
Add a Distributed VLAN Connection

You can add a distributed VLAN connection to connect the transit gateway directly to a VLAN in the physical infrastructure.

The VLANs used for external connections are available on ESX hosts on the same uplinks (physical NICs) as the NSX overlay VMkernel adapter.

1. In the *vSphere Client* home page, select the **Networking** tab and click **Configure.**
2. On the **Configure** tab, expand **Networking** and select **External Connection**.

Figure 397: External Connection



3. Add **External Connection**. Select **Distributed Connection** and enter the name for the connection.

Field Name	Description
Dedicated to Subnet	This option allows you to allocate an external IP block for a specific subnet.
Subnet Extension Connection	You can create this connection for VLAN extension by turning the **Dedicated to Subnet** toggle on and selecting the option from the drop-down menu. • Layer 2: The overlay VM traffic does not go through VPC and Distributed Transit Gateway (DTGW) to the external network. • Layer 2 and Layer 3: In Layer 3, the extended subnet is connected to the VPC gateway and traffic can go through VPC and Distributed Transit Gateway (DTGW) to the external network. **Note:** This option is displayed only when **Dedicated to Subnet** is set to **On**. When **Dedicated to Subnet** is set to **Off**, only **VLAN ID** and **Gateway CIDR IPv4 Address** are displayed.
VLAN ID	Enter the VLAN ID to which the traffic is to be routed. The VLAN must be dedicated for external connectivity of workload deployment in VPCs. It must be stretched across all clusters where VPCs are deployed. If you are creating this connection for VLAN extension, provide the ID of the VLAN that is to be extended. **Note:** The External IP block must be a subset of VLAN CIDRs and any in-use IP address on the physical network must be pre-allocated on NSX to avoid duplicate IP assignment.

Field Name	Description
Gateway CIDR IP Address	Enter the gateway address and netmask through which the traffic should be routed to the external network.
VLAN/Subnet Block	The CIDR block that is associated with the physical infrastructure is obtained from the Gateway CIDR IP address.

4. Click **Save**.

Transit Gateways

When a new tenant (NSX Project) is created, a default Transit Gateway is created automatically in vCenter, for the Project.

A Transit Gateway (TGW) provides both inter-VPC communication and connection to the outside network through External Connections. This model keeps an inter-VPC communication inside a tenant. With a Transit Gateway, there is no need for sharing a Tier-0 directly with an NSX Project. Instead, an External Connection is shared to the project, to which the TGW is connected. A tenant admin can connect a TGW to one or more external External Connections. For more information about connection types, see External Connections.

A TGW is a transit hub that interconnects tenant's virtual private clouds (VPCs), external connections, and tenant's remote networks. It enables inter-VPC connectivity between external / project-scoped subnets of connected VPCs. It reduces operational complexity and provides a more intuitive, centralized interface for managing network resources. Also, this integration lays the foundation for the TGW span in VCF 9.1, offering a complete user experience.

A Transit Gateway can be created with a centralized connectivity or it can be created with a distributed connectivity. A default TGW for every tenant, including Default is created automatically. Based on your network requirements, you can also create more TGWs (both CTGWs and DTGW) for a project. With multiple TGWs in a project, a VPC can be re-attached or made to point to another TGW as and when required. You can move a VPC to another TGW by changing the VPC connectivity profile. When you move a VPC to another TGW, the system runs validation checks at different levels, such as IPAM validation and VPC span validation. When you move a VPC to another TGW, ensure that the requirements for dependent configurations are met.

A TGW can also have multiple External Connections to support complex routing, such as separating Internet and Intranet paths. All external connections for a TGW must be of the same type, that is, either all centralized or all distributed. A TGW cannot have a mixed type of connections.

Centralized Transit Gateway

A centralized transit gateway (CTGW) is created by attaching a TGW with a centralized external connection. Previously, a CTGW shared a similar High Availability (HA) mode and placement on the Edge cluster as the Tier-0/VRF gateway to which it connected. In this case, if the Tier-0/VRF had an active-active HA, CTGW VRF was not able to support stateful services (auto SNAT).

Now, a CTGW can be deployed on an Edge cluster which can be different from the Edge cluster of the Tier-0/VRF gateway to which it connects. Also, the CTGW can now have an independent HA mode that supports configuring multiple topologies and assigning different HA modes per tenant. Note that CTGW sharing its HA mode with the Tier-0 gateway to which it connects is deprecated now and post upgrade any new CTGW that you create will have indepedent an HA mode.

A CTGW can have the following HA modes:

- Active active - If you need a better traffic throughput, define the HA mode as active-active.
- Active standby - If you require services such as NAT, define the HA mode as active-standby. If there are multiple CTGWs in a project, they can have different HA modes. The following image shows multiple centralized transit gateways in a project.

Distributed Transit Gateway

A distributed transit gateway (DTGW) is created by attaching a TGW with a distributed external connection. If there are multiple DTGWs in a project, each DTGW can use the same or a different VLAN but subnets must not overlap within different DTGWs.

VPCs can be migrated between DTGWs by changing the connectivity profile of VPCs. Validations are triggered at the policy layer when a VPC is moved across DTGWs. Ensure that the prerequisites are met before moving a VPC across DTGWs. You can also move a VPC with a VLAN extendable subnet to a different DTGW.

Create a Transit Gateway with Centralized Connection

This task helps you configure an external connection for a Transit Gateway. For a Centralized connection, the Transit Gateway has its own HA Mode (Active/Standby or Active/Active). In Active-Active mode, it does not support NAT including Default Outbound NAT.

1. In the *vSphere Client*, navigate to **Virtual Private Clouds** folder in your datacenter.
2. From the **Actions** drop-down menu, select **New Transit Gateway**.
3. Enter the name for transit gateway and select **Centralized Connection** as your **Transit Gateway Connectivity**.
4. Under the **Additional Settings**, set the **Span**, **HA Mode**, and **TGW Edge Cluster**.
5. Select the default span from the **Span** drop-down list. Click the vertical ellipsis (three dots) to create a span. For more information, see Configure Network Spans.
6. Set the **High Availability mode**. **Active-Standby** is set by default.
7. Select the transit gateway edge cluster from the **TGW Edge Cluster** drop-down menu.
8. Click **Next**.
9. In the **Workload Domain Connectivity** page, set the type of external connection.

   Click **Create New** to add a connection.

   Use Existing option allows you to use the existing connection to create the transit gateway. If **Use Exisitng** is selected, then select an **External Gateway** from the drop-down list.
10. Select the **Tier-0 Gateway** from the drop-down menu. Select the tier-0 gateway through which you want to route the traffic to external networks.
11. Under **VPC Network Configuration**, set the **VPC External IP Blocks**, **IP Blocks**, and **Outbound NAT**.

Option	Description
VPC External IP Blocks	Enter external IP blocks that can be used to create public subnets in the VPC. The entered IP blocks allow outside connectivity to VPC workloads either on public subnets or through external IPs.
Private - Transit Gateway IP Blocks	Enter private IP blocks to create private subnets in the VPC. The entered IP blocks are available for inter-VPC communication and they are not advertised by the Transit Gateway to the outside datacenter.

Option	Description
Default Outbound NAT	Turn on the toggle to ensure that workloads on private subnets have outbound access outside of VPC. One SNAT rule per VPC is auto-plumbed with IP used for translation that is taken from an External IP block for this VPC. **Note:** This toggle is available only if the **N-S Services** toggle is turned on. Turn on the toggle to enable service gateway for the connected subnets and to support centralized services such as N-S firewall, NAT, or gateway QoS profile in the VPC. To configure N-S Services, 1. In *vSphere Client*, click **Networking**. 2. Select **Virtual Private Clouds** and click **Configure > Settings > Connectivity Profiles**. 3. Edit the VPC connectivity profile used by the new VPC. 4. Turn on the toggle for **N-S Services**.

12. Click **Done**.

You transit gateway with centralized connection is created.

Edit Transit Gateway

You can view and edit the current transit gateway (TGW) to add more external connections to your transit gateway.

1. In the *vSphere Client*, navigate to **Networks** under **Inventory**.
2. Click **Virtual Private Clouds** folder in your datacenter.
3. Click **Networks** > **Transit Gateways**.
4. Select a transit gateway from the list under **Virtual Private Clouds** folder.
5. Click **Edit** from the three-dot menu of the transit gateway that you want to edit.
6. To add more external connections to the TGW, in the **External Connection** field select existing connections from the drop-down list or click the three-dot menu and click **Create New**. Fill the required information and click **Save**.
7. To edit a transit gateway with a centralized connection:
   a) Select the **Centralized** or **None** from the **Connection** drop-down list.
   b) Edit the name for the connection.
   c) Select the **HA Mode**, **Network Span** from the drop-down menu.
   d) Enter the **VPC Transit Subnets**.
   e) Select **Transit Gateway Edge Cluster** from the drop-down menu.
   f) Click **Save**.
8. To edit a transit gateway with a distributed VLAN connection:
   a) Enter the **VLAN ID** and **Gateway CIDR IPv4 Address**.
   b) Edit the advanced settings and click **Save**.
9. You can view the properties and can view the **Advertised Rules**. The number of external connections selected while creating a TGW is the same as the number of advertised rules displayed.
10. To view and edit, click **View Configuration**. In the **Type** field, you can select the types of network the gateway can advertise across the connection. If no selection is made, it is set to **None**.

Configure a Static Route

You can configure a static route on the tier-0 gateway to external networks. After you configure a static route, there is no need to advertise the route from tier-0 to tier-1. The tier-1 gateways have a static default route to their connected tier-0 gateway.

1. In the *vSphere Client*, navigate to **Virtual Private Clouds** folder in your datacenter.
2. Select a transit gateway from the list under **Virtual Private Clouds** folder.
3. Select **Configure > Routing and Forward > Static Routes** and click **Add**.
4. Enter a name and network address in CIDR format in the **Network** field. Static routes that are based on IPv6 are supported. IPv6 prefixes can only have an IPv6 next hop.

   Example: 10.10.10.0/23
5. Click **Add Next Hop** to add next-hop information. Select the centralized transit gateway from the **Scope** from the drop-down list.
6. Enter the value for **Admin Distance**.
7. Click **Save**.

Configure a NAT Rule for Transit Gateway

You can add a NAT rule for a Transit Gateway.

SNAT rule should be configured at the furthest point until the Match IP is visible - VPC for Private IP and CTGW for Project IP. For more information, see Network Address Translation (NAT).

1. In the *vSphere Client*, navigate to **Virtual Private Clouds** folder in your datacenter.
2. Select a transit gateway from the list under **Virtual Private Clouds** folder. Alternatively, select a VPC from the list of VPCs are under **Virtual Private Clouds** folder.
3. Select **Configure > Services > NAT** and click **Add NAT Rule**.

Option	Description
Name	Enter a name for your NAT Rule.
Action	Select an action. The following actions are available: • SNAT • DNAT • Reflexive • NO SNAT • NO DNAT
Match Source IP	Enter the source network address. Specify an IP address, or an IP address range in CIDR format. For **SNAT**, **NO_SNAT** and **Reflexive** rules, this is a required field and represents the source network of the packets leaving the network. For **DNAT** and **NO_DNAT** rules, optionally it can contain a source network of incoming packets. For a NULL value, this field represents any network.
Match Source Port	Enter a value for the source port.
Match Destination IP	For **DNAT** and **NO_DNAT** rules, it is a required field and represents the source network of the packets leaving the network. This field is not applicable for **Reflexive**.
Match Destination Port	Enter a value for destination port.
Protocol	Select the required protocol from the drop-down list. • TCP • ICMPv4 • UDP Protocol is not available if **Action** is set to **Reflexive**, **NO SNAT**, or **NO DNAT**.
Translated IP	Enter a value for translated IP. Specify an IPv4 address or an IP address range in CIDR format. If translated IP is less than the match IP for, **SNAT** it works as PAT.
Translated Port	Enter a value for the translated port.
Apply To	Select the same TGW external connection from the drop-down list.
Enabled	Turn on the toggle to enable the rule.
Logging	Turn on the toggle to enable logging.
Priority	A lower value means a higher priority. The default is 0. A N**o SNAT** or **No DNAT** rule should have a higher priority than other rules.

Option	Description
Firewall	The available settings are: • Match External Address - The firewall is applied to the external address of a NAT rule. – For SNAT, the external address is the translated source address after NAT is done. – For DNAT, the external address is the original destination address before NAT is done. – For REFLEXIVE, to egress traffic, the firewall is applied to the translated source address after NAT is done. For ingress traffic, the firewall is applied to the original destination address before NAT is done. • Match Internal Address - is applied to the internal address of a NAT rule. – For **SNAT**, the internal address is the original source address before NAT is done. – For **DNAT**, the internal address is the translated destination address after NAT is done. – For **REFLEXIVE**, for egress traffic, the firewall is applied to the original source address before NAT is done. For ingress traffic, the firewall is applied to the translated destination address after NAT is done. • Bypass - The packet bypasses firewall rules.

4. Click **Save**.

Create a Transit Gateway with Distributed VLAN Connection

This task helps you configure a distributed VLAN connection for a Transit Gateway.

1. In the *vSphere Client*, navigate to **Virtual Private Clouds** folder in your datacenter.
2. From the **Actions** drop-down menu, select **New Transit Gateway**.
3. Enter the name for transit gateway and select **Distributed VLAN Connection**.
4. Under the **Additional Settings**, set the **Span**.
5. Select the **Span** from the drop-down list.
6. In the **Workload Domain Connectivity** page, set the type of external connection.

   Click **Create New** to add a connection.

   Use existing option allows you to use the existing connection to create the transit gateway. If **Use Exisitng** is selected, then select an **External Gateway** from the drop-down list.
7. Add the **VLAN ID**.
8. Add **Gateway CIDR IPv4 Address**.
9. Under **VPC Ntework Configuration**, set the external IP blocks, IP blocks, and outbound NAT.

Option	Description
VPC External IP Blocks	Enter external IP blocks that can be used to create public subnets and also the External IP binding for 1:1 NAT in the VPC. The entered IP blocks allow outside connectivity to VPC workloads either on public subnets or through external IPs.
Private - Transit Gateway IP Blocks	Enter private IP blocks to create private subnets in the VPC. The entered IP blocks are available for inter-VPC communication. Networks created from the Private - Transit Gateway IP blocks are not advertised.

10. In the **VPC Services** page, set the **Virtual Network Appliance Cluster** from the drop-down list and turn the toggle on to enable or disable **Default Outbound NAT**.

    If you have not selected a **Virtual Network Appliance Cluster**, then **Default Outbound NAT** is not enabled.

11. Click **Done**.

You transit gateway with distributed connection is created. To view and edit, select a transit gateway from the list under **Virtual Private Clouds** > **Configure > Settings > Properties**. For more information, see Edit a Transit Gateway.

View the Summary of Transit Gateway

Learn how to view the summary of TGWs in *vCenter*.

1. From your browser, log in to the *vSphere Client* to view the summary of a Transit Gateway.
2. In the *vSphere Client*, navigate to the **Transit Gateway** under **Virtual Private Clouds**.
3. Click **Summary**.

Option	Description
Overview	
Connectivity	Displays the type of connectivity that is used in TGW.
Status	Displays the status of the TGW.
Span	Displays the type of network span that is used by the TGW.
TGW Edge Cluster	Displays the details of the TGW cluster.
External Connections	Displays the number of external connections available for this TGW.
Connectivity Profiles	Displays the number of connectivity profiles that are used in this TGW.
Connected VPCs	Displays the number of VPCs connected to this TGW.
Connectivity Policies	
By Status	Displays the status of the connectivity policy.
By Type	Displays the type of connectivity policy applied.
North-South Traffic	
North-South Traffic	Displays the statistics for all traffic in the north-south service.
Received(Rx)	Total number of packets that are received successfully by the receiving network.
Transmitted (Tx)	Total number of packets that were transmitted successfully.
Rx Dropped Packets	Total number of packets that are dropped by the receiving network.
Tx Dropped	Total number of packets that were dropped before transmission.
Rx Blocked Packets	Total number of packets that are blocked by the receiving network.
East-West Traffic	
East-West Traffic	Displays the statistics for all traffic in the east-west service.
Received(Rx)	Total number of packets that are received successfully by the receiving network.
Transmitted (Tx)	Total number of packets that were transmitted successfully.
Rx Dropped Packets	Total number of packets that are dropped by the receiving network.
Tx Dropped	Total number of packets that were dropped before transmission.
Rx Blocked Packets	Total number of packets that are blocked by the receiving network.

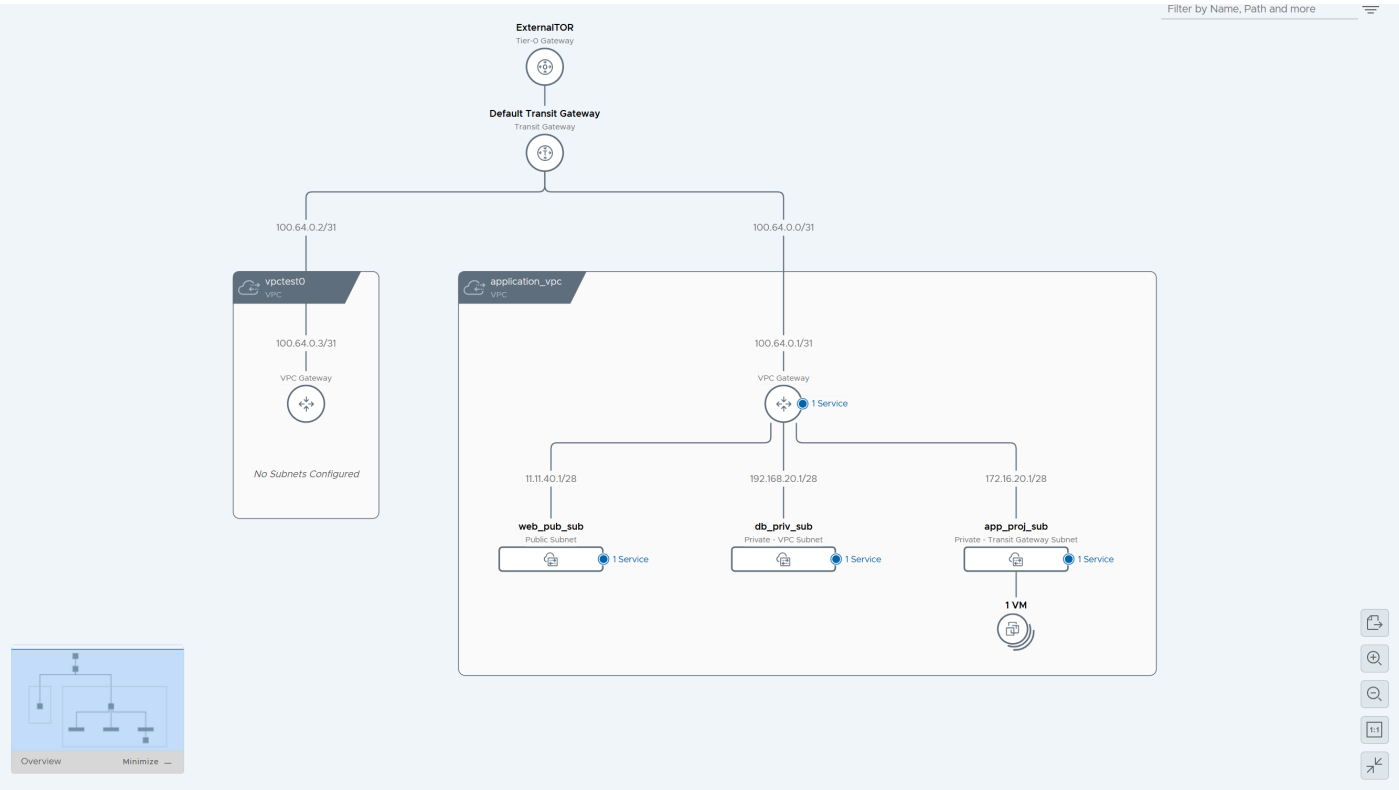
View the Topology of Transit Gateway

View the structure and components of a Virtual Private Cloud by using its topology diagram. The topology diagram of a VPC provides a visual representation of the VPCs and Subnets that are connected to the switch.

1. In the *vSphere Client*, navigate to **Virtual Private Clouds** folder in your datacenter.
2. Select a transit gateway under **Virtual Private Clouds** folder.
3. Select **Configure > Settings > Topology.**
4. The topology diagram of a transit gateway is displayed.

Figure 398: Transit Gateway Topology

Visual description of transit gateway



Configure Connectivity Profiles

A connectivity profile defines a common northbound connectivity configuration for VPCs. When a project is created, a default connectivity profile is also created for it.

1. In the *vSphere Client*, navigate to **Virtual Private Clouds** folder in your datacenter.
2. Select **Configure > Settings > Connectivity Profiles.**

Option	Description
Name	Enter a name for the VPC connectivity profile.
Transit Gateway	By default, the system-created default transit gateway is already selected.
External IP Blocks	Enter external IP blocks that can be used to create public subnets and also the External IP binding for 1:1 NAT in the VPC. The entered IP blocks allow outside connectivity to VPC workloads either on public subnets or through external IPs.

Option	Description
Private-Transit Gateway IP Blocks	Enter private IP blocks to create private subnets in the VPC. The entered IP blocks are available for inter-VPC communication. Networks created from the Private - Transit Gateway IP blocks are not advertised.
VPC Service Gateway Configurations	
Edge Cluster	Select an edge or Virtual Network Appliance (VNA) cluster for this profile. If the transit gateway has a centralized connection, select an edge cluster. If the transit gateway has a distributed connection, select a VNA cluster.
N-S Services	Turn on the toggle to enable service gateway for the connected subnets and to support centralized services such as N-S firewall, NAT, or gateway QoS profile in the VPC.
Default Outbound NAT	Turn on the toggle to ensure that workloads on private subnets have outbound access outside of VPC. One SNAT rule per VPC is auto-plumbed with IP used for translation that is taken from an External IP block for this VPC. **Note:** This option is available only if the **N-S Services** toggle is turned on.
Auto SNAT IP Block	Enter external IP blocks that can be used for default outbound NAT while connecting to external networks.

Configure a VPC Service Profile

A VPC service profile is a collection of services (DHCP and subnet profiles). It simplifies the configuration of common services on multiple VPCs. When a project is created, a default service profile is created for it.

1. In the *vSphere Client*, navigate to **Virtual Private Clouds** folder in your datacenter.
2. Select **Configure > Settings > Service Profiles** and click **Edit**.

Option	Description
Enable DHCP Server Config	Turn on the toggle to enable the DHCP server. If you create a subnet and select the DHCP server option for the subnet, the settings that are configured here are applied to the subnets. A DHCP server can either be a Centralized or a Distributed. By default, the DHCP server mode is Distributed. In case the VPC has a Distributed connectivity, the DHCP server mode can only be Distributed DHCP. If the VPC has a Centralized connectivity, the DHCP server mode can either be a Centralized DHCP server or a Distributed server with the default mode being a Distributed server. You can also use the NSX API to change the DHCP server mode. For more information about the API, see the **NSX API Guide**.
DNS Server IPs	Enter the IPs of DNS servers that are provided to clients by the DHCP server. **Note:** If you require to configure DNS servers, the **Enable DHCP Server Config** toggle must be on.
Lease Time (Seconds)	Enter the time for which clients can lease IP addresses from the DHCP server.
NTP Server IPs	Enter the IPs of NTP servers that are provided to clients by the DHCP server.

Option	Description
Enable DHCP Relay Config	Turn on the toggle to enable the relay-type DHCP configuration for relaying DHCP client requests to external DHCP servers deployed outside the VPC. If you create a public subnet and select the DHCP relay option for the subnet, the settings that are configured here are applied to the subnet's DHCP relay. **Note:** DHCP relay has the following support limitations: • DHCP client requests can only be relayed to external DHCP servers with IP addresses located outside the NSX environment. • DHCP relay is supported for public subnets only. It is not supported for private subnets. • In version 9.1.1 and later, DHCP relay is supported for transit gateways with centralized, distributed VLAN, and distributed VXLAN connections. • In version 9.1.0, DHCP relay is supported only for transit gateways with centralized connections.
External DHCP Server Addresses	Enter the addresses of external DHCP servers. **Note:** If you require to configure external DHCP servers, the **Enable DHCP Relay Config** toggle must be on.

Add an IP Block

You can configure IP address blocks for use by other components.

1. In the *vSphere Client*, navigate to **Virtual Private Clouds** folder in your datacenter.
2. Select **Virtual Private Clouds** from the inventory or a VPC from the list of VPCs are under **Virtual Private Clouds** folder.
3. Select **Configure > Settings > IP Blocks** and click **Add**.
4. Enter a name and other information that is required for adding an IP address block.

Option	Description
CIDRs	Enter an IP block in CIDR format.
IP Address Ranges	Enter an IP address ranges in CIDR format.
Visibility	This option allows you to determine the visibility of the contents of IP blocks for tenants. Select the option from the drop-down menu. • Private • External
Excluded IPs	The IPs that are excluded from the list of IP blocks.
Reserved for Specific Subnet	You can reserve certain IP addresses from an IP block by not allowing them to be allocated to tenants as subnets. You can use this option when a TGW has a distributed or direct VLAN connection, where some IPs of a block may already be in use on the VLAN network.

5. Click **Save**.

Manage IP Allocation

You can see, manage, and allocate the IP blocks that are available for VPCs and it displays the IP/CIDRs that are allocated to other services and subnets.

1. In the *vSphere Client*, navigate to **Virtual Private Clouds** folder in your datacenter.
2. Select **Virtual Private Clouds** from the inventory or a VPC from the list of VPCs are under **Virtual Private Clouds** folder.
3. Select **Configure > Settings > IP Allocation .**
4. Click **Request IP/CIDR**.
5. Select an **IP Address Block** from the drop-down list.
6. Click the toggle to **Auto allocate from IPAM**. If **Auto-allocate from IPAM** is set to **No**, provide the **IP Address** in the field.
7. Enter an **IPv4 IP address**.
8. Click **View IPS** to view the IP block usage.

   You can view the overall usage, visibility of CIDR/Ranges, and usage by CIDR/Ramge. You can also search for available IPs.

Create a Virtual Private Cloud in *vCenter*

Learn how to create a VPC in *vCenter*.

A VPC connects to a Transit Gateway (recommended) and can have the following types of subnets:

- **Private - VPC**: A subnet scoped to the VPC and routed with other subnets inside the same VPC, but not to the outside network. NAT is needed if outside communication is required.
- **Private - Transit Gateway**: Subnets allowing inter-VPC connectivity below the Transit Gateway without NAT. It requires IP translation if workloads must be reachable to an outside network.
- **Public**: A subnet that is accessible from outside the environment, from other VPCs but also from customers and workload above the Transit Gateway.

In case you have to grant access to specific workloads on private networks from the outside environment, you can use External IPs.

External IPs: IPs that allow external connectivity for a VM on a private network by performing IP translation. It is implemented with 1:1 NAT on the workload picking an IP from the External IP block and assigning it to the VM.

In addition to External IPs, it is also possible to configure Static Routes and Groups in a VPC.

Regardless of External Connection type (Centralized or Distributed), it is possible to configure Static Routes, Groups, and DHCP Server in a VPC.

1. From your browser, log in to the *vSphere Client*.
2. In the *vSphere Client*, navigate to **Virtual Private Clouds** folder in the *vCenter* Networking inventory. From the **Actions** drop-down menu, select **New VPC**.
3. On the **Basic Info** page, enter the following information.

Option	Description
Name	Enter a name for the VPC.
Description	Enter a description for the VPC.
Private- VPC IP CIDRs	Enter a block of IP CIDRs that can be assigned to private subnets of this VPC. These CIDRs are local to VPCs and can overlap between the VPCs.

Option	Description
Connectivity Profile	Select a connectivity profile for this VPC. For more information about the VPC Connectivity Profile, see Add a VPC Connectivity Profile. The connectivity profile is used to connect VPCs through a transit gateway and does not require you to share a Tier-0 gateway with the project. If your connectivity profile has some missing configurations, you can configure them from the links in the warning that is displayed in the topology.
Connectivity Policy	Select a connectivity policy for this VPC. You can view the transit gateway, connection type, and span information. For example: Transit Gateway: Default Transit Gateway | Centralized | Span: Default Span To create a connectivity policy, enter a name for the policy and select the type of policy. • Isolated: VPCs that have the isolated policy do not communicate with each or outside the group span within a VPC. However, it can communicate with VPCs with promiscuous policy. • Community: VPCs that have the community policy can communicate to the VPCs in the same community or VPCs with a promiscuous policy. These VPCs do not communicate with other communities or isolated VPCs. • Promiscuous: VPCs with promiscuous policy can communicate with one another and all the other VPCs that are connected. To add more VPCs to this policy, see Configure a Connectivity Policy

4. Click **Save**.

   A new VPC is created in *vCenter*.

To edit a VPC in vCenter, navigate to the **Networking** tab, select a VPC under the **Virtual Private Clouds** folder, and click **Configure > Settings > Properties**. Click **Edit** to modify the properties of your VPC.

Add a VPC Subnet in *vCenter*

Learn how to add a VPC subnet in your *vCenter*.

1. From your browser, log in to the *vSphere Client*.
2. Navigate to **Virtual Private Clouds** folder in the *vCenter* Networking inventory and select a VPC.
3. Right-click on the VPC and from the **Actions** drop-down menu select **New Subnet**.
4. On the **Basic Info** page, enter the following information.

Option	Description
Name	Enter the name of the subnet.
VLAN Extension	Enables the VPC subnet to connect at Layer 2 with a VLAN in the physical infrastructure. For more information about extending a VPC subnet to VLAN, see Extending VPC Subnet to VLAN and Onboarding DVPG VMs to VPC Subnets.

Option	Description
Access Mode	Select an access mode for the subnet. If **VLAN Extension** is set to **Yes**, select an option from the drop-down menu. • L2 Only • Public If **VLAN Extension** is set to **No**, select an option from the drop-down menu. • Private VPC • Private Transit Gateway • Public
VLAN ID	Enter the VLAN ID to which the traffic is to be routed.
Gateway CIDR IPv4 Address	Enter the gateway address and netmask through which the traffic should be routed to the external network.
IP Block	Select an IP block with VLAN extension from the drop-down list to allocate a complete CIDR block to be used within this subnet.
Auto allocate Subnet CIDR from IP Blocks	Turn on the toggle **Auto allocate Subnet CIDR from IP Blocks** to auto allocate the IPs and select a subnet size. Based on whether you create a private or a public subnet, IPs are allocated from the private or public CIDRs you defined in the connectivity profile of the VPC. By clicking the **View IPs** link, you can see the list of IP blocks with current IP addresses in use.
IP CIDR	If you turn the **Auto allocate Subnet CIDR from IP Blocks** toggle off, then enter the IP CIDRs.
Subnet Size	The number of IP addresses in a network which is specified by the subnet mask.
Advanced Settings	
VPC Gateway Connectivity	Turn on this toggle to connect this subnet to the other subnets in the VPC.
DHCP Configuration - The **DHCP Configuration** options are available only if the DHCP settings are configured in the **Service Profile** linked to the VPC.	
DHCP Config	Select one of the following options: • **None**: DHCP is not enabled in the subnet and all IP addresses are set statically. • DHCP Server: The DHCP server that is specified in the service profile automatically assigns IP addresses and network settings to VMs connected to the subnet. You must set **Generic DHCP Options**, and **Static Bindings** if you select this option. If the **Auto allocate Subnet CIDR from IP Block** toggle is off, then you must provide IP CIDRs for allocation in the **Reserved IP Ranges** field. For more information, see the NSX DHCP reference, NSX DHCP Configuration Settings: Reference. • **DHCP Relay**: The DHCP relay that is specified in the service profile forwards DHCP messages from VMs connected to the subnet to an external DHCP. **Note:** DHCP relay is supported only for public subnets that are connected to a transit gateway with the following connection types: • In version 9.1.1 and later, centralized, distributed VLAN, and distributed VXLAN connections are supported. • In version 9.1.0, only centralized connections are supported.

Option	Description
External DHCP Server Addresses	This option visible only if **DHCP Config** is set **DHCP Relay**. Displays the IP addresses configured in the default VPC service profile. You can edit the value of this field using NSX UI.

5. Check **Advanced Settings** to customize default DHCP configuration for the subnet.
6. For **DHCP Server**, in the **DHCP Configuration** page, enter the following information.

Option	Description
DNS Server IPs	IPs of DNS servers are displayed. Read-only.
Lease Time (seconds)	The time during which clients can use IP addresses from the DHCP server. Read-only.
NTP Server IPS	The IPs of NTP servers. Read-only.
Reserved IP Ranges	Reserved IP range is used for eliminating conflicts and providing network stability.
Classic Stateless Routes	To set Classic Stateless Routes, click **Add** to enter the Network, then click **Next**, and click **Add Classless Static Route.** 1. Enter **CIDR.** 2. Enter an `IPv4 address`. 3. Click **Add.** 4. Click **Apply.**
Generic DHCP Options	To set generic **DHCP Options**, click **Add** to enter the code and values. Click **Add Generic Option**. 1. Enter **Code** . 2. Enter **Value.** 3. Click **Add.** 4. Click **Apply.**

Option	Description
Static Bindings	To **Set Static Bindings**, click **Add IPV4 Static Binding** to enter the following information • **Name**: Enter a unique display name to identify each static binding. • **MAC Address**: Enter the MAC address of the DHCP client to which you want to bind a static IP address. The following validations apply to MAC address in static bindings: The MAC address must be unique in all the static bindings on a segment that uses a Segment DHCP server. The MAC address must be unique in all the static bindings across all the segments that are connected to the gateway and which use the Gateway DHCP server. • **IP Address**: Required for IPv4 static binding. Enter a single IPv4 address to bind to the MAC address of the client. • **Gateway Address**: Enter the default gateway IP address that the DHCP for IPv4 server must provide to the DHCP client. • **Host Name**: Enter the host name • **Lease Time**: Enter the amount of time in seconds for which the IP address is bound to the DHCP client. When the lease time expires, the IP address becomes invalid and the DHCP server can assign the address to other DHCP clients on the segment. The valid range of values is 60–4294967295. The default value is 86400. • **DHCP Options**: Set DHCP Options • **Description**: Add a description • **Tags**: Add tags to label static bindings so that you can quickly search or filter bindings, troubleshoot and trace binding-related issues, or do other tasks. 1. Click **Add**. 2. Click **Apply**.

7. Click **Next.**
8. On the warning window, read the warning and proceed.
9. For **DHCP Relay**, enter the **External DHCP Server Address**.
   a) Click **Next.**
   b) Click **Finish**.
10. If **DHCP** is set to **None**, click **Next**.
11. Click **Finish**.

To edit a VPC subnet in vCenter, navigate to the **Networking** tab, select a subnet under the **Virtual Private Clouds** folder, and click **Configure > Settings > Properties**. Click **Edit** to modify the properties of your VPC subnet.

Connect Virtual Machines to a Subnet

Learn how to connect virtual machines to a subnet. You can also connect a VM to subnet by editing the VM.

1. From your browser, log in to the *vSphere Client*.
2. Navigate to **Networking > VPC > VPC Subnets**
3. Select a VPC Subnet from your inventory.
4. From the **Actions** drop-down menu, select **Connect VMs.**
5. You can filter using a **Quick Filter** or an **Advanced Filter**.
6. Select virtual machines to connect to the subnet.
7. Select a network adapter from the **Network Adapter** drop-down menu.
8. On the **Ready to Complete** page, click **Finish**.

Add a NAT Rule for VPC

You can add a NAT rule for a VPC for external IP binding for mapping between internal and external networks.

For creating a NAT rule for CTGW, consider the following conditions.

- DNAT rule should be configured at the nearest point to the translated IP, that is, the VPC.
- SNAT rule should be configured at the furthest point till the Match IP is visible - VPC for Private IP and CTGW for Project IP.

1. In the *vSphere Client*, navigate to **Virtual Private Clouds** folder in your datacenter.
2. Select a VPC from the list of VPCs are under **Virtual Private Clouds** folder.
3. Select **Configure > Services > NAT > Add NAT Rule.**

Option	Description
Name	Enter a NAT name.
Action	Select an action. The following actions are available: • SNAT • DNAT • Reflexive • NO SNAT • NO DNAT
Source IP | Port	Enter the source network address and source port. Specify an IP address, or an IP address range in CIDR format. For SNAT, NO_SNAT and Reflexive rules, this is a required field and represents the source network of the packets leaving the network. For DNAT and NO_DNAT rules, optionally it can contain a source network of incoming packets. For a NULL value, this field represents any network.
Destination IP | Port	Enter a destination IP address or an IP address range in CIDR format and destination port. For DNAT and NO_DNAT rules, this is a required field and represents the source network of the packets leaving the network. This field is not applicable for Reflexive.
Select Protocol	Select the required protocol.
Translated IP | Port	Enter a value for translated IP. Specify an IPv4 address, or an IP address range in CIDR format. If the translated IP is less than the match IP for SNAT, it works as PAT.
Enabled	Turn on the toggle to enable the rule.
Logging	Turn on the toggle to enable logging.
Priority	A lower value means a higher priority. The default is 0. A **No SNAT** or **No DNAT** rule should have a higher priority than other rules.

Option	Description
Firewall	The available settings are: • Match External Address - The firewall is applied to the external address of a NAT rule. – For **SNAT**, the external address is the translated source address after NAT is done. – For **DNAT**, the external address is the original destination address before NAT is done. – For **REFLEXIVE**, to egress traffic, the firewall is applied to the translated source address after NAT is done. For ingress traffic, the firewall is applied to the original destination address before NAT is done. • Match Internal Address - Indicates that the firewall is applied to the internal address of a NAT rule. – For **SNAT**, the internal address is the original source address before NAT is done. – For **DNAT**, the internal address is the translated destination address after NAT is done. – For **REFLEXIVE**, for egress traffic, the firewall is applied to the original source address before NAT is done. For ingress traffic, the firewall is applied to the translated destination address after NAT is done. • Bypass - The packet bypasses firewall rules.
Description	Enter a description for NAT.
Tag	Enter the tag and scope for NAT.

4. Click Save.

Assign an External IP

Learn how to assign an external IP to your virtual machines.

1. In the *vSphere Client*, navigate to VM. Alternatively, navigate to a VPC or a Subnet from the **Virtual Private Clouds** folder and click **Configure > External IPs**.
2. From the **Actions** drop-down menu, select **Assign External IP**. In the **Assign External IP** page, enter the settings.

Option	Description
External IP	Click **Set** to set external IPs for the VMs attached to subnets. Select the VM, the NIC, and the option of whether to auto assign the IPs, alternatively provide IPs from an IP block.
Virtual Machine	Shows the virtual machines that are connected to the private subnets in the VPC.
Network Adapter	Shows the network adapters that are not assigned an external IP and connected to private subnets in the VPC.

Unassign an External IP

Learn how to unassign an external IP from your virtual machines.

1. In the *vSphere Client*, navigate to a VPC or a Subnet.
2. Click **Configure > External IPs**.
3. Select the IP listed under **External IP**.
4. Click **Unassign**.
5. Confirm the prompt by clicking **Unassign**.

Configure Network Spans

Network Span provides the ability for the administrative user to limit the availability of VPC networks on a specific set of vSphere clusters. An enterprise administrator can configure the layout of the span that is based on the vSphere clusters. A span is applied to a TGW which makes VPC subnets that are connected to that TGW available across the clusters associated with that span

Cluster-based span is a collection of vSphere clusters. This span is applied at the transit gateway. When a gateway is created, by default the TGW is associated with the default span.

- Exclusive Span: By default vCenter clusters can be shared across multiple spans. Span allows overlapping of vSphere Clusters. You can apply an exclusive span when you want to specify a set of vCenter clusters for a dedicated workload. Clusters that are part of an exclusive span configuration are not shared or overlapped with other span configuration including default span.
- Default Span: Default Span includes all the VTEP vCenter clusters and standalone hosts in the environment excluding the clusters that are part of exclusive span.

1. In the *vSphere Client* home page, select the **Networking** tab and click **Configure.**
2. On the **Configure** tab, expand **Networking** and select **Network Spans**.
3. Alternatively, in the *vSphere Client*, navigate to **Virtual Private Clouds** folder in your datacenter and select **Configure > Settings > Network Spans.**
4. Click `Add Network Span`.
5. Enter a name for the new network span.
6. Toggle **Yes** or **No** for **Exclusive Span**.
7. Select the clusters to be added to the span.
8. Click **Ok**.

For a global span view in vCenter, select the **Networking** tab, and click **Configure.** On the **Configure** tab, expand **Networking,** and select **Network Spans**. This view provides information on spans across different clusters, datacenters, and other objects. You can also view the clusters that are outside the selected vCenter.

Configure a Connectivity Policy

VPC connectivity policy simplifies VPC-to-VPC connectivity management for users with limited networking experience. Administrators can configure VPC-centric policies with a few simple steps and the system translates these policies into forwarding rules that are based on the current VPC state. You can define how VPCs communicate with each other by applying a specific traffic pattern to a VPC. You can define how your VPCs communicate with each other based on workload connectivity requirements.

1. In the *vSphere Client*, select the **Networking** tab and navigate to the **Virtual Private Clouds** folder in your datacenter.
2. Select a transit gateway from the list under the **Virtual Private Clouds** folder.
3. Go to **Configure > Routing and Forward > Connectivity Policies**.
4. Click **Add Policy**.

   For each policy, they belong to a group that is assigned internally. You can add only VPC members to a group. To see the VPCs added to the group, click the number count under VPC Members.

5. Enter a name for the policy. You can reuse the templated policy and can apply it to multiple VPCs.
6. Select the type of policy from the available options.
   - **Isolated**: VPCs with the isolated policy cannot communicate with each or outside the group span within the same VPC. However, they can communicate with VPCs that have a promiscuous policy.
   - **Community**: VPCs that have the community policy can communicate to the VPCs in the same community or VPCs with a promiscuous policy. These VPCs do not communicate with other communities or isolated VPCs.
   - **Promiscuous**: VPCs with promiscuous policy can communicate with one another and all the other VPCs that are connected.
7. Select one or more VPCs to the group to apply the connectivity policy.
8. Select **View the Ungrouped VPCs** to view the VPCs, that are not added to any connectivity policy. These VPCs can connect with each other and can connect to Promiscuous VPCs.
9. Click **Save**.

Manage Groups

Groups that are used in different connectivity policies are displayed. Using vSphere client, you cannot edit any groups.

1. In the *vSphere Client*, navigate to **Virtual Private Clouds** folder in your datacenter.
2. Select **Configure > Settings > Groups.**

   You can view the groups that are used in connectivity policies. If a group is also used by other NSX services, you can navigate to the NSX UI to view more information about the usage of the group.

Monitor your Virtual Private Cloud in *vSphere Client*

Learn how to monitor VPCs and information that is related to the current state of each VPC.

1. To monitor VPCs in your *vSphere Client*, select Virtual Private Clouds, datacenter, a cluster, or a host from the inventory and navigate to **Networks > VPCs** folder.
2. Use the drop-down menu to select a **Quick Filter** or an **Advanced Filter**.
   a) To use **Quick Filter**, enter value in the field.
   b) To use an advanced filter, select **Advanced Filter** from the drop-down menu and click **Add New** to enter the parameters for search.

View the Topology of Virtual Private Cloud in vCenter

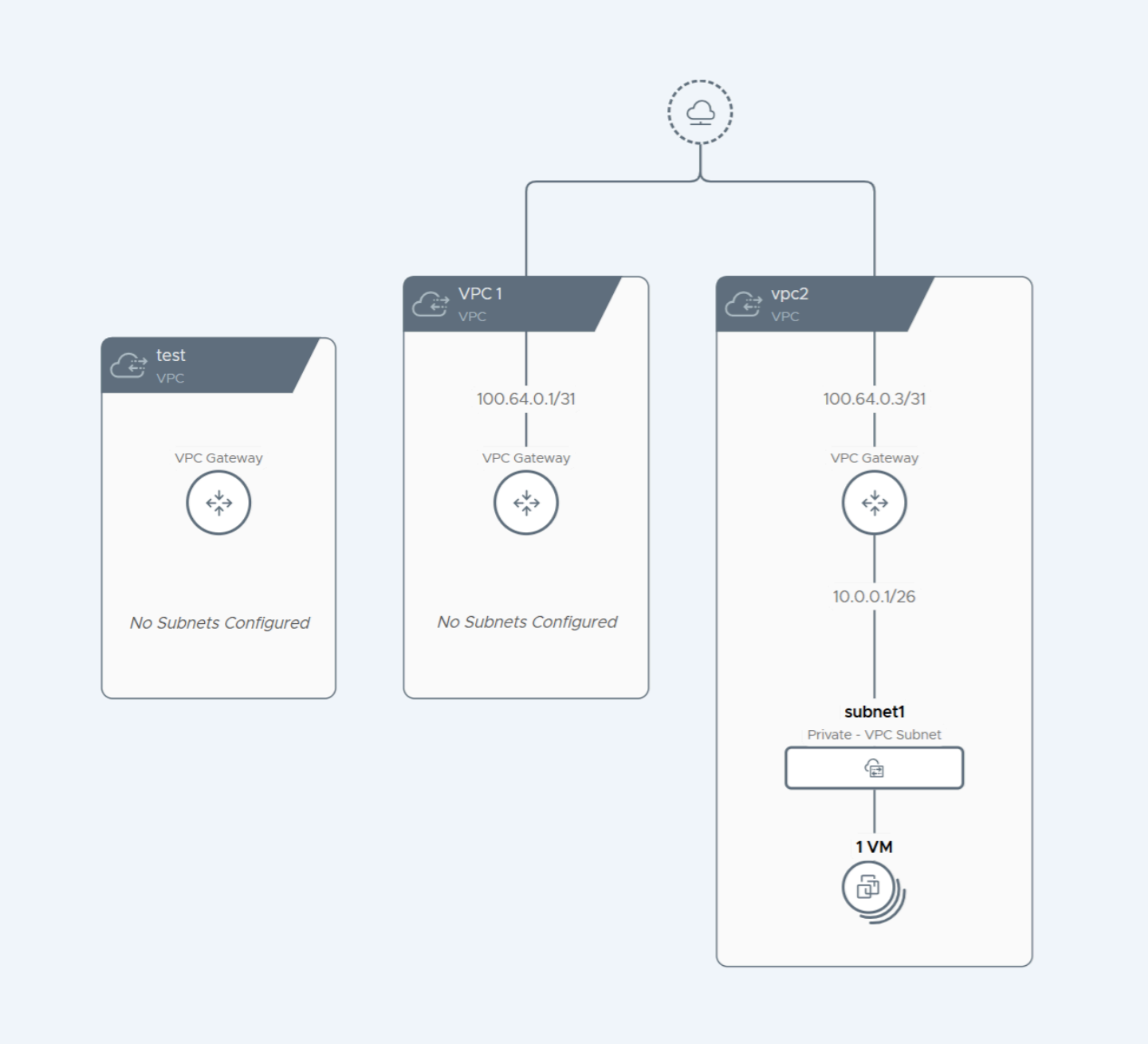
View the structure and components of a Virtual Private Cloud by using its topology diagram. The topology diagram of a VPC provides a visual representation of the VPCs and Subnets that are connected to the switch.

1. In the *vSphere Client*, navigate to **Virtual Private Clouds** folder in your datacenter.
2. Select **Virtual Private Clouds** from the inventory or a VPC from the list of VPCs are under **Virtual Private Clouds** folder.
3. Select **Configure > Settings > Topology.**

The topology diagram of a VPC is displayed.

Figure 399: VPC Topology

Visual representation of VPCs and subnets.



View the Summary of Virtual Private Cloud in *vCenter*

Learn how to view the summary of VPCs in *vCenter*. You can view the information summary for all the VPCs or a specific VPC.

1. To view the summary of a single VPC,
   a) From your browser, log in to the *vSphere Client*.
   b) In the *vSphere Client*, navigate to the **VPC** under **Virtual Private Clouds**.
   c) Click **Summary**.

Option	Description
VPC Details	
Status	Displays the status of the VPC.
VPC Subnets	Displays the number of subnets of this VPC.

Option	Description
External IPs	Displays the number of external IPs assigned to the VM across all datacenters and *vCenter* instances.
VMs	Displays the total number of VMs connected to this VPC across all datacenters and *vCenter* instances.
Private - VPC IP CIDRs	Block of IP CIDRs that are assigned to private subnets of this VPC. These CIDRs are local to VPCs and can overlap between the VPCs.
Network Span	Displays the network span for this VPC.
Connectivity and Services	
External IP Blocks	External IPs for the VMs attached to subnets. Indicates if IP blocks are set.
Private - Transit Gateway IP Blocks	Indicates if the transit gateway IP blocks are set. Private IP blocks to create private subnets in the VPC.
Transit Gateway	Default connectivity profile. Indicates if transit gateway connectivity is configured.
N-S Services	Displays the status of N-S services. Service gateway for the connected subnets and to support centralized services such as N-S firewall, NAT, or gateway QoS profile in the VPC.
Default Outbound NAT IP	Indicates if the traffic from workloads on private and transit gateway private subnet is routed outside the NSX VPC.
Connectivity Policy	Displays the connectivity policy that is applied to this VPC.
Subnets and Workloads	
By Status	Shows the subnets of this VPC by status.
By Access Mode	Indicates the access mode for the subnet. • Private VPC • Private Transit Gateway • Public
Top IP Blocks by Usage	
Name	Name of the IP block.
CIDRs	Enter an IP block in CIDR format.
IP Address Ranges	Enter an IP address ranges in CIDR format.
Visibility	The visibility of the contents of IP blocks for tenants. • Private • External
Excluded IPs	The IPs that are excluded from the list of IP blocks.
Reserved for Specific Subnet	Certain IP addresses that are reserved from an IP block by not allowing them to be allocated to tenants as subnets. This option is used when a TGW has a distributed or direct VLAN connection, where some IPs of a block may already be in use on the VLAN network.
Used by this VPC	Percentage of the IP block used by this VPC.
Overall Usage	Percentage of overall usage of this IP block.
Status	The status of this IP block.
View Statistics	Click to view IP block statistics for the selected block.
Statistics	
North-South Traffic	Displays the statistics for all traffic in the north-south service.
Rx Dropped Packets	Displays the total number of packets that are dropped by the receiving network.

Option	Description
Transmitted (Tx)	Displays the total number of packets that were transmitted successfully.
Tx Dropped	Displays the total number of packets that were dropped before transmission.
Received(Rx)	Displays the total number of packets that were received successfully.
East-West Traffic	Displays the statistics for all traffic in the east-west service.
Rx Dropped Packets	Displays the total number of packets that are dropped by the receiving network.
Transmitted (Tx)	Displays the total number of packets that were transmitted successfully.
Tx Dropped	Displays the total number of packets that were dropped before transmission.
Received(Rx)	Displays the total number of packets that were received successfully.

2. To view the aggregated summary of all the VPCs,
   a) From your browser, log in to the *vSphere Client*.
   b) In the *vSphere Client*, navigate to **Virtual Private Clouds** in the *vCenter* inventory.
   c) Click **Summary**.

Option	Description
Overview	
Transit Gateways	Displays the number of transit gateways.
VPCs	Displays the number of VPCs.
VPC Subnets	Displays the number of subnets of this VPC.
VMs	Displays the total number of VMs connected to the VPC across all datacenters and vCenter Servers.
VPCs	
Top VPCs by number of subnets	Shows the number of subnets of this VPC.
VPCs	Shows the status and available numbers of VPCs.
Top IP Blocks by Usage	Displays the IP blocks used in VPCs. For more information, see #unique_1176/unique_1176_Connect_42_id-2e685d9f-187b-49aa-8300-39c96a059e91.
Transit Gateways	
Transit Gateways	Shows the status and the number of transit gateways that are available.
Transit Gateway Per External Connection	Shows the available number of transit gateways per external connection. • Centralized • Distributed • Distributed VXLAN

Managing your Virtual Private Cloud Subnets in *vCenter*

Learn how to manage VPC subnets in *vCenter*.

Monitor your VPC Subnets in *vCenter*

Learn how vSphere monitors VPC subnets and how it provides information about the current state of each VPC subnet.

1. To monitor VPCs in your *vSphere Client*, you can
2. Navigate to inventory and navigate to **Networks > VPC Subnet**

   Or
3. Select a host, datacenter, cluster, VM, VPC, or a NSX-Managed folder from the inventory and navigate to **Networks > VPC Subnet**
4. Use the `drop-down menu to select a **Quick Filter** or an **Advanced Filter**.
5. To use the advanced filter, select **Advanced Filter** from the drop-down menu and click **Add New** to enter the parameters for search.

View the Summary of VPC Subnet in *vCenter*

Learn how to view the summary of VPC subnet in vCenter.

1. From your browser, log in to the *vSphere Client*.
2. Navigate to the **VPC Subnet**.
3. Click **Summary**.
4. To view the statistics, click `View More`.
5. Cumulative statistics for all traffic through the ports on the subnet are displayed.
6. Click **VPC Gateway Interface** to view more details on data, total packets that are received and transmitted, and dropped packets that are received and transmitted.

   You can view the details of the VPC Subnet.

Option	Description
Access Mode	One of the following options is present: • Private VPC • Private Transit Gateway • Public
VPC Gateway Connectivity	It displays if this subnet is connected to the other subnets in the VPC. One of the options is present: • Yes • No
IP Block	Based on whether you create a private or a public subnet, IPs are allocated from the private or public CIDRs you defined in the connectivity profile of the VPC.
Gateway	The default gateway address.
Network Span	Displays the network span of the VPC subnet.
VMs	Shows the total number virtual machines that are attached to this subnet across all data centers and vCenters.
DHCP Type	One of the following options is present: • **None**: DHCP is not enabled in the subnet and all IP addresses are set statically. • **DHCP Server**: The DHCP server that is specified in the service profile automatically assigns IP addresses. • **DHCP Relay**: The DHCP relay that is specified in the service profile forwards DHCP messages from VMs connected to the subnet to an external DHCP.

Configure Overlay VMKernel Adapter

In VCF 9.1, overlay Tunnel EndPoint (TEP) can now be configured directly from vCenter like other VMkernel NICs. This provides a central place to configure all the services running on ESX hosts. The overlay configuration was supported in NSX and SDDC Manager to provide a configuration for VMK interface like vSAN or vMotion which is configured in vCenter . VI Admin can now configure VMkernel setting in vCenter which will be used by NSX.

1. To create a VMkernel Network Adapter,
2. In the **vSphere Client**, navigate to the host and click **Add Networking**.
3. On the **Select connection** type page, select **VMkernel Network Adapter** and click **Next**.
4. From the **Select an existing network** option, select a distributed port group and click **Next**.
5. On the **Port properties** page, configure the settings for the VMkernel adapter.

Option	Description
Network label	The network label is inherited from the label of the distributed port group.
IP settings	Select IPv4, IPv6, or both. **Note:** The IPv6 option does not appear on hosts that do not have IPv6 enabled.
MTU	Choose whether to get MTU for the network adapter from the switch or to set a custom size. You cannot set the MTU size to a value greater than 9000 bytes.
TCP/IP stack	Select an TCP/IP stack from the list. Once you set an TCP/IP stack for the VMkernel adapter, you cannot change it later. If you select the vMotion or the Provisioning TCP/IP stack, you are able to use only these stacks to handle vMotion or Provisioning traffic on the host. All VMkernel adapters for vMotion on the default TCP/IP stack are deactivated for future vMotion sessions. If you set the Provisioning TCP/IP stack, VMkernel adapters on the default TCP/IP stack are deactivated for operations that include Provisioning traffic, such as virtual machine cold migration, cloning, and snapshot migration. If you select the vNetworking stack, no other service can run on the same VMkernel adapter and a dedicated VMkernel adapter for TEP Traffic is available.

Option	Description
Available services	You can enable services for the default TCP/IP stack on the host. Select from the available services: • **vMotion**. Enables the VMkernel adapter to advertise itself to another host as the network connection where vMotion traffic is sent. The migration with vMotion to the selected host is not possible if the vMotion service is not enabled for any VMkernel adapter on the default TCP/IP stack, or there are no adapters using the vMotion TCP/IP stack. • **Provisioning**. Handles the data that is transferred for virtual machine cold migration, cloning, and snapshot migration. • **Fault Tolerance logging**. Enables Fault Tolerance logging on the host. You can use only one VMkernel adapter for FT traffic per host. • **Management**. Enables the management traffic for the host and vCenter. Typically, hosts have such a VMkernel adapter that is created when the ESX software is installed. You can create another VMkernel adapter for management traffic on the host to provide redundancy. • **vSphere Replication**. Handles the outgoing replication data that is sent from the source ESX host to the vSphere Replication server. • **vSphere Replication NFC**. Handles the incoming replication data on the target replication site. • **vSAN**. Enables the vSAN traffic on the host. Every host that is part of a vSAN cluster must have such a VMkernel adapter. • **vSphere Backup NFC**. VMkernel port setting for dedicated backup NFC traffic. NFC traffic goes through the VMkernel Adapter when vSphereBackup NFC service is enabled. • **NVMe over TCP**. VMkernel port setting for dedicated NVMe over TCP storage traffic. NVMe over TCP storage traffic goes through the VMkernel Adapter when NVMe over TCP adapter is enabled. For more information, see vSphere Storage Guide. • **NVMe over RDMA**. VMkernel port setting for dedicated NVMe over RDMA storage traffic. NVMe over RDMA storage traffic goes through the VMkernel Adapter when NVMe over RDMA adapter is enabled. For more information, see vSphere Storage Guide. • **vNetworking**. To integrate NSX with other VCF components in vCenter, a VMkernel setting for dedicated vNetworking is added to support the overlay configuration in vCenter.

6. On the IPv4 settings page, select an option for obtaining IP addresses.

Option	Description
Obtain IPv4 settings automatically	Use DHCP to obtain IP settings. A DHCP server must be present on the network.
Use static IPv4 settings	Enter the IPv4 IP address and subnet mask for the VMkernel adapter. The VMkernel Default Gateway and DNS server addresses for IPv4 are obtained from the selected TCP/IP stack. Select the Override default gateway for this adapter check box and enter a gateway address, if you want to specify a different gateway for the VMkernel adapter. **Note:** Selected netstack should have the default gateway explicitly on each host.

7. On the IPv6 settings page, select an option for obtaining IPv6 addresses.

Option	Description
Obtain IPv6 addresses automatically through DHCP	Use DHCP to obtain IPv6 addresses. A DHCPv6 server must be present on the network.
Obtain IPv6 addresses automatically through Router Advertisement	Use a router advertisement to obtain IPv6 addresses. In ESX 6.5 and later, router advertisement is enabled by default and supports the M and O flags in accordance with RFC 4861.
Static IPv6 addresses	1. Click Add IPv6 address to add a new IPv6 address. 2. Enter the IPv6 address and subnet prefix length, and click OK. 3. To change the VMkernel default gateway, click Override default gateway for this adapter. The VMkernel Default Gateway address for IPv6 is obtained from the selected TCP/IP stack.

8. Review your settings selections on the **Ready to complete** page and click **Finish**.

Configure vSphere Clusters with Overlay configuration

To integrate NSX closely with other VCF components, VCF 9.1 introduces overlay VMkernel (VTEP) lifecycle management in vCenter. A new VMkernel NIC vNetworking is introduced for NSX overlay VTEPs. NSX applies vNetworking enabled VMkernel (vcenterVtep) when preparing a cluster for overlay networking. Overlay Tunnel EndPoint (TEP) can now be configured directly from vCenter like other VMkernel NICs.

If clusters are imported, upgraded, or if they are non-VCP clusters without an overlay configuration, then they are not VPC ready clusters. You can prepare vSphere clusters with an overlay configuration to support the cases.

- For a vSphere cluster not prepared for NSX, and for a vSphere cluster with VLAN backed NSX configuration, configure the cluster with an overlay on management interface and dedicated overlay TEP.
- For a vSphere Cluster with Overlay on Management Interface enabled, prepare a cluster with a dedicated overlay.

1. In the **vSphere Client** inventory, navigate to the cluster.
2. From the **Actions** drop-down menu, select **vNetworking > Overlay and Network Configuration**.

   Alternatively, you can navigate to the cluster and click **Configure > vNetworking > Overlay and Network Configuration**.
3. Click **Configure** to specify the overlay network settings to prepare this cluster for NSX.

   If **Configure** is disabled, see NSX UI for advanced configurations.
4. Select the vSphere Distributed Switch from the drop-down menu.

   The distributed switches that are configured with overlay transport zones are available in the drop-down menu.
5. Select the **IPv4 Assignment** from the drop-down menu.

   Select **IPv4 assignment** as `Management Kernel Adapter` to configure this cluster with overlay on management interface and select **DHCP** or **IP Pool** for dedicated overlay TEP.

   DHCP: DHCP Server provides IP addresses to the hosts. The external DHCP server allocates the TEP IP.

   IP Pool: Transport Nodes are configured with tunnel endpoint (TEP pool) IP addresses.

   Management VMkernel Adapter: Configure this cluster with an overlay on management interface
6. Select an **IP Pool** from the drop-down menu if **IPv4 Assignment** is set to **IP Pool**.

   To add a new IP Pool, click **Create New**. For more information, see Create an IP Pool for Host Tunnel Endpoint IP Addresses.

7. Enter the value for **VLAN**.
8. Use the toggle to enable or disable **NSX on DVPGs**.
9. Select the **Network Span** from the drop-down menu. For more information, see Configure Network Spans.
10. Click **Configure**.

    The cluster status and host status show **In Progress** while preparing the cluster. Once the cluster and hosts are ready, you can view the details.

    If the cluster preparation has failed, an error message is displayed.

    You can monitor the VMkernel adapters that are reserved for overlay networking are displayed in the relevant list for each host in the NSX-ready cluster in the vCenter.

Traceflow

Traceflow allows you to inject a packet into the network and monitor its flow across the network. This flow allows you to monitor your network and identify disruptions.

Traceflow allows you to identify the path (or paths) a packet takes to reach its destination or, conversely, where a packet is dropped along the way. Each entity reports the packet handling on input and output, so you can determine whether issues occur when receiving a packet or when forwarding the packet. The vCenter graphically displays the trace route, which is based on the parameters that you set (IP address type, traffic type, source, and destination). This display page also enables you to edit the parameters, retrace the trace flow, or create a one.

1. In the *vSphere Client*, navigate to the vCenter for the Workload Domain and select Monitor > Troubleshooting > Traceflow
2. Select an IPv4 or IPv6 address type.
3. Select a traffic type.

   For IPv4 addresses the traffic type choices are Unicast, Multicast, and Broadcast. For IPv6 address the traffic type choices are Unicast or Multicast.

4. Specify the source and destination information according to the traffic type.
5. If **Traffic Type** is set to **Unicast** under **Source**, select a VM or a logical port. For a VM,
   a) Select a VM from the drop-down list.
   b) Select a virtual interface.
   c) The IP address and MAC address are displayed if VMtools is installed in the VM, or if the VM is deployed using OpenStack plug-in (address bindings will be used in this case). If the VM has more than one IP address, select one from the drop-down list.
   d) If the IP address and MAC address are not displayed, enter the IP address and MAC address in the text boxes.
   e) For a logical port, Select an attachment type: **VIF**, **DHCP**, **Edge Uplink**, or **Edge Centralized Service**.
   f) Select a port.
6. If **Traffic Type** is set to **Unicast** under **Destination**, select a VM or a logical port. For a VM,
   a) Select a VM from the drop-down list.
   b) Select a virtual interface.
   c) The IP address and MAC address are displayed if VMtools is installed in the VM or if the VM is deployed using OpenStack plug-in (address bindings will be used in this case). If the VM has more than one IP address, select one from the drop-down list.
   d) If the IP address and MAC address are not displayed, enter the IP address and MAC address in the text boxes.
   e) For a logical port, Select an attachment type: **VIF**, **DHCP**, **Edge Uplink**, or **Edge Centralized Service**.
   f) Select a port.
   g) For IP-MAC, select the trace type (layer 2 or layer 3). For layer 2, enter an IP address and a MAC address. For layer 3, enter an IP address.
7. If the Traffic Type is set to Multicast under Source, performs the steps from 5a to 5f.
8. If the Traffic Type is set to Multicast under Destination, enter an IP Address. It must be a multicast address from 224.0.0.0 - 239.255.255.255.
9. If the Traffic Type is set to Broadcast under Source, performs the steps from 5a to 5f.
10. If the Traffic Type is set to Broadcast under Destination, enter a subnet prefix length.
11. Select a protocol and provide related information.

Option	Description
TCP	Specify a source port, a destination port, and TCP flags.
UDP	Specify a source port and a destination port.
ICMPv6	Specify an ICMP ID and a sequence.
ICMP	Specify an ICMP ID and a sequence.
DHCPv6	Select a DHCP message type: **Solicit**, **Advertise**, **Request**, or **Reply**.
DHCP	Select a DHCP OP code: **Boot Request** or **Boot Reply**.
DNS	Specify an address and select a message type: **Query** or **Response**.

12. Click **Advanced** to set the values per trace session.
    a) In the left column, enter the desired values or input for the following fields:

Option	Description
Hop Limit	This value should be between 2-255.
Next Header	This value should be between 2-255.
Frame Size	The default is 128.
Timeout (seconds)	The default is 10000.
Ethertype	The default is 2048.

Option	Description
Payload Type	Select **Base64**, **Hex**, **Plaintext**, **Binary**, or **Decimal**.
Payload Data	Payload formatted based on the selected type.

b) Under **Global Settings**, enter the values for the following fields.

Option	Description
INT DSCP value for Virtual Network Appliances (VNA)	In-band Network Telemetry (INT) header for the traffic between a Virtual Network Appliance and the ESXi host.
INT DSCP value for VLAN-backed entities.	In-band Network Telemetry (INT) header for traffic for VLAN-backed entities.

13. Click **Trace**.

Note:

Information about the connections, components, and layers is displayed. The output includes a table that lists the details when unicast and logical switch are selected as a destination.

- **Observation Type** (**Delivered**, **Dropped**, **Received**, **Forwarded**)
- **Transport Node**
- **Component**
- A graphical map of the topology

You can apply a filter (**All**, **Delivered**, **Dropped**) on the observations that are displayed. If there are dropped observations, the **Dropped** filter is applied by default. Otherwise, the **All** filter is applied. The graphical map shows the backplane and router links. Bridging information is not displayed.

SDDC Manager Workflows in VMware Cloud Foundation 9.1

With VMware Cloud Foundation 9.0 the SDDC Manager UI is being deprecated. SDDC Manager workflows can now be found in VCF Operations and vSphere Client.

- Managing Licenses in VMware Cloud Foundation
- Managing Certificates in VMware Cloud Foundation
- Managing ESX Hosts in VMware Cloud Foundation
- Managing VCF Domains in VMware Cloud Foundation
- Converging Existing Virtual Infrastructure to a VCF or a vSphere Foundation Platform
- Managing Network Connectivity in vCenter
- Managing Infrastructure Services for VMware Cloud Foundation
- Lifecycle Management of VCF Components
- Managing Passwords for VMware Cloud Foundation Components
- Component Backup and Restore of VMware Cloud Foundation